Correcting Control in “Common Ownership around the World”: Data and Exhibits

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July 27, 2026

Contents

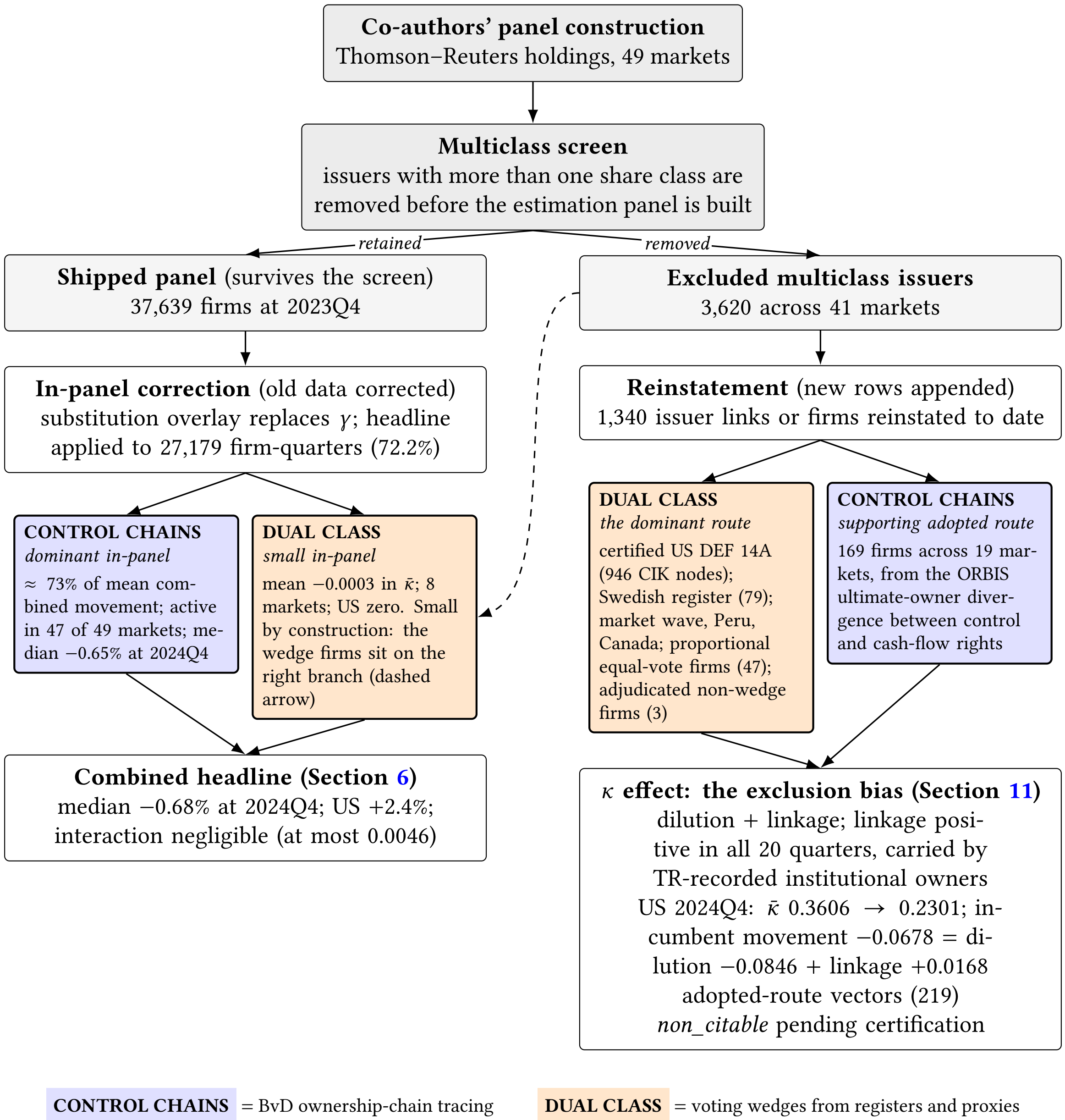
1 Executive summary

This document accompanies two datasets built to correct the control side of the profit weights κ in "Common Ownership around the World." The first reconstructs control chains from Bureau van Dijk ownership filings across the 49 covered markets, tracing pyramids through intermediate holding layers to ultimate owners. The second corrects dual-class voting rights across 37 markets outside the United States. Because each jurisdiction publishes voting rights in its own format, register-specific adapters parse the local disclosures—statutes, prospectuses, annual reports, and regulatory filings—and map them into a common per-holder schema. The most developed instance derives US voting control from SEC DEF 14A proxy statements. This route is needed because the standard US ownership disclosures—13F, 13D/G, and insider filings—report holdings rather than per-holder voting power by class, and because the multiple-class firms themselves are excluded from the co-authors' panel. Voting control for those firms therefore had to be computed from the proxies' beneficial-ownership tables, charter terms, and footnotes. Together, the two datasets supply corrected control vectors for 54,488 firms. They change the vectors of 20,833 of them—38.2% by firm count, and roughly half of covered market capitalisation in the average market. At 2023Q4 the two series cover almost the same share of the co-authors' Thomson Reuters (TR) panel: the headline series applies a corrected vector to 27,179 firms (72.2%), and the demoted permissive variant to 27,685 (73.6%).

Two features of the co-authors' panel are corrected here. The first is the proportional-control assumption behind their κ, under which a holder's control over a firm equals its cash-flow stake, one share carrying one vote. That description is adequate for a widely held firm without a dominant blockholder, while it misstates control in a pyramid, where a family or a state entity commands voting power at each layer well beyond the cash-flow claim obtained by multiplying stakes down the chain, and under a dual-class charter, where a super-voting class concentrates votes in a small economic position. The second is the multiclass screen applied during sample construction, which removed the 3,620 issuers carrying more than one share class before the estimation panel was built. The screen therefore sets a boundary that no in-panel correction can cross, and the firms it removed are precisely those in which votes and cash-flow rights are most likely to diverge.

The screen runs before any κ is computed, which makes it the first branching event and the organizing principle of every exhibit in this report. Firms that survive it form the shipped panel, where the *in-panel* layer corrects the control vectors of firms the co-authors already carry; every shipped-universe number in this report refers to that layer. Firms the screen removed sit outside the panel, and the *panel-enlargement* layer restores them as new panel rows, each carrying its own control vector under per-firm provenance. Because their ownership links were already present in the co-authors' Thomson–Reuters records, the enlargement adds excluded firms without introducing new ownership data. Series computed on the enlarged universe are always labelled as such and never blend into the shipped-universe headline. Figure 1 maps the resulting hierarchy, and the rest of this summary walks it, taking the left branch first and then the right.

Figure 1: The architecture of the control correction: the screen partitions the universe, the mechanisms decompose each branch



Note: Series, bases, and caveats as defined in Section 2.6. The diagram reads top down. The co-authors' multiclass screen partitions the corrected universe before any κ is computed; the left branch is the shipped universe on which the three graded series and the channel decomposition of Section 6 are defined, and the right branch is the enlargement universe of Section 11. The mechanism split on the right branch describes the provenance of each reinstated firm's control vector (which route measured it); the κ movement on that branch is generated by panel membership, with the dilution term vector-free and the linkage term carried entirely by owners already present in the Thomson–Reuters data. In-panel figures are 2024Q4 values from the channel-decomposition and attribution exhibits; the United States enlargement figures are the 2024Q4 sample-edge values of Section 11 on the frozen pre-verification netting basis, and their 2023-annual-mean companions are smaller.

On the left branch the correction enters the co-authors' computation as a substitution overlay: for each treated firm the control vector γ is replaced by the corrected one, the holdings data and the other ingredients of the panel are retained, and untreated firms remain unchanged. The exhibits display three series that differ only in which graded records they admit. *The headline series*[1] retains the discovered-controller records accepted by the grading, namely the *verified* discovered controllers and the *vendor-sourced* controllers, and holds out those it cannot corroborate.[2] *The strict series* admits only the verified discovered controllers. *The demoted permissive variant*, the former shipped estimator, additionally retains the held-out records. Under the headline series the median market barely moves, although individual markets move in both directions; the correction that appears large is confined to the demoted permissive variant.

Within the shipped panel, ownership-chain tracing is the dominant source of the κ changes: the chains channel carries about 73% of the mean combined in-panel movement and is active in 47 of the 49 markets. The in-panel dual-class channel is an order of magnitude smaller, active in eight markets with a cross-market mean effect of -0.0003 in $\bar{\kappa}$, and it is small by construction, because the screen had already removed most firms carrying a genuine voting-power wedge before the panel was built. For the United States the screen removed all of them, so the US correction runs through the chains channel alone. The dual-class data therefore matter for κ chiefly through the panel-enlargement layer of the right branch, where the excluded wedge firms re-enter and the exclusion bias is measured.

- The all-market median change in $\bar{\kappa}$ (a market's average profit weight) is -0.68% at 2024Q4 and -0.29% at the composition-stable 2023 annual mean; within the 2024Q4 figure the control chains alone account for -0.65%. The direction of the change varies across markets. It runs downward in the controlling-shareholder markets, including Korea (-18.0%), Mexico (-16.5%), Taiwan (-14.4%), Brazil (-12.7%), Sweden (-12.4%), New Zealand (-12.1%), and Malaysia (-11.2%), and it runs upward in 16 markets. Russia and Saudi Arabia are unchanged at 2024Q4; at the 2023 annual mean Russia is again unchanged while Saudi Arabia moves -0.29%.

- In the United States the correction is positive, consistent with net controller additions: $\bar{\kappa}$ *rises* from 0.3525 (baseline) to 0.3608 at 2024Q4, a +2.4% change (44th of 49 markets by signed movement). The strict series, which admits only verified discovered controllers, moves the all-market median +0.71%, so the median sits just above zero under the most conservative reading.

- The large declines occur under the demoted permissive variant, which retains every discovered-controller record, including the held-out ones, and moves the median -38.26%. Because the attribution exhibit assigns nearly all of that movement to the held-out records the headline series drops, the exhibits retain the variant only as a labeled series outside the headline.

- The headline is a conservative lower bound on the coverage of newly discovered con-

[1] Internally *net_main_secondary* in the machine-readable outputs, named for the two record tiers it admits.

[2] *Held out* means excluded from every displayed series and kept at the co-authors' baseline vector. The held-out records are not all rejected: under the current reason-code convention, 62.0% of their mass at 2024Q4 lies in the two default unresolved classes and 38.0% falls outside them; those codes do not constitute a row-level adjudication of refutation.

trollers: because a record that fails verification is held at the co-authors' baseline vector, any correction it would carry stays outside the headline until later verification promotes it. That unverified but potentially genuine correction remains parked in the machine-readable country-quarter cells. The bound is on coverage of discovered controllers only and does not establish a directional bound on $\bar{\kappa}$.

- Each correction channel is also reported in isolation, so either channel can be quoted without the other. The control-chain channel carries about 73.2% of the mean combined in-panel movement. Measured instead by mean absolute 2024Q4 movement, the dual-class channel is about six times smaller than the control-chain channel, and it operates mainly through the reinstated firms.

The right branch measures the bias that the screen itself introduces into the shipped panel. The screen removes multiple-class firms disproportionately, among them 360 adjudicated genuine-wedge firms out of 702 scoped US candidates. Restoring these firms both lowers the panel-average level of $\bar{\kappa}$, which the exclusion had left overstated, and adds real common-ownership linkage among the firms that remain. The weights of the two mechanisms invert here. Dual class is the dominant route of reinstatement: the certified DEF 14A build supplies 946 United States CIK nodes and the Swedish register build 79 firms, followed by the audited market wave and the Peruvian and Canadian increments, a proportional equal-vote route for 47 Chinese firms whose votes equal capital by identity, and a route carrying the built two-leg vectors of three adjudicated non-wedge firms. A supporting adopted route derived from the control chains adds 169 firms across 19 markets from the ORBIS ultimate-owner divergence between control and cash-flow rights. To date, 1,340 issuer links or firms have been reinstated, most through the certified United States and Swedish builds and the rest through the later audited increments; the 219 firms entering through the three adopted measurement routes carry vectors that remain *non_citable* pending the next certification cycle.

The United States shows both effects at 2024Q4. On the shipped universe the average $\bar{\kappa}$ is 0.3606 under the headline series, quoted on the reinstatement exhibits' frozen pre-verification netting basis (the regenerated levels exhibits of Section 6 report 0.3608), against a baseline of 0.3525. Once the 946 reinstated firms enter, it falls to 0.2301. The reinstatement dilutes the incumbent average by 0.0846 and, separately, adds 0.0168 of genuine common-ownership linkage, a contribution that is positive in all twenty quarters. The mechanics compress into one image: a reinstated multiclass firm is commonly owned while not commonly controlled, so it enters the κ matrix as a full column — the institutions hold its cash flows, which supplies the linkage — and as a nearly empty row, because its founder holds no rival stakes, which is why dilution dominates. Because the correction is reported on both universes, the exclusion bias can be read directly as the difference between them. That the United States holds 1,603 of the 3,620 excluded issuers is systematic: super-voting structures are the founders' defense against the same deep index ownership that makes measured common ownership highest there, so the screen's bias concentrates in the flagship market. The excluded firms also have a documented anatomy: the DEF 14A derivation records 360 genuine-wedge firms among the US multiple-class issuers the co-authors' panel excludes, and among the 353 of them with a positive computed wedge, the median controller holds 33.9 percentage points more of the votes than of the cash flows, while the widest separation reaches 93.9 points.

The exclusion is panel-wide: 3,620 issuers across 41 markets, censused in Section 10.1, with the complete disposition of every excluded firm reported in Section 11.2.

The headline series rests on record-level grading. Its first component re-weights holdings that TR already records. Beyond that, the chain tracing sometimes discovers a controller that TR does not carry, and each such discovery is a single candidate record, graded by the strength of its evidence into the verified, vendor-sourced, and held-out tiers defined in Section 2. Rebuilding $\bar{\kappa}$ across these grades traces the range the attribution exhibit reports: +0.71% at the median with verified records only, −0.65% once the vendor-sourced tier is added (control chains alone), and −38.26% for the demoted permissive variant. The vendor-sourced tier has since been verified record by record, leaving 997 of the 1,061 records (94%) individually verified and 64 unresolved; Section 8 and Appendix B document that verification and the attribution evidence. The estimator choice is recorded as taken, with the headline set as the verified-plus-vendor-sourced series pending co-author ratification (the decision stands as recorded until the full co-author team formally adopts or revises it). Sensitivity to the modelling rules is likewise slight: the pyramid-propagation rule is immaterial for the full corrected series (median absolute difference 0.0001 in $\bar{\kappa}$), the dual-class attribution rule is immaterial outside two markets (Brazil and Sweden) in the standalone dual-class satellites, and the loyalty quarantine moves the sensitivity previously found in Italy and the Netherlands into the loyalty satellite comparison.

1.1 Roadmap

The report follows the two branches of Figure 1, and within each branch it describes the data before drawing their consequences for κ. Three foundational sections come first. Section 2 introduces the control correction and fixes the series, bases, and caveats used throughout; then Sections 3 and 4 construct the two correction datasets and summarise them. The left branch runs from Section 5 to Section 9, covering in turn the country-by-country coverage and treatment census (Section 5), the corrected levels of common ownership with their decomposition by control channel (Section 6), the time-series results (Section 7), the attribution of the correction by record grade (Section 8), and the sensitivity of the correction to the two consequential modelling rules (Section 9). The right branch occupies Sections 10 and 11. Section 10 describes the excluded universe, giving the market-by-market scope of the exclusion (Section 10.1) and the voting–cash-flow anatomy of the excluded US firms from the DEF 14A proxies (Section 10.2); Section 11 then measures the κ effects of reinstating those firms, together with the panel-enlarged variant (Section 11.1) and the complete disposition of every excluded issuer (Section 11.2). Section 12 lists the open decisions and the pending work. Each section is self-contained, with its own tables, figures, and source notes.

2 Overview of the control correction

2.1 The proportional-control assumption

The co-authors' profit weights κ are computed from Thomson Reuters (TR) institutional holdings under a proportional-control assumption: a holder's control over a firm equals its cash-flow stake, one share carrying one vote. Although that description is adequate for a widely held firm without a dominant blockholder, it does not describe firms in which control exceeds cash-flow rights. In a pyramid, a family or a state entity controls a listed firm through intermediate holding companies, and the controller's voting power at each layer can substantially exceed the cash-flow claim obtained by multiplying stakes down the chain. Under a dual-class charter, a super-voting share class can similarly concentrate votes in a small economic position. In either case, proportional-control κ assigns too much influence to institutional investors and omits the common-ownership links created by the controller's other holdings. The exhibits document a correction to the control side of κ for these firms and describe its data and empirical implications.

2.2 Control-chain and dual-class data

The correction uses two new datasets, whose full construction census is reported in Section 4. The first reconstructs control chains from Bureau van Dijk ownership filings. It traces 755,944 firm–owner links across the 49 covered markets to their ultimate owners, and it identifies a controlling owner—most often a named family or individual—for 48,344 of the 72,807 listed firms. The second dataset corrects dual-class voting rights. It covers 1,701 candidate multiple-class firms in 37 markets outside the United States, of which 897 (52.7%) carry a genuine voting-power wedge. It also covers the US multiple-class issuers that the co-authors' panel excludes altogether. For those US issuers, voting control is derived from SEC DEF 14A proxy statements, and 360 of the 702 in-scope candidates resolve to a wedge.

2.3 Discovered-controller records and grading

The correction first re-weights holdings that TR already records when the traced control structure reassigns the firm's votes; this component rests entirely on holdings the paper already trusts. The ownership chains also sometimes reveal a controller that TR does not carry, such as a family, founder, or state entity holding a large stake in some company. Each such discovery is a single record, asserting that one particular owner controls one particular company. Across the 49 markets, the correction generates 47,392 candidate records of this kind.

The grading of every candidate record determines which records the headline series carries. The *verified discovered controllers* (165 records) were each confirmed against primary sources and have survived repeated adversarial audits. The *vendor-sourced controllers* were graded at 1,381 records drawn from the ownership vendor's chain data. Their measured error rate of 26.0% (one-sided 95% upper bound 38.1%) prompted a record-by-record web-verification. That verification leaves 997 of the 1,061 records in the 2024Q4 roster (94%) individually verified and 64 unresolved (Section 8 and Appendix B). The remaining 45,846 candidates could not be corroborated and are held out of the headline. Under the current reason-code convention, 62.0% of their mass at 2024Q4 lies in the two default unresolved classes, and the other 38.0% falls outside them. These grades attach to individual records, never to firms or countries, so a single firm can carry a verified record alongside a held-out one. The three series differ in which of these grades they carry. The headline series retains the verified and vendor-sourced records and adds a synthetic cash-flow (denominator) leg for each. The strict series admits only the verified tier. The demoted permissive variant additionally retains the 45,846 held-out records. The choice of which series the paper ultimately ships is recorded as taken pending co-author ratification. The exhibits present the headline series alongside the strict series and the demoted permissive variant, with the attribution evidence for the choice.

2.4 The substitution overlay

The correction enters the co-authors' computation as a substitution overlay. For each treated firm the control vector γ is replaced by the corrected one, with the holdings data and the other ingredients of the panel retained; untreated firms remain unchanged. On the cash-flow side, the TR vector β is kept as recorded for holdings TR already carries. For each discovered controller absent from TR, however, the headline series writes a synthetic cash-flow (denominator) leg, so only the strict series retains the TR cash-flow data verbatim. The overlay computes corrected control vectors for 54,488 firms and changes them for 20,833 (38.2%). In the average market, the treated firms account for about half of covered market capitalisation.

The corrected γ is a measured voting share throughout: each holder's fraction of the votes actually attached to its holdings, read from the register or proxy where the figure is published and recomputed from class terms or consolidated chain stakes where it is not. No voting-power index enters the computation. The κ formula retains the co-authors' functional form, in which control is proportional to votes, and the correction changes only the measurement of the votes, so the movement in every exhibit is attributable to measurement rather than to a change of control model. Shapley–Shubik power appears in one auxiliary role: the Aminadav–Papaioannou control classification, under which a firm counts as controlled when its largest ultimate-owner block holds at least 20% of voting rights or is pivotal. That classification drives the census typology and the validation against their published control rates; it never supplies a κ weight. A power-index variant of κ (Banzhaf or Shapley–Shubik weights computed from the same per-holder vote distributions) would be a different estimator family; it can be built from the shipped vectors as a robustness axis but is not part of the current exhibit set. The choice is directionally disclosed: proportionality in votes leaves institutional holders a residual control weight even at firms with a majority or pivotal controller, whereas a model that absorbs that residual into the controller would move each treated firm's profit weights further toward the controller's own portfolio — for United States founder firms further down, because those controllers hold no rival stakes, and for diversified controllers such as the Swedish investment spheres toward stronger intra-group links. The reported reductions at founder-controlled firms are therefore floors with respect to this modelling margin.

Measured common ownership depends on the control assumption used to construct it; the estimator choice is recorded as taken pending co-author ratification, and the exhibits that follow report the results on levels, dynamics, provenance, and robustness.

2.5 The architecture of the correction: screen first, then mechanism

Figure 1 in Section 1 maps the hierarchy that organizes every exhibit in this report: the co-authors' multiclass screen partitions the corrected universe into the shipped panel and the excluded issuers, and the decomposition by mechanism, control chains against dual-class voting rights, applies within each branch. Three properties of that structure carry more technical weight than the summary conveys.

The relative weight of the two mechanisms inverts across the branches. Control chains dominate the correction inside the shipped panel, while dual class dominates the reinstatement on the right branch, and a single event produces both facts. The screen moved the firms carrying a genuine voting-power wedge out of the panel and onto the right branch, which leaves the mechanism that measures those wedges with little to correct in the panel and with nearly everything to supply in the reinstatement; the dashed arrow in Figure 1 marks that transfer. The inversion is thus a property of the co-authors' sample construction rather than of the correction data, and it explains why the in-panel exhibits and the enlargement exhibits draw on the two datasets in such different proportions.

The mechanism split also means something different on each branch, and the distinction governs how the right-branch numbers may be read. On the left branch the split is causal for the movement: a treated firm's κ changes because its control vector changes, and the channel decomposition attributes that change to the chains leg or to the dual-class leg. On the right branch the split records provenance, naming the route that measured each reinstated firm's control vector, whereas the κ movement there is generated by panel membership. The dilution term is vector-free, arising from the enlarged averaging denominator alone, so it would appear even if the entrants brought no common-ownership links whatever; the linkage term is carried by owners already present in the Thomson–Reuters data, the institutional cross-holders that hold incumbents and reinstated firms alike. Reading the right-branch route labels as a decomposition of the exclusion bias would therefore attribute a membership effect to a measurement channel.

The additive reading of the mechanism split holds to a small residual. Within the panel, the combined headline movement equals the sum of the chains and dual-class channels plus an interaction whose median across markets is zero and which reaches at most 0.0046 in $\bar{\kappa}$, in Brazil. The same question arises across the branches, where composing the in-panel correction with the reinstatement is likewise inexact, because correcting the incumbents' control vectors also changes their links to the entering firms. At 2024Q4 the naive sum implies a United States $\bar{\kappa}$ of 0.2331 against the exact joint recompute of 0.2301, an interaction of −0.0030.

2.6 Series, bases, and caveats

The exhibits share the following series, reporting bases, and caveats, stated here once and referenced by the individual exhibit notes. Three series are displayed throughout: the *headline series* (internally *net_main_secondary*), admitting the verified and vendor-sourced discovered controllers; the *strict series*, admitting the verified discovered controllers only; and the *demoted permissive variant* (the former shipped estimator), which additionally retains the held-out records. At 2024Q4 the applied roster carries 165 verified, 1,061 vendor-sourced, and 46,129 held-out records. These come from 165, 1,381, and 45,846 graded records respectively, and Appendix B traces the grading-to-applied progression. Under the current two-code convention, 62.0% of held-out mass lies in the default unresolved classes and 38.0% falls outside them. Record grading is orthogonal to universe membership. The three series are computed on the co-authors' shipped firm universe, which excludes the multiclass firms of Section 10 even though their reinstated voting vectors are per-firm adjudicated. A firm can therefore be verified and still be absent from every displayed series, because the shipped sample excludes it. Exhibits that restore those firms are labelled enlarged and reported in Section 11. Levels are reported on two bases: the point-in-time 2024Q4 quarter and the composition-stable 2023 annual mean. Because firms that have left the panel by 2024Q4 drop out of the applied counts, that quarter carries a roster-attrition caveat, and the 2023 annual mean is the more stable basis for comparisons across time. Three alternative voting-control routes — a control-chains route, a proportional equal-vote route, and an adjudicated non-wedge route, described in Section 11.2 — were adopted on 22 July 2026 as admissible measurements, with per-firm route provenance retained. Every adopted-route vector remains *non_citable* pending the next certification cycle. The route-audit ledger is reported in Section 11.2 and Appendix A. All series are anchored to a single reconstruction vintage, whose per-market coverage and overfull profile are documented in Section 3. All numbers reported in these exhibits remain subject to the verification cycle.

3 Data construction

The two correction datasets (a control-chain reconstruction and a non-US dual-class panel), together with a separate US DEF 14A derivation, were built from primary and vendor sources across the 49 covered markets and audited before any number entered a κ computation. This section gives the design principle common to every pipeline (Section 3.1), a one-paragraph summary of each dataset and its verification (Sections 3.2–3.5), and the integration with the co-authors' panel (Section 3.6); the construction mechanics are deferred to Appendix B.

3.1 Deterministic processing and model-assisted judgment

The construction must be reproducible, with every number regenerable by committed code from archived inputs, while accommodating judgments that vary by jurisdiction and cannot be reduced to a universal rule. Examples include whether two names in different scripts denote one family and whether a register entry is a named holder or a dispersed pool. Hand-curated keyword lists cannot make such judgments because a list calibrated on the jurisdictions already inspected fails on the next. The pipelines therefore assign everything jurisdiction-invariant to a deterministic engine (the Shapley–Shubik power computation and voting cutoff, ownership-tree traversal, the pyramid cash-flow product, code parsing, and every screen and bounds check), while confining the judgment tier to jurisdiction-varying decisions grounded in a written per-jurisdiction brief. The model never runs at build time. Instead, each decision is made offline and frozen into a provenance-tagged cache that the engine consumes, so a rebuild reproduces every value and the caches become the population the audits sample. Four constraints bound the tier. Outputs are closed-set values the engine can validate. Defaults are conservative, admitting a controller only on positive evidence, so failure costs coverage rather than inventing numbers. Candidate decisions are consequence-ranked, which concentrates scrutiny on the highest-stakes cases. Finally, provenance tags make any decision reversible in isolation. Table 1 shows how this division applies to each task.

Table 1: Division of labour in the construction pipelines

Stage	Deterministic engine (runs at build time)	Judgment tier (decided offline, cached with provenance)
Acquisition	Vendor pulls and register harvests; immutable raw archives with checksums and pull dates	—
Extraction	Table walking, footnote-to-row linkage, locale and script handling, percentage and type-code parsing	Bounded extraction of narrow fragments where deterministic rules fail or report low confidence
Identity	Surname aggregation, ultimate-owner roll-up, vehicle merging by rule	Name-to-family keying; dispersed-pool classification; same-entity and concert adjudication
Control arithmetic	Shapley–Shubik and cutoff classification, pyramid products, denominators, wedges	—
Sample gating	Screens, bounds guards, quarantine of violating rows	Wedge-status adjudication of candidate multiple-class firms
Verification	Embedded assertions, named anchors, population-level movement guards	Adversarial audits; cross-model for the highest-stakes datasets

3.2 Control-chain reconstruction

The control-chain reconstruction begins from Bureau van Dijk (BvD) ownership filings. It pulls each market's listed-firm universe with its first-level shareholders, and it expands every significant corporate holder recursively through the global ownership database, so that a chain passes through a private holding company to the family or state entity behind it. On the consolidated ownership blocks, the engine computes control from the 20% voting cutoff and Shapley–Shubik pivotality, reconstructed from the worked examples in Aminadav and Papaioannou's appendix. Cash-flow rights β are the pyramid stake product, capped at the control stake γ, so the difference $\gamma - \beta$ measures pyramid attenuation. Owner identity is central, since one owner fragmented across nominees and holding vehicles would omit the cross-firm links κ depends on. The resulting control and family shares are validated country by country against those Aminadav and Papaioannou publish. The consolidation rules, the concert tier, the three identity passes, and the movement guards are given in Appendix B.

3.3 Dual-class voting rights outside the United States

The non-US dual-class dataset measures who holds which share class of each multiple-class firm and how many votes each class carries. Its sources are primary ownership registers scattered across dozens of national regimes with no common format. Register-specific adapters map each regime into a common schema, from which the build computes voting power, cash-flow rights, and the wedge. The per-jurisdiction conventions are fixed: statutory votes-per-share are taken from law where the law fixes them, cash-flow legs are par-weighted where votes attach to nominal capital, and loyalty-share regimes are flagged as a satellite set. Candidate firms are identified from Capital IQ share-class records under the same multiclass screen family the co-authors apply, so identification recall is bounded by that source's class coverage and the counts here are lower bounds on the treatable wedge set. Because counting share classes is an unreliable screen abroad, every candidate firm was adjudicated before entering the panel: of 1,701 candidate multiple-class firms across 37 of the 48 covered non-US markets, 897 (52.7%) carry a genuine voting-power wedge, 389 (22.9%) are adjudicated as carrying no wedge, and 415 (24.4%) yield no usable disclosure and stay untreated. The eleven markets without adjudicated candidates carry recorded coverage reasons — a harvest gated on a disclosure-API key (Japan, with 30 candidate firms pending), a bot-walled register with disclosure withholding (Russia), multiple-class counts that reflect trust rather than dual-class equity structures (Australia, New Zealand), and zero or near-zero candidate counts (the remaining seven) — itemized in Appendix B. Because nothing is imputed, an unpublished leg stays empty and a missing controller is never invented. The construction stages, the bounds and statute-envelope guards, the named controller anchors, and the round-trip gate against the US wedge set are given in Appendix B.

3.4 The United States: deriving voting control from DEF 14A proxy statements

The United States required a separate derivation, because the co-authors' panel excludes US multiple-class issuers outright and US proxy statements rarely report a holder's voting power directly. The pipeline derives it from the Item 403 beneficial-ownership share counts, the charter's votes-per-share terms, and the footnote attribution across affiliated trusts and vehicles, computing the wedge at the consolidated control-group level. This consolidation matters because a parser reading only the ownership table would miss the super-voting class through which a founder-and-trusts group's combined vote exceeds its economic stake. The universe is the excluded-issuer set, bridged to EDGAR by CIK and collapsed to 946 representative CIK nodes. From these, the derivation resolves 360 genuine wedges, and every firm in scope receives a documented verdict. The three-tier extraction, the two denominator conventions, and the held-out golden benchmark are given in Appendix B.

3.5 Verification and reproducibility

Every dataset above was audited before its numbers were admitted downstream. The audit followed four rules. First, review was independent of the build and conducted adversarially; for the highest-stakes datasets, a model family different from the one used in construction checked the results against primary documents, and the US derivation in particular was audited cross-model against the proxy statements before it entered any κ computation. Second, remediation was class-based: each audit finding is enumerated over the full population and corrected as a class rather than case by case. Third, record grading sorts the discovered-controller records into the evidence tiers defined in Section 2.6. Fourth, every exhibit number is reproducible, generated by committed code from provenance-tagged artifacts. The overlap audit's de-duplication round moved 8 vendor-sourced records into the held-out set, so the frozen decision cache retains 1,373 vendor-sourced controllers (from the 1,381 graded) against 45,854 held out. The word *certified* remains reserved for individually verified records and named prior-campaign artifacts, never for a current series, basis, or result.

Verification itself has a fixed per-record meaning throughout this report. A record asserting that owner X controls firm Y at stake γ is *confirmed* only when all of the following hold: at least one primary document independent of the vendor's chain data — a regulator filing, register entry, exchange disclosure, or annual report, read in the local language where necessary — names the same owner after identity normalization, names the same issuer, and supports the control relationship at a stake consistent with γ; and the verdict is cached with its evidence quotation and source reference. A record is *refuted* only by positive contradicting primary evidence, never by the absence of evidence, and a record meeting neither threshold is *unresolved.* The chain traversal that generates a candidate record carries no evidentiary weight toward confirming it, the engine's arithmetic gates can disqualify a record but never confirm one, and a language model's reading of a document is bounded to these conditions and audited cross-model. No record is therefore admitted on the strength of the chain arithmetic, a model assertion, or an engine gate alone.

3.6 Integrating the correction with the co-authors' panel

The co-authors' panel already embeds its own hand corrections, including fund-family aggregation, BlackRock–BGI consolidation, a single consolidated Chinese state owner, and family blocks internal to Thomson Reuters (TR). Because a discovered controller added naively could duplicate an adjustment already made in the panel, four guards prevent this. *Replacement*: a discovered controller matching a position TR carries replaces the matched conduit row. *Absence*: a controller with no TR counterpart is admitted only after a per-name check confirms its absence from the full holder population. *Conservative default*: a record that neither matches nor clears the absence check is quarantined, so ambiguity resolves toward the co-authors' baseline. The panel's showcased Hermès row (66.7%) stands alone, while a competing depth-two conduit fragment was quarantined and nothing added. *Arithmetic backstop*: any issuer whose retained synthetic legs plus TR-recorded cash-flow rights exceed unity reverts to baseline, swept across all 49 markets. In $\gamma \times \beta$ units at 2024Q4 the integration therefore mostly re-attributes rather than adds: 50.5% of discovered-controller mass substitutes for a TR position already present, 48.6% is quarantined and never enters $\bar{\kappa}$, and only 0.9% enters as genuinely new mass under the absence guard. China is the closest overlap case (their consolidated state owner against our tracing through provincial vehicles); Appendix B carries the vehicle-level detail.

Quarantine is a deliberate under-correction: an unverified record absent from TR that violates no guard stays out on its baseline vector until affirmative evidence promotes it, so the headline understates the controllers TR missed. Of the quarantined mass at 2024Q4, 62.0% carries one of the two quarantine reason codes that mark a record unresolved by default and 38.0% falls outside them; those two codes do not by themselves establish row-level refutation. The cross-model overlap audit confirmed the design: the matched assignments substitute rather than sum, and the one confirmed hand-correction overlap, a Spanish family aggregate absorbing a spouse already in TR, was quarantined with seven other flagged records. The same audit establishes that the demoted permissive series double-counts even the verified anchor controllers against their retained TR rows, as the attribution exhibit reports (Section 8). The backstop's firing profile measures the growing arithmetic incompatibility of the 2020–2024 control vintage with earlier cash-flow records: it fires for 6 issuers at 2024Q4, for 127 one quarter earlier, and for roughly 452 per quarter through 2019, across 25,896 issuer-quarters in all. This incompatibility is why the corrected headline is anchored in the 2020–2024 window.

4 Summary statistics for the correction data

The exhibits in this section describe the correction datasets before any profit weight (κ) is computed. Table 2 reports their contents, Table 3 the divergence between corrected control vectors and the cash-flow vectors they replace, and Table 4 the types of identified controllers. Table 21 in Appendix A reports their overlap with the co-authors' Thomson Reuters (TR) holdings panel by market. Every number is computed from the audited pipeline artifacts; counts that also appear in the coverage census (Table 5) are machine-checked against it.

4.1 Construction census

The correction draws on two newly constructed datasets. The first reconstructs corporate control chains from Bureau van Dijk ownership filings across 49 markets. For each of 72,807 listed firms it assigns a control status, tracing the firm's ownership upward through intermediate holding layers to the ultimate owners. The full reconstruction follows 755,944 firm–owner links. Two thirds of the firms have an identified controlling owner, most often a named family or individual; that comes to 48,344 firms, or 66.4% of the total. The reconstruction also records, for each firm's owners, both a traced control stake γ and a cash-flow stake β. Across 54,248 firms this yields 619,853 owner-level records, most of which come straight from the filings: 72.7% are extracted deterministically, while the remainder are resolved by the vendor's ultimate-owner roll-up, nominee look-through, or a cached, provenance-tagged LLM adjudication of the hardest name-resolution cases.

The second dataset covers dual-class voting rights. Outside the United States, we adjudicate candidate multiple-class firms against statutes, prospectuses, and annual reports. The adjudication covers 1,701 firms across 37 of the 48 covered non-US markets, of which 897 (52.7%) carry a genuine voting-power wedge, 389 (22.9%) are adjudicated as carrying no wedge, and 415 (24.4%) yield no extractable determination and stay untreated. The eleven markets without adjudicated candidates are accounted for in Section 3.3 and Appendix B. A subset of the adjudicated firms then enters the frozen integration overlay, each with per-holder voting and cash-flow stakes: 616 firms across 22 markets. Because the co-authors' panel excludes US multiple-class firms, the approach for the United States differs: voting control is derived directly from DEF 14A proxy statements. The derivation starts from 946 representative CIK nodes bridged from the excluded issuers, of which 702 are in-scope multiple-class candidates, and it resolves 360 of these as genuine wedges. The remainder fall into three dispositions: 261 are proportional multiclass structures with no normalized wedge, 66 are single-class artifacts, and 15 are documented as unresolvable. Within the 261 proportional structures, 25 have unequal nominal votes across their economic legs, yet their normalized voting and economic entitlements remain proportional.

Table 2: Construction census of the correction datasets

	N	Markets	Share (%)
Panel A: Control-chain reconstruction (BvD ownership tracing)			
Listed firms with a reconstructed control assignment	72807	49	–
controlled firm: named family or individual	29480	–	40.5
controlled firm: private holding vehicle	15849	–	21.8
controlled firm: state	3015	–	4.1
widely held (with or without a blockholder)	24463	–	33.6
controlled through SS pivotality alone	5635	–	7.7
Ultimate-owner links traced (firm × owner)	755944	49	–
Firms entering the κ overlay universe	54248	49	–
Owner-level corrected records (γ, β legs)	619853	49	–
extracted deterministically	450639	–	72.7
vendor GUO roll-up	168335	–	27.2
nominee look-through	740	–	0.1
LLM-adjudicated (cached, provenance-tagged)	139	–	0.0
Panel B: Dual-class voting rights (non-US markets)			
Candidate multiple-class firms adjudicated	1701	37	–
genuine voting-power wedge	897	31	52.7
equal-vote multiclass	154	–	9.1
non-equity instrument artifact	226	–	13.3
board-representation wedge only	9	–	0.5
no usable disclosure	415	–	24.4
Firms entering the overlay with per-holder γ/β	616	22	–
Holder-level records	3854	22	–
Panel C: United States DEF 14A derivation (proxy statements)			
Excluded-universe CIK bridge	946	1	–
screened out (not multiple-class)	244	–	25.8
scoped multiple-class candidates	702	–	74.2
genuine voting–cash-flow wedge	360	–	51.3
proportional multiclass (no wedge)	261	–	37.2
single-class artifact	66	–	9.4
documented unresolvable	15	–	2.1

Note: Series, bases, and caveats as defined in Section 2.6. The table gives the construction census of the two correction datasets. Panel A reports the control-chain reconstruction from Bureau van Dijk ownership filings over the 49 covered markets (the reference-only Amadeus duplicate of France is excluded). A firm is *controlled* when its largest ultimate-owner block holds at least 20% of the voting rights or is Shapley–Shubik pivotal: 42,709 controlled firms meet the 20% criterion and 5,635, displayed separately, are controlled through pivotality alone; controller categories follow the Aminadav–Papaioannou typology. *Share* is a percentage of the immediately enclosing total, using firms for the typology rows and owner-level records for the extraction rows. Panel B reports the non-US dual-class adjudication (*wedge_status/verdicts.csv*); its share column gives percentages of adjudicated firms. Panel C reports the US DEF 14A derivation of voting control for issuers excluded from the co-authors' panel, with shares as percentages of the 946-node CIK bridge for the screening rows and of the 702 scoped candidates for the verdict rows. The 616-firm/3,854-holder overlay is frozen at its pre-Ericsson integration scaffold; Ericsson is a known integration seam carried in the current verdict census but not the frozen overlay, described in Appendix B.

4.2 Overlap with the co-authors' TR universe

Table 21 in Appendix A reports the overlap with the co-authors' TR holdings panel, market by market. We measure the overlap at a single composition-stable reference quarter, 2023Q4. In that quarter the TR panel carries 37,639 firms across the 49 markets, while the correction datasets cover 54,488. Of the covered firms, 36,793 (67.5%) match a TR issuer through the audited firm crosswalk, on an ISIN or name match. Because coverage is broader than the TR roster in most markets, the overlap is high when viewed from the panel's side. Two series record how many panel firms the correction reaches, and they differ only slightly: the demoted permissive variant reaches 27,685 TR-panel firms (73.6% of the panel), while the headline series reaches 27,179 (72.2%). Both series are defined in Section 2.6, and the gap of 506 firm-quarters between them comprises the firms whose record grading reverts to baseline. The correction reaches more of the panel by value than by firm count: in the average market the applied firms hold 71.5% of TR-panel market capitalisation, so by value the correction covers the bulk of the co-authors' universe. By firm count its reach is narrower, because at many applied firms the corrected vector coincides with the baseline; the control vector actually changes at 20,833 treated firms, well below the applied count.

4.3 Control–cash-flow divergence before κ

Table 3 summarises how far the corrected control vector γ departs from the cash-flow vector β it replaces. It covers the 20,833 treated firms and separates them by treatment channel. For each firm the divergence is the largest gap at any single owner, $|\gamma - \beta|$. This gap is sizeable at the typical treated firm: the median is 0.083, which means 8.3 percentage points of control reassigned at one owner. The upper tail is much larger, reaching 0.375 at the 90th percentile, and the mean across treated firms is 0.144. Divergence is larger in the dual-class channels than in the chains-only channel. Firms treated by both mechanisms have the largest divergence at the mean and median, whereas dual-class-only firms carry the heavier upper tail at the 75th and 90th percentiles. The channel medians are 0.081 for chains-only firms, 0.174 for dual-class-only firms, and 0.231 for the both-mechanisms group. The divergence almost always runs the same way, toward more control than cash flow. The signed $\gamma - \beta$ at the largest-divergence owner is positive at 99.7% of treated firms. For the chains mechanism this holds by construction, because the pyramid cash-flow product cannot exceed the control stake. The dual-class mechanism can attach to inferior-voting blocks as well, so it can take either sign; it is positive at 83.6% of dual-class firms. The correction also changes which owner is the largest controller at 3,672 treated firms, or 17.6%. Summed over all treated firms, the reallocated control mass comes to 1,646 firm-control equivalents, of which the bulk—1,543 equivalents—arises in the chains-only channel.

4.4 Controller types and record grades

Table 4 describes the controllers the reconstruction identifies. Named families and individuals are the most common controller worldwide, controlling 29,480 firms, or 40.5% of the 49-market universe. Their prevalence varies sharply by region: they control 46.5% of firms in Asia–Pacific but only 26.9% in the Americas. The Americas instead hold many more widely-held firms with no blockholder at all, 28.1% of firms there against 6.1% in Asia–Pacific. Private holding vehicles,

Table 3: Pre-κ divergence between corrected control and cash-flow rights, by channel

	Chains only	Dual class only	Both	All treated
Treated firms	20315	329	189	20833
Largest owner-level $\lvert\gamma - \beta\rvert$ per firm				
mean	0.142	0.220	0.269	0.144
25th percentile	0.031	0.048	0.136	0.031
median	0.081	0.174	0.231	0.083
75th percentile	0.199	0.356	0.354	0.203
90th percentile	0.370	0.508	0.500	0.375
Signed $\gamma - \beta$ at that owner				
mean	0.142	0.187	0.258	0.143
share positive (%)	100.0	83.6	96.3	99.7
Identity of the largest controller				
firms whose largest controller changes	3579	48	45	3672
share of treated firms (%)	17.6	14.6	23.8	17.6
Control mass reallocated				
mean $\sum_o \lvert\gamma_o - \beta_o\rvert$ per firm	0.152	0.340	0.498	0.158
aggregate, firm-control equivalents	1543	56	47	1646

Note: Series, bases, and caveats as defined in Section 2.6. The table reports the distribution of control–cash-flow divergence among the 20,833 treated firms, split by treating mechanism (chains only, dual class only, or both; disjoint and summing to the total, with definitions as in Table 20). The firm-level divergence is the largest owner-level $\lvert\gamma - \beta\rvert$ across the firm's owner rows, γ the traced control (chains) or voting-power (dual-class) stake and β the corresponding cash-flow stake, both firm shares in $[0, 1]$. *Signed* $\gamma - \beta$ takes the mechanism with the larger divergence, nonnegative by construction for chains (the pyramid cash-flow product is capped at the control stake) and either sign for dual class. The *largest controller changes* when the largest-γ and largest-β owners differ within the treating mechanism (the 20 dual-class rows with a missing cash-flow leg are excluded here only). *Control mass reallocated* is $\sum_o \lvert\gamma_o - \beta_o\rvert$ over the firm's overlay owner rows, halved in the aggregate row so one unit equals one firm's entire control changing hands; quantiles use the nearest-rank convention.

both traced and opaque, control a further 15,849 firms, or 21.8%. State owners control fewer firms, 3,015 or 4.1%, and concentrate in Asia–Pacific and the Middle East.

Panel B grades the owner-level records that the reconstruction adds to the TR data. Each record asserts that a controller absent from the TR data holds at least 3% of a listed firm. The discovery audit examined 47,392 such candidate records and sorted them into three grades. The smallest group, 165 records, are verified discovered controllers confirmed against primary sources. A further 1,082 are vendor-sourced controllers, which carry a disclosed, measured error rate. The remaining 46,145, or 97.4% of the audited candidates, are held out of the headline. Grade definitions and the reason-code convention are given in Section 2.6. The held-out records are excluded from the headline and strict series, although the demoted permissive variant retains them in full; the attribution exhibit measures what that inclusion does to the results. The note to Table 4 reconciles the 1,381 initially graded vendor-sourced records down to 1,082 and decomposes the rest of the overlay mass.

Table 4: Controller anatomy: typology of identified controllers and record tiers

	N	All (%)	Am. (%)	Eur. (%)	AP (%)	MEA (%)
Panel A: Firm-level controller typology, 49 markets						
Named family or individual	29480	40.5	26.9	42.4	46.5	34.0
Private vehicle, ownership traced	4649	6.4	8.1	9.1	4.6	14.9
Private vehicle, owners unidentified	11200	15.4	11.5	14.3	17.1	20.6
State	3015	4.1	0.8	5.1	5.2	8.8
Widely held, with a blockholder	16052	22.0	24.6	24.6	20.5	19.4
Widely held, no blockholder	8411	11.6	28.1	4.6	6.1	2.4
All firms	72807	100.0	19021	8977	42742	2067
Panel B: Record tiers of discovered controllers and audit coverage						
Candidate discovered records (audited)	47392	7.6	–	–	–	–
verified discovered controllers	165	0.3	–	–	–	–
vendor-sourced controllers	1082	2.3	–	–	–	–
rejected on investigation (quarantined)	46145	97.4	–	–	–	–
Records outside the discovery audit	576315	92.4	–	–	–	–
re-weighting TR-known holdings	143599	24.9	–	–	–	–
owner absent, unaudited candidates	35682	6.2	–	–	–	–
owner below 3% floor or ambiguous	250484	43.5	–	–	–	–
firm not matched to a TR issuer	142696	24.8	–	–	–	–
dual-class records (own channel)	3854	0.7	–	–	–	–

Note: Series, bases, and caveats as defined in Section 2.6. Panel A classifies the 72,807 control-chain firms (Table 2, Panel A) by controlling-owner type on the Aminadav–Papaioannou typology; *All (%)* is the 49-market share and the regional columns each region's share, the final row each region's firm count. Panel B grades the owner-level overlay records by their standing in the discovery audit of the 47,392 owner-absent candidates, on the post-verification headline basis. Overlap-audit and web-verification demotions reduce the 1,381 graded vendor-sourced records to 1,082 (Section 8), the audited total unchanged. The remaining 576,315 overlay records fall outside the audit: 143,599 re-weight holdings TR already carries, 35,682 are further owner-absent candidates, 250,484 name owners below the 3% floor or with an ambiguous match, 142,696 belong to firms unmatched to TR, and 3,854 are dual-class holder records. Panel B percentages are shares of the enclosing total: the 623,707 overlay records (group rows), the 47,392 audited candidates (tier rows), and the 576,315 unaudited records (component rows).

The two audited datasets span the 49 markets and overlap substantially with the co-authors' panel. Within these data, treated firms have economically large control–cash-flow divergences, and the record grading separates verified discoveries from vendor material with a disclosed error rate. The following exhibits report the effects on measured common ownership.

5 Coverage and treatment under the control correction

This section opens the left branch of Figure 1, which corrects control within the panel the co-authors shipped. The control correction replaces each firm's baseline one-share-one-vote control vector with a corrected vector built from ownership-chain tracing and, where they apply, dual-class voting rights. Table 5 reports how many listed firms the correction covers, how many it changes, through which mechanism, and how large the induced change is. It shows the ten markets with the most covered firms together with the 49-market pooled total; the full per-market table is Table 20 in Appendix A. A firm is *covered* when the correction computes a corrected control vector for it; a firm is *treated* when that correction alters its control vector, either because ownership-chain tracing reassigns control away from the direct registered holder or because dual-class voting rights separate votes from cash-flow rights. The mechanism buckets (chains only, dual-class only, and both) are disjoint and sum to the total treated count.

Across the 49 markets, the correction covers 54,488 firms and treats 20,833 of them, or 38.2% by firm count. In the average market, treated firms account for 50.3% of covered-firm market capitalisation, so treatment is concentrated among larger firms. The control-chain channel accounts for nearly all treated firms: of the 20,833 treated, 20,315 are chains-only, 329 are dual-class-only, and 189 are treated by both mechanisms. Treatment rates and the composition of mechanisms vary substantially across markets. These counts are cross-sectional and are computed on the demoted permissive variant defined in §2.6; the basis of these counts, how many firms the headline series actually treats, and the sample screens that bound the census universe are set out after the table.

Table 5: Coverage and treatment of the control correction: ten markets with the most firms covered and pooled total

	Firms covered	Treated firms (by mechanism)			Total treated	Treated firms (%)	Treated mkt. cap (%)	Mean $\lvert\gamma - \beta\rvert$
		Chains	Dual-cl.	Both				
China	11979	3505	78	0	3583	29.9	52.5	0.158
United States	7195	2064	0	0	2064	28.7	17.4	0.085
India	6149	2464	3	2	2469	40.2	71.1	0.125
Japan	4024	1222	0	0	1222	30.4	25.1	0.122
Korea	2745	1172	17	32	1221	44.5	84.9	0.149
Canada	2475	426	1	1	428	17.3	12.9	0.124
Taiwan	2122	772	2	0	774	36.5	13.4	0.093
Australia	1763	731	0	0	731	41.5	16.7	0.080
Vietnam	1574	775	1	0	776	49.3	77.7	0.151
United Kingdom	1506	886	0	1	887	58.9	39.8	0.098
All countries	54488	20315	329	189	20833	38.2	50.3	0.144

Note: Series, bases, and caveats are as defined in §2.6. Markets are ordered by firms covered; the ten with the most covered firms are shown here, with the full 49-market table, complete column definitions, and provenance in Table 20 of Appendix A. *Firms covered* counts distinct listed firms for which the correction computes a corrected control vector; *treated firms* are those whose control vector it alters, from an owner-level wedge $\gamma - \beta$ above a 10^{-6} tolerance, split into chains-only, dual-class-only, and *both*. Within each market, *Treated mkt. cap (%)* is the treated firms' share of covered-firm capitalisation; the pooled row is the unweighted mean across the 47 joinable markets. *Mean* $|\gamma - \beta|$ is the mean over treated firms of the largest within-firm control-share reallocation. The United States enters through ownership-chain tracing only; its DEF 14A dual-class voting vectors are reported separately in the reinstatement analysis of Section 11 and are excluded from this overlay. Counts are cross-sectional on the demoted permissive variant and do not use a quarter filter.

5.0.0.1 Basis of the treated counts.

The covered and treated counts in Table 5 are cross-sectional and are computed on the demoted permissive variant. A firm is counted as treated whenever any candidate owner record moves its control vector, including records the headline series subsequently holds out under record grading, so the count of firms the headline series actually treats is smaller. At the 2023Q4 reference quarter, now counting firm-quarters rather than firms, the headline series applies the overlay to 27,179 of the 37,639 TR-panel firm-quarters (72.2%), against 27,685 (73.6%) under the permissive variant. Record grading thus reverts 506 firm-quarters to baseline, just under 2% of the permissive variant's applied set, and the corresponding 2024Q4 reversion is 148 of 22,172. Grading has little effect on the count of firms reached; its main effect is on the magnitude of each firm's correction. The median market's $\bar{\kappa}$ falls by 0.37% under the headline series against 26.7% under the permissive variant at 2023Q4 (0.68% versus 38.3% at 2024Q4), with the country-level movements developed in Sections 6 and 8.

5.0.0.2 Scope of the sample screens.

The census covers the firms present in the co-authors' estimation panel, which applies several sample screens documented in their *main_OSOV.tex* besides the multiclass exclusion: depositary receipts are excluded; firm-quarters in which a single holder holds more than 98.5% are dropped; firm-quarters whose ownership ratio exceeds 120% are excluded and those between 100% and 120% are re-weighted; firm-quarters with less than 5% ownership coverage are excluded; and low-coverage countries are dropped entirely. The reinstatement analysis of Section 11 undoes *only* the multiclass screen: firms removed by the other screens remain outside both the baseline and the enlarged panels, so the census and the enlargement pertain only to the multiclass-excluded universe. Two of these screens bear on the composition of the correction and warrant the co-authors' attention. The single-holder $>$ 98.5% screen and the $<$ 5% coverage screen may alter the composition of tightly held firms, but the direction of the resulting change in measured common ownership has not been established. The co-authors' screen implementation has not been independently audited.

Across the pooled covered-firm universe, the correction reaches about two firms in five and half of covered capitalisation. Sections 6 and 7 report the resulting changes in measured common ownership, and Section 4 describes the underlying datasets in detail.

6 Corrected levels of common ownership

After accounting for control chains and dual-class voting rights, the corrected country-average profit weight $\bar{\kappa}$ moves only modestly at the centre of the cross-section while remaining sizeable in the concentrated-ownership tail. The unweighted cross-market average change is −0.6% at the composition-stable 2023 annual mean and −2.7% at the 2024Q4 cross-section, and the median market moves by −0.3% and −0.7% respectively. The dispersion across markets is wide: the change ranges from −18.0% to +16.7% at 2024Q4 and from −19.4% to +35.5% at the 2023 annual mean. Throughout, the displayed series is the headline series defined in §2.6 (the graded discovered-controller assignment combined with the dual-class voting correction); the demoted permissive variant appears only as a labeled robustness leg in the channel-decomposition summary. The attribution exhibit decomposes the movement by record grade. The country panel below reports the levels market by market, the figure that follows ranks the markets by the size of the change, and the closing decomposition splits the correction into its two channels.

6.1 Country panel of $\bar{\kappa}$, baseline vs. corrected

Under the headline series the correction generally lowers measured common ownership. It reduces $\bar{\kappa}$ in 28 of the 49 markets at the composition-stable 2023 annual mean and in 31 markets at the terminal 2024Q4 cross-section, while it raises $\bar{\kappa}$ in 20 and 16 markets respectively, a movement consistent with net controller additions. Russia is unchanged at the 2023 annual mean while Saudi Arabia moves −0.29%; both are unchanged at 2024Q4. The 2024Q4 range is −18.0% to +16.7%, and the 2023-mean range is −19.4% to +35.5%. Table 6 reports the country-average profit weight $\bar{\kappa}$ under the baseline one-share-one-vote assignment and under the correction, at both the composition-stable 2023 annual mean and the terminal 2024Q4 cross-section.

Table 6: Country panel: levels of $\bar{\kappa}$, baseline vs. corrected

	2023 annual mean			2024Q4		
	Baseline	Corrected	Δ%	Baseline	Corrected	Δ%
Australia	0.009	0.009	−5.9	0.035	0.036	4.1
Austria	0.010	0.012	14.3	0.019	0.022	16.7
Belgium	0.015	0.015	0.8	0.023	0.023	−1.3
Brazil	0.009	0.008	−8.9	0.081	0.071	−12.7
Canada	0.005	0.005	−5.2	0.031	0.028	−8.4
Chile	0.019	0.018	−0.9	0.020	0.018	−9.4
China	0.048	0.048	1.3	0.044	0.045	1.3
Colombia	0.014	0.014	−2.5	0.016	0.015	−1.0
Croatia	0.030	0.029	−1.3	0.041	0.040	−0.7
Denmark	0.020	0.020	3.0	0.026	0.026	0.7
Egypt	0.064	0.064	0.0	0.069	0.069	0.3
Finland	0.039	0.036	−6.4	0.044	0.041	−7.6

	2023 annual mean			2024Q4		
	Baseline	Corrected	Δ%	Baseline	Corrected	Δ%
France	0.007	0.007	2.6	0.013	0.013	0.2
Germany	0.012	0.011	−10.4	0.018	0.016	−9.3
Greece	0.004	0.004	−1.1	0.004	0.004	−1.9
Hong Kong	0.007	0.007	−0.3	0.016	0.016	−0.3
India	0.001	0.001	1.6	0.001	0.001	0.8
Indonesia	0.000	0.000	2.5	0.000	0.000	2.2
Ireland	0.128	0.131	2.8	0.203	0.216	6.5
Israel	0.014	0.014	0.9	0.017	0.016	−2.4
Italy	0.006	0.006	2.2	0.032	0.031	−0.9
Japan	0.027	0.027	0.5	0.213	0.212	−0.6
Jordan	0.035	0.034	−0.1	0.026	0.026	−0.0
Korea	0.002	0.002	−19.4	0.002	0.002	−18.0
Kuwait	0.028	0.028	−1.3	0.031	0.032	0.7
Malaysia	0.003	0.003	−3.3	0.026	0.023	−11.2
Mexico	0.039	0.033	−15.8	0.040	0.033	−16.5
Morocco	0.018	0.024	35.5	0.020	0.018	−10.3
Netherlands	0.044	0.045	2.1	0.070	0.070	0.8
New Zealand	0.040	0.039	−4.8	0.112	0.098	−12.1
Norway	0.017	0.017	2.4	0.021	0.021	−2.2
Oman	0.117	0.111	−4.8	0.117	0.111	−5.6
Peru	0.007	0.008	1.1	0.016	0.016	−1.7
Philippines	0.003	0.003	2.5	0.003	0.003	2.7
Poland	0.004	0.004	−2.4	0.003	0.003	−0.7
Romania	0.018	0.017	−3.5	0.020	0.020	−2.1
Russia	0.010	0.010	0.0	0.009	0.009	0.0
Saudi Arabia	0.026	0.026	−0.3	0.037	0.037	0.0
Singapore	0.002	0.002	3.1	0.009	0.009	−0.5
South Africa	0.124	0.124	−0.0	0.180	0.178	−0.8
Spain	0.007	0.006	−2.7	0.006	0.006	−4.3
Sweden	0.043	0.041	−6.5	0.041	0.036	−12.4
Switzerland	0.074	0.082	11.1	0.096	0.097	1.2
Taiwan	0.007	0.007	−6.7	0.037	0.031	−14.4
Thailand	0.002	0.002	−7.6	0.003	0.003	−6.4
United Arab Emirates	0.032	0.031	−3.0	0.029	0.029	0.2
United Kingdom	0.039	0.039	−0.6	0.059	0.061	3.8
United States	0.191	0.199	3.9	0.353	0.361	2.4
Vietnam	0.003	0.003	−0.1	0.003	0.003	−0.1

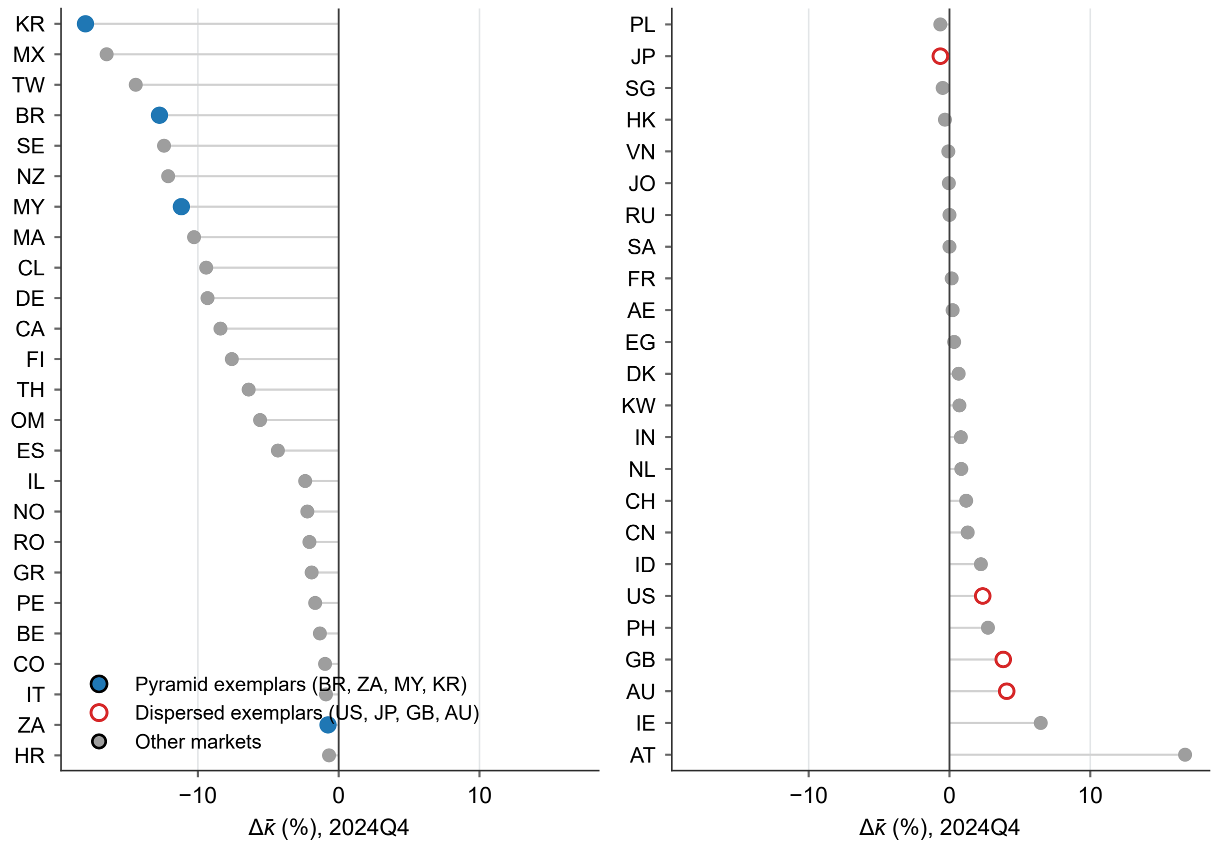
	2023 annual mean			2024Q4		
	Baseline	Corrected	Δ%	Baseline	Corrected	Δ%
Unweighted mean	0.029	0.029	−0.6	0.048	0.047	−2.7

Note: Series, bases, and caveats are as defined in §2.6. Country-average profit weights $\bar{\kappa}$: *Baseline* is the one-share-one-vote assignment and *Corrected* is the headline series, each shown at the 2023 annual mean (the simple average of the four 2023 quarters) and at 2024Q4, with $\Delta\% = 100 \times (\text{corrected/baseline} - 1)$, calculated from unrounded values. Rows are the 49 covered markets, all with full 2023 coverage and present at 2024Q4; the final row is the unweighted cross-country average of each column. The 2024Q4 cross-section carries the roster-attrition caveat of §2.6: in aggregate the covered universe shrinks by about 23% between 2023Q4 and 2024Q4 (from 37,639 to 29,037 firms, with Japan falling by roughly a third in the final quarter alone, −32% from 2024Q3, and per-country rates ranging widely from about −2% to −79%), so the terminal-quarter levels rest on a smaller and probably differently composed roster while the 2023 annual mean is the composition-stable companion.

6.2 Δ% by market

Figure 2 ranks all 49 markets by the change at 2024Q4 and summarises the cross-sectional pattern. Korea, Mexico and Taiwan head the ranking, each losing between about 14 and 18 percent of baseline common ownership, while Malaysia sits a little further back at roughly 11 percent. Two of the four pyramid exemplars, Korea and Brazil, fall among the five deepest reductions, and a third, Malaysia, ranks seventh. The fourth, South Africa, sits in the middle of the ranking (24th), because its large permissive reduction rests on discovered-controller records that the composed headline does not carry. The dispersed-ownership exemplars cluster at the opposite end, where the correction raises measured common ownership: the United States rises 2.4% (44th of 49), the United Kingdom 3.8% and Australia 4.1%. These positive movements are consistent with net controller additions; no owner-flow decomposition is imposed here. The exemplar groups are illustrative anchors along a continuous ordering, as the note details.

Figure 2: Corrected-vs-baseline $\Delta\bar{\kappa}$ (%) by market, 2024Q4



Note: Series, bases, and caveats are as defined in §2.6. Percent change in the country-average $\bar{\kappa}$ from the baseline one-share-one-vote assignment to the headline series at 2024Q4, all 49 markets sorted from the largest reduction. Filled markers are the pyramid/concentrated-control exemplars (Brazil, South Africa, Malaysia, Korea), open markers the dispersed-ownership exemplars (United States, Japan, United Kingdom, Australia), and gray markers the remaining markets; these exemplar groups are illustrative anchors on a continuous ordering without defining a taxonomy, while a formal regime classification remains a pending co-author decision. The 2024Q4 basis carries the roster-attrition caveat of §2.6; at the 2023 annual mean the cross-market average change is −0.6% against −2.7% at 2024Q4, and the country ordering is only partly preserved across the two bases (rank correlation 0.64, largest rank shift 41 places, 19 markets showing a larger reduction at the 2023 mean than at 2024Q4).

6.3 Decomposition by control channel

The control correction combines two data layers that differ in construction. Ownership-chain tracing reassigns control from registered holders to ultimate owners through the cross-jurisdiction control engine, while the dual-class layer separates votes from cash-flow rights using per-firm adjudicated voting terms read from statutes and primary disclosures. Because a reader may weigh the two kinds of evidence differently, the decomposition reports each layer's effect on average common ownership separately and alongside the combined effect. The displayed decomposition satisfies the additive identity $\Delta_{\text{combined}} = \Delta_{\text{chains}} + \Delta_{\text{dual-class}} + \Delta_{\text{interaction}}$ by construction. The interaction is the committed permissive-base approximation rather than a separately estimated headline-plus-dual-class counterfactual, a basis choice disclosed in the generated decomposition artifacts. A reader who accepts only the control-chain evidence can read the chains channel, while one who accepts only the dual-class evidence can read the dual-class channel. The combined channel recovers the headline correction. The interaction channel records the difference between the combined effect and the sum of the two single-channel effects. Table 7 summarises the channels across the 49 markets at 2024Q4, and Table 8 reports the same decomposition market by market.

Across the 49 markets at 2024Q4 the control-chain channel is dominant, carrying about 73.2% of the mean combined movement. Its median effect is a reduction in average $\bar{\kappa}$ of 0.0001, about 0.7 percent of baseline common ownership in the median market, and it is active in 47 of the 49 markets. The dual-class channel, by contrast, is active in 8 markets, with an across-market mean effect of -0.0003 in $\bar{\kappa}$. That footprint is a floor: the screen removed most wedge firms from the panel before it was built (Section 10 measures them on the right branch), and the frozen overlay covers 616 of the 897 genuine-wedge firms, so the channel registers only the residual multiclass firms the screen left in the panel. The interaction, also in $\bar{\kappa}$, is negligible in most markets: its median is zero, it exceeds 10^{-4} in only three markets (Brazil, Sweden, Germany), and it reaches at most 0.0046, in Brazil. The combined correction is therefore close to the sum of its two parts everywhere. At the composition-stable 2023 annual mean the two channels are smaller again. The control-chain channel is the larger of the two in 47 of the 49 markets, and the dual-class channel is larger only in Brazil and Korea. The interaction reaches at most 0.0003 in $\bar{\kappa}$ (in Sweden), while the standalone dual-class channel never exceeds 0.0013 in $\bar{\kappa}$ (in Sweden and Austria). Beneath these headline channels the summary table reports the demoted permissive chains leg together with the other robustness legs. At 2024Q4 the permissive chains channel moves the median market by -0.0091 in $\bar{\kappa}$, or -38.2 percent of baseline, against -0.7 percent of baseline for the netted headline channel, a reduction roughly fifty-five times larger. Accordingly, the demoted permissive variant is no longer the display. For the United States, the separation shows that the correction in this integration vintage operates through ownership-chain tracing only. Its 2024Q4 baseline average $\bar{\kappa}$ of 0.3525 rises by 0.0083 to 0.3608 under the netted headline, an effect entirely attributable to the chains channel with no dual-class contribution. The reinstatement section reports the United States dual-class evidence in its panel-enlargement exhibit.

Table 7: Channel movements in average $\bar{\kappa}$ across the 49 markets (2024Q4)

	Change in $\bar{\kappa}$		% of baseline		Markets active
	Median	Mean	Median	Mean	
Chains (headline)	−0.0001	−0.0006	−0.7	−1.8	47
Dual-class (replacement)	0.0000	−0.0003	0.0	−1.2	8
Interaction	0.0000	0.0001	0.0	0.2	8
Combined	−0.0002	−0.0008	−0.7	−2.7	47
Chains (permissive)	−0.0091	−0.0174	−38.2	−34.7	47
Chains (weak-link)	−0.0087	−0.0173	−36.2	−34.1	47
Chains (strict)	0.0001	0.0015	0.7	2.1	47
Dual-class (overwrite)	0.0000	−0.0001	0.0	−0.4	7

Note: Series, bases, and caveats are as defined in §2.6. The table summarises across the 49 markets the per-market channel changes reported in Table 8. Each row is a correction channel or robustness leg; the columns give the median and mean change in average $\bar{\kappa}$ across markets, the same two statistics as a percentage of baseline, and the count of markets moving $\bar{\kappa}$ by more than 10^{-6}. The four rows above the gap are the composed-headline channels (the control-chain channel under the headline netting, the dual-class channel under the full-replacement denominator convention, their interaction, and the combined headline); the four below are robustness legs. *Chains (permissive)* is the demoted former display series, which retains every discovered-controller candidate record including the held-out tier; *Chains (weak-link)* recomputes that permissive chains leg under the weakest-link control-propagation rule in place of the ultimate-owner trace; *Chains (strict)* admits only the verified discovered controllers; and *Dual-class (overwrite)* applies the holder-overwrite attribution rule, which retains the baseline register and overwrites only the disclosed dual-class holders, in place of the headline's full-replacement rule. Section 9 reports each rule comparison in detail, and the series definitions are those of §2.6. The percentage columns restate the change as the per-market ratio to baseline $\bar{\kappa}$, in percent, with the median and mean taken across all 49 markets; for a channel active in few markets the median ratio is zero even where the mean is not. Because *Markets active* counts movement above 10^{-6} rather than treatment, the two dual-class rows tally differently: the same eight markets are dual-class-treated under both attribution rules, and Italy's movement under the overwrite rule (-9×10^{-7}) falls just below the counting threshold, leaving that row at seven. The interaction is the permissive-base approximation described in the text; components may not sum at displayed precision because of rounding, and it is small throughout, reaching at most 0.0046, in Brazil.

Table 8 carries the same decomposition to the market level, one row per market with the two summary rows repeated at the foot. Reading it row by row shows how differently the mechanisms combine across jurisdictions: New Zealand's −12.1% runs entirely through the chains channel, Brazil is the one market where both channels are large (chains −0.0049, dual class −0.0101) and where the interaction peaks at 0.0046, and Korea's −18.0% is dual-class-led on a baseline of 0.0019.

Table 8: Channel-isolated decomposition of the control correction, by market (2024Q4)

	Baseline $\bar{\kappa}$	Change in $\bar{\kappa}$ by channel				Combined (% of base)
		Chains	Dual-cl.	Inter.	Combined	
New Zealand	0.1116	−0.0135	0.0000	0.0000	−0.0135	−12.1
Brazil	0.0811	−0.0049	−0.0101	0.0046	−0.0103	−12.7
Mexico	0.0399	−0.0066	0.0000	0.0000	−0.0066	−16.5
Oman	0.1173	−0.0065	0.0000	0.0000	−0.0065	−5.6
Taiwan	0.0368	−0.0053	0.0000	0.0000	−0.0053	−14.4
Sweden	0.0406	−0.0034	−0.0021	0.0005	−0.0050	−12.4
Finland	0.0442	−0.0031	−0.0003	0.0000	−0.0034	−7.6
Malaysia	0.0264	−0.0029	0.0000	0.0000	−0.0029	−11.2
Canada	0.0307	−0.0026	0.0000	0.0000	−0.0026	−8.4
Morocco	0.0204	−0.0021	0.0000	0.0000	−0.0021	−10.3
Chile	0.0196	−0.0018	0.0000	0.0000	−0.0018	−9.4
Germany	0.0179	−0.0012	−0.0008	0.0003	−0.0017	−9.3
Japan	0.2133	−0.0014	0.0000	0.0000	−0.0014	−0.6
South Africa	0.1796	−0.0014	0.0000	0.0000	−0.0014	−0.8
Norway	0.0212	−0.0005	0.0000	0.0000	−0.0005	−2.2
Romania	0.0199	−0.0004	0.0000	0.0000	−0.0004	−2.1
Israel	0.0165	−0.0002	−0.0002	0.0000	−0.0004	−2.4
Korea	0.0019	0.0001	−0.0005	0.0001	−0.0003	−18.0
Belgium	0.0234	−0.0003	0.0000	0.0000	−0.0003	−1.3
Italy	0.0317	−0.0003	−0.0001	0.0000	−0.0003	−0.9
Spain	0.0065	−0.0003	0.0000	0.0000	−0.0003	−4.3
Croatia	0.0407	−0.0003	0.0000	0.0000	−0.0003	−0.7
Peru	0.0163	−0.0003	0.0000	0.0000	−0.0003	−1.7
Thailand	0.0034	−0.0002	0.0000	0.0000	−0.0002	−6.4
Colombia	0.0156	−0.0002	0.0000	0.0000	−0.0002	−1.0
Greece	0.0038	−0.0001	0.0000	0.0000	−0.0001	−1.9
Hong Kong	0.0165	−0.0001	0.0000	0.0000	−0.0001	−0.3
Singapore	0.0094	0.0000	0.0000	0.0000	0.0000	−0.5
Poland	0.0028	0.0000	0.0000	0.0000	0.0000	−0.7
Jordan	0.0260	0.0000	0.0000	0.0000	0.0000	−0.0
Vietnam	0.0030	0.0000	0.0000	0.0000	0.0000	−0.1
Russia	0.0089	0.0000	0.0000	0.0000	0.0000	0.0
Saudi Arabia	0.0372	0.0000	0.0000	0.0000	0.0000	0.0
Indonesia	0.0004	0.0000	0.0000	0.0000	0.0000	2.2
India	0.0013	0.0000	0.0000	0.0000	0.0000	0.8
France	0.0129	0.0000	0.0000	0.0000	0.0000	0.2
United Arab Emirates	0.0289	0.0001	0.0000	0.0000	0.0001	0.2
Philippines	0.0027	0.0001	0.0000	0.0000	0.0001	2.7
Denmark	0.0258	0.0002	0.0000	0.0000	0.0002	0.7
Kuwait	0.0313	0.0002	0.0000	0.0000	0.0002	0.7
Egypt	0.0688	0.0002	0.0000	0.0000	0.0002	0.3
China	0.0443	0.0006	0.0000	0.0000	0.0006	1.3
Netherlands	0.0698	0.0006	0.0000	0.0000	0.0006	0.8

	Baseline $\bar{\kappa}$	Change in $\bar{\kappa}$ by channel				Combined (% of base)
		Chains	Dual-cl.	Inter.	Combined	
Switzerland	0.0957	0.0011	0.0000	0.0000	0.0011	1.2
Australia	0.0345	0.0014	0.0000	0.0000	0.0014	4.1
United Kingdom	0.0588	0.0022	0.0000	0.0000	0.0022	3.8
Austria	0.0188	0.0045	−0.0013	0.0000	0.0031	16.7
United States	0.3525	0.0083	0.0000	0.0000	0.0083	2.4
Ireland	0.2026	0.0131	0.0000	0.0000	0.0131	6.5
All 49 (median)	0.0260	−0.0001	0.0000	0.0000	−0.0002	−0.7
All 49 (mean)	0.0476	−0.0006	−0.0003	0.0001	−0.0008	−2.7

Note: Series, bases, and caveats are as defined in §2.6. For each of the 49 markets in the κ panel at 2024Q4, the table reports the baseline one-share-one-vote average of the firm-level common-ownership measure and the change attributable to each correction channel taken separately. *Chains* is ownership-chain tracing alone under the headline netting (*net_main_secondary*); *dual-class* is dual-class voting rights alone under the full-replacement rule; and *interaction* is the combined change minus the two single-channel changes. The combined column therefore equals chains plus dual-class plus interaction by construction, and the final column expresses that combined change as a percentage of baseline. Because each firm is routed to a single mechanism per variant, the applied treated sets are disjoint, and the interaction measures only the cross-channel κ links between a chains-treated and a dual-class-treated firm. The interaction is the committed permissive-base approximation rather than a separately estimated headline-plus-dual-class counterfactual, so displayed components may miss the combined cell by rounding. Markets are ordered by the size of the combined correction, and the two summary rows report the cross-market median and mean. The United States enters through ownership-chain tracing alone in this vintage, so its dual-class and interaction entries are an identified zero. Full per-market provenance is recorded in *decomp_numbers.json*.

Under the headline series the correction at 2024Q4 moves the median market by less than one percent and the average market by a few percent, operating through the control-chain channel in all but a handful of dual-class markets. Positive movements are consistent with, but do not independently establish, net controller additions. The demoted permissive variant, which moved the median market by about −38% of baseline, appears only as a labeled robustness leg. The attribution exhibit reports the movement attributable to each grade of discovered-controller record, and the choice among the series is a pending co-author decision. The survival exhibit reports the corresponding time-series results.

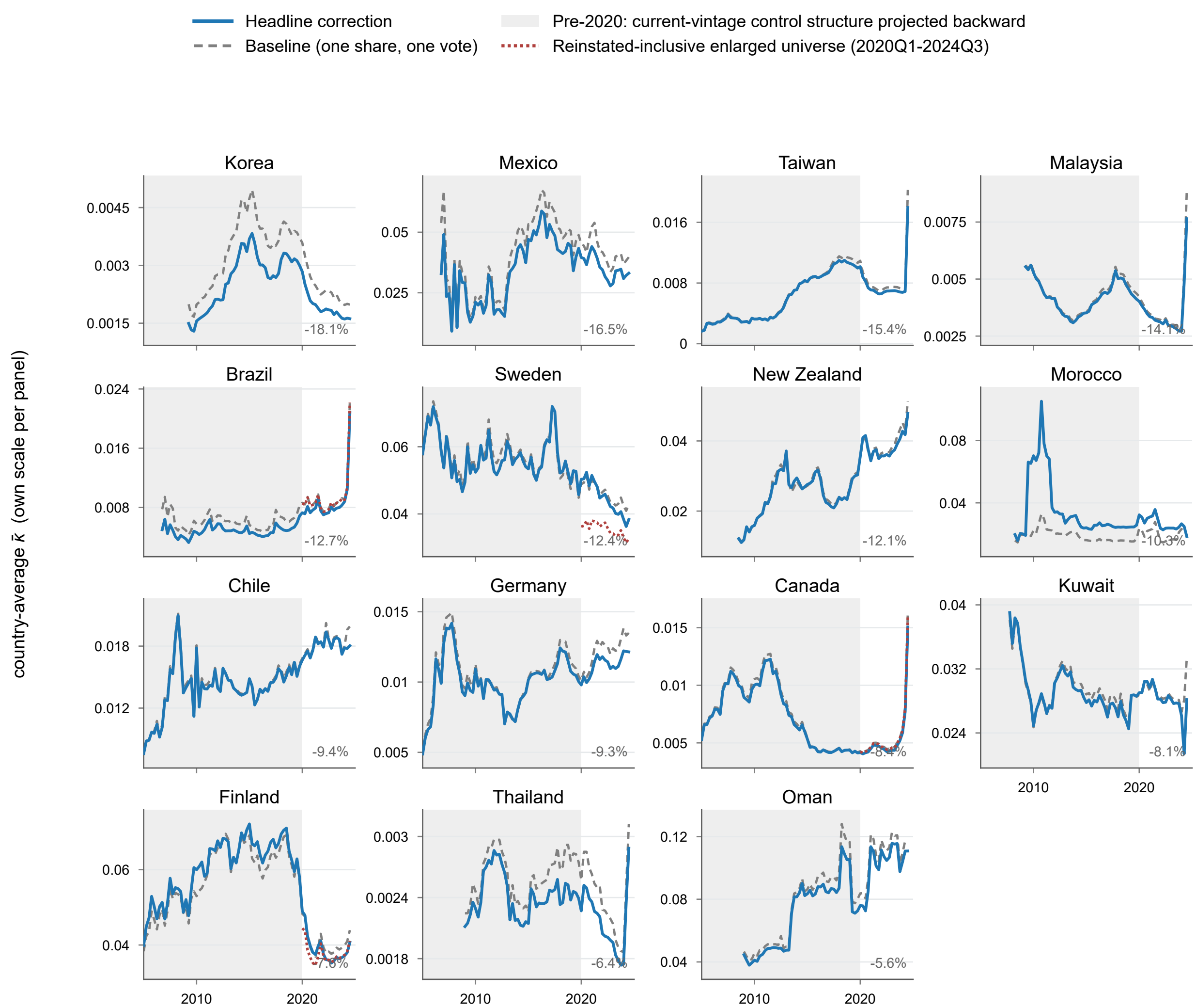
7 Growth of common ownership under the control correction

7.1 Time-series results

The levels exhibits show that the control correction changes measured common ownership modestly and heterogeneously, lowering it at the all-market median while raising it in the United States. The time-series analysis finds that the baseline rise survives the correction across the tabulated markets over each market's available horizon. Figures 3–5 plot the quarterly country-average profit weight $\bar{\kappa}$ over 2005–2024, one panel for each of the 49 jurisdictions, under the baseline one-share-one-vote assignment and under the headline series defined in Section 2.6. The panels are grouped across three figures by the sign and size of the 2024Q4 headline correction: markets where the correction lowers $\bar{\kappa}$ substantially (Figure 3), where it barely moves $\bar{\kappa}$ (Figure 4), and where it raises $\bar{\kappa}$ substantially (Figure 5); the largest positive 2024Q4 correction falls in Austria rather than the United States. For every market with a validated enlarged-universe source (Section 11) — twenty-one panels, of which seventeen carry reinstated firms — each panel also draws that market's reinstated series over its 2020–2024 build window as a dotted red line. For the United States, restoring the excluded firms places this reinstated line about a third below the corrected in-panel line, so the in-panel rise must be read together with the exclusion bias that Section 11 measures. Across the groups the corrected trajectory tracks the shape of the baseline path and preserves its trend. On the shipped universe the US corrected line stays above the baseline throughout, and even the pyramidal and family-controlled markets that take the largest negative corrections keep the direction of their baseline path.

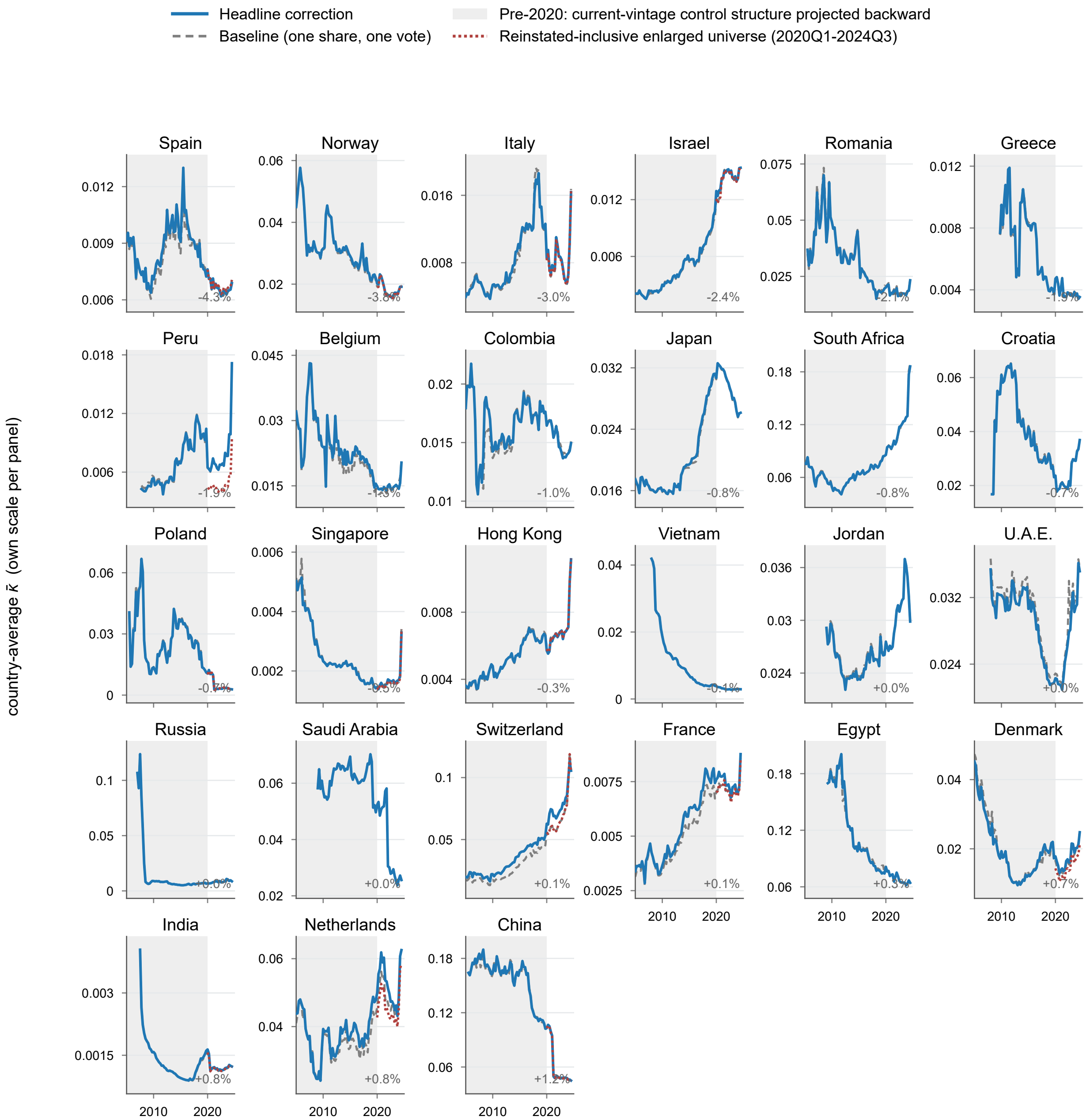
Table 9 reports, for each market and group, the cumulative growth of the country-average profit weight $\bar{\kappa}$: the percent change in its annual mean from a base year (2005 for the earliest markets) to 2023, on the baseline series and again on the corrected headline series. Wherever the baseline rise was real, the correction leaves it substantially intact. For the United States the growth and the level are two separate facts: in growth the corrected series rises 401% against the baseline's 425% (a −24pp difference), while in level the corrected line sits above the baseline in every US quarter. The two are consistent, because the correction lifts the US level in the base year as well as later, so the larger base makes the same rise register as a smaller percentage rather than a weaker rise. The pyramid group's rise instead steepens (130% to 135%), and Japan's is essentially unchanged (+64.7% against +63.7%). Where the baseline rise was small, the correction keeps its sign and shifts its magnitude by at most a few percentage points, even when the proportional change is large: Korea's rise more than doubles (+1.9% to +4.7%), Australia's shrinks by roughly two-thirds (+9.4% to +3.0%), and the dispersed-group average edges from +5.4% to +4.0%. The United Kingdom's and Malaysia's baselines themselves fell over the window, so neither had a rise to preserve. The demoted permissive variant (the former shipped estimator), which retained the discovered-controller records the roster could not corroborate, behaves differently on exactly these markets: under it the same small rises turn into declines (Korea −6.1%, Australia −15.0%, the dispersed group −8.2%). The attribution exhibit decomposes the movement by record grade and traces those reversals to the uncorroborated records the headline series excludes.

Figure 3: Country-average $\bar{\kappa}$ by jurisdiction: markets where the control correction lowers common ownership, 2005–2024



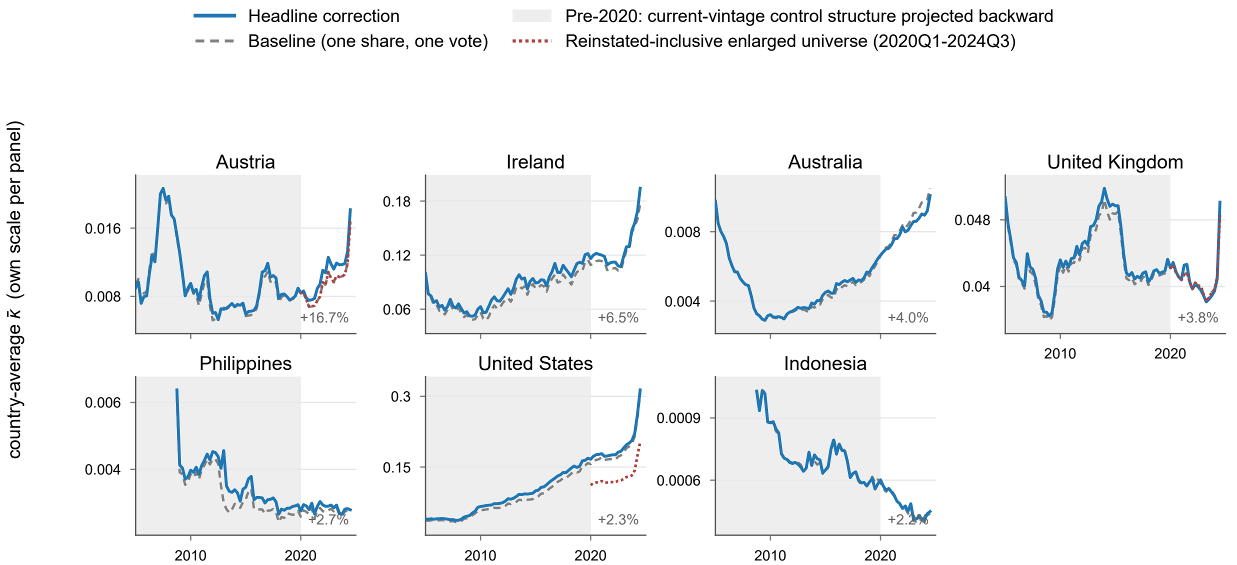
Note: Quarterly country-average $\bar{\kappa}$ under the baseline one-share-one-vote assignment (dashed grey) and the headline series (solid), one panel per jurisdiction on its own vertical scale, for the markets whose 2024Q4 headline correction is below -5% (annotated in each panel, ordered from most negative). Dotted red lines mark the reinstated-inclusive enlarged-universe series in the four panels in this figure where a validated source is available (the baseline with excluded multiclass firms restored), drawn over 2020Q1–2024Q3. Construction, the pre-2020 shaded region, the grouping cutoffs, and the series not drawn are described in the note to Figure 5.

Figure 4: Country-average $\bar{\kappa}$ by jurisdiction: markets where the control correction barely moves common ownership, 2005–2024



Note: Quarterly country-average $\bar{\kappa}$ under the baseline one-share-one-vote assignment (dashed grey) and the headline series (solid), one panel per jurisdiction on its own vertical scale, for the markets whose 2024Q4 headline correction lies between −5% and +2% (annotated in each panel, ordered from most negative to most positive). Construction, the pre-2020 shaded region, the grouping cutoffs, and the series not drawn are described in the note to Figure 5. Dotted red lines mark the reinstated-inclusive enlarged-universe series in the 14 panels in this figure where a validated source is available, drawn over 2020Q1–2024Q3.

Figure 5: Country-average $\bar{\kappa}$ by jurisdiction: markets where the control correction raises common ownership, 2005–2024



Note: Series, bases, and caveats as defined in Section 2.6. Quarterly country-average $\bar{\kappa}$ by jurisdiction under the baseline one-share-one-vote assignment (dashed grey) and the headline series (solid), each panel on its own vertical scale (levels span orders of magnitude), which limits cross-panel comparisons to shape. The 49 jurisdictions are split across the three figures by the 2024Q4 headline correction $100(\bar{\kappa}^{\text{headline}}/\bar{\kappa}^{\text{baseline}} - 1)$, the level-correction magnitude reported in the levels exhibit: above +2% here, between −5% and +2% in Figure 4, and below −5% in Figure 3, the two cutoffs sitting at natural gaps in the sorted distribution; each panel is annotated with its correction. The shaded pre-2020 region applies the 2020–24 control-structure vintage backward, and the pre-2020 display remains under audit because current-vintage correction evidence is citable only for 2020Q1–2024Q4. Each trajectory ends at 2024Q3; 25 of the 49 panels begin at 2005Q1 and the other 24 begin between 2005Q3 and 2009Q4. The terminal quarter (2024Q4) is excluded for the roster-attrition reason in the note to Table 9 but still defines group membership, matching the level exhibits. Dotted red lines mark the reinstated-inclusive enlarged-universe series wherever a validated source is available: 3 panels here and 21 panels across the three figures, drawn over 2020Q1–2024Q3. The United States series applies the headline correction and reinstates the excluded multiclass firms under the certified DEF 14A vectors; the other 20 enlarged series add their per-market reinstated firms to the baseline. The solid and dashed lines in every panel are otherwise computed on the shipped universe, which excludes the reinstated firms. The control-chain-only series, the loyalty-restored satellite, and the earlier group-average lines are omitted. The control-chain-only line differs from the headline in 46 of 49 panels somewhere over the plotted span, in 12 panels post-2020, and in 9 markets at 2024Q4; applying a 10^{-6} activity threshold gives 8 dual-class-active markets at 2024Q4. The growth table reports the channel split and retains the country groups.

Table 9: Growth of $\bar{\kappa}$: baseline vs. corrected, by country group

	Base year	Baseline (OSOV)			Corrected			Δ growth (pp)
		Base	2023	Growth (%)	Base	2023	Growth (%)	
United States	2005	0.0364	0.1913	425.3	0.0397	0.1987	401.1	−24.1
Pyramid group	2010	0.0150	0.0345	129.7	0.0146	0.0342	134.9	5.1
Brazil	2007	0.0081	0.0087	7.6	0.0054	0.0080	48.1	40.5
South Africa	2005	0.0766	0.1243	62.2	0.0766	0.1242	62.3	0.1
Malaysia	2010	0.0050	0.0030	−41.0	0.0050	0.0028	−42.5	−1.5
Korea	2010	0.0021	0.0021	1.9	0.0016	0.0017	4.7	2.8
Dispersed group	2005	0.0239	0.0252	5.4	0.0240	0.0249	4.0	−1.4
United Kingdom	2005	0.0464	0.0388	−16.3	0.0468	0.0386	−17.5	−1.2
Japan	2005	0.0167	0.0274	63.7	0.0167	0.0275	64.7	1.0
Australia	2005	0.0085	0.0093	9.4	0.0085	0.0087	3.0	−6.4

Note: Series, bases, and caveats as defined in Section 2.6. Annual means of the quarterly country-average $\bar{\kappa}$. "Base" is each row's first fully covered calendar year (all four quarters present; Base-year column); for a group it is the first year *every* member is fully covered (2010 for the pyramid group, since Malaysia and Korea enter in 2009Q2). Growth is the percent change in the annual mean from that base year to 2023; for the pyramid group this is a 13-elapsed-year horizon, not two decades. Δ growth is corrected minus baseline, in percentage points; 2024 is excluded for the roster-attrition reason given in Section 2.6. The rise survives the correction in every group: the United States grows 401% against a 425% baseline while remaining above the baseline in all 80 panel quarters (by up to 0.014), the pyramid group's rise steepens (135% against 130%), and the dispersed group's is modestly trimmed (4.0% against 5.4%); where the baseline rise was small the correction moves it without reversing it (Korea +1.9% → +4.7%, Australia +9.4% → +3.0%), while the United Kingdom's and Malaysia's baselines themselves fell over the window. The pre-2020 dual-class treatment matters only for the dual-class-active markets, where the control-chain channel alone gives Brazil +13.0% (versus +48.1%), the pyramid group +130.1% (versus +134.9%), and Korea +2.8% (versus +4.7%), while every other row moves by at most 0.6pp. Where the base $\bar{\kappa}$ is near zero, percent growth is sensitive to the base level, so Δ growth is reported alongside: for Korea the base and 2023 means are $\bar{\kappa} = 0.00209 \rightarrow 0.00213$ (baseline) and $0.00164 \rightarrow 0.00171$ (corrected).

The correction therefore lowers the level of common ownership in most markets and raises it in the United States, while preserving the baseline rise in the United States, the pyramid group over 2010–2023, and the dispersed group; no baseline rise becomes a decline. These trajectories come from the headline series, which excludes the uncorroborated discovered-controller records retained by the demoted permissive variant. The attribution exhibit reports the movement attributable to each record grade, the robustness exhibits test the modelling choices behind it, and the sections that follow turn to the multiclass firms the screen removed.

8 Attribution of the control correction by record grade

This section stays on the left branch of Figure 1, asking which records carry the movement reported in Sections 6 and 7. The control correction alters country-average profit weights $\bar{\kappa}$ by re-weighting recorded holdings and by adding newly discovered controllers. Re-weighting occurs when a firm's votes are reassigned to the controller identified by the traced ownership chain, even though every holding was already present in the Thomson Reuters data. Discovery adds controllers omitted from the Thomson Reuters roster that appear only after following an ownership chain to an ultimate owner the vendor did not link to the firm. Re-weighting uses holdings the paper already trusts, whereas each newly introduced owner identity requires separate verification.

The series displayed throughout this document is the headline series, which admits only discovered controllers that survive the grading rules. It re-weights holdings through the traced control chains, folds in the dual-class voting leg, and carries the verified and vendor-sourced discovered-controller tiers graded below. Across the 49 markets its 2024Q4 median movement is −0.68%, while in the United States it raises $\bar{\kappa}$ from 0.353 to 0.361, a +2.36% move opposite in sign to that all-country median. Earlier drafts instead displayed a permissive construction that retained every discovered-controller candidate record irrespective of grade; this draft demotes it to a labeled variant, the demoted permissive variant. Two independent findings support the demotion. First, the record-grade decomposition below shows that candidate records the Thomson Reuters roster cannot corroborate account for almost all of the demoted permissive variant's large movement. Second, the arithmetic audit that follows shows that the permissive construction double-counts even the verified controllers against their own Thomson Reuters rows.

Each candidate record asserts that a named owner controls a named company. Counts progress through three explicit stages, the first two measured on the full graded set and the third on the 2024Q4 cross-section. The grading ledger holds 165/1,381/45,846 verified/vendor-sourced/quarantined records, and the frozen cache holds 165/1,373/45,854 after the eight overlap-audit demotions. Restricted to 2024Q4 and then netted by the web-verification layer's 291 refuted records, the applied roster holds 165/1,061/46,129. These are internally MAIN, SECONDARY, and QUARANTINE, as defined in the record-grade conventions (2.6). Measured now in denominator mass (weight in the cash-flow denominator) rather than record counts, the verified and vendor-sourced tiers the headline admits carry a combined discovered-controller mass of about 111, against roughly 1,494 for the quarantined tier. The mass the headline declines to admit is thus more than ten times the mass it admits.

8.1 Attribution by record grade

Table 10 admits a different set of discovered-controller records into the 2024Q4 correction in each row and reports the all-country median movement of $\bar{\kappa}$. Differences across rows identify the movement attributable to each record grade.

Table 10: Attribution of the $\bar{\kappa}$ correction, by record grade (2024Q4, all 49 countries)

	Discovered records	Median 2024Q4 $\bar{\kappa}$ move (%)	Status
Demoted permissive variant (chains leg)	47,355	−38.24	demoted variant
No-discovery outer bound: TR-mapped controllers only	0	+0.71	outer bound
The strict series: verified discovered controllers	165	+0.71	verified tier
+ vendor-sourced controllers	1,226	−0.65	headline CC leg; disclosed error

Note: Each row admits a different set of discovered-controller records into the 2024Q4 control-chain correction; the movement column is the all-49-country median percent change in country-average $\bar{\kappa}$ relative to the one-share-one-vote baseline, on the post-verification netting basis. Rows add records cumulatively: *the no-discovery outer bound* admits only controllers carrying a hard Thomson Reuters mapping, *the strict series* adds the 165 verified anchors, and the headline control-chain leg further adds the 1,061 vendor-sourced records; *the demoted permissive variant* instead retains all ~47,000 residual records ungraded. The no-discovery outer bound and the strict series, the two rows that admit no vendor-sourced records, move the median only +0.71%; the verified-plus-vendor-sourced control-chain series moves it to −0.65%. Both lie inside the control-chain bracket [−0.6511%, +0.7057%], whereas the ~46,100 quarantined records the roster could not corroborate account for nearly all of the demoted variant's chains-leg −38.24% movement.

8.2 The permissive construction double-counts the discovered controllers

The record-grade decomposition explains why the demoted permissive variant moves so far; an independent adversarial audit of the netting engine establishes a second, arithmetic reason to demote it. When a discovered controller is admitted, the intended correction is to *replace* the controller's existing Thomson Reuters rows with a single traced control leg, without enlarging the cash-flow denominator. The two constructions carry out this step differently, and the difference shows in the denominator mass each one touches across the 165 verified anchors on 155 issuers. The headline netting engine performs the replacement, removing the matched Thomson Reuters rows and installing the traced leg, so it touches a denominator mass of 79.96, a net change of +4.10 over the baseline those rows had carried. The permissive construction instead adds each discovered controller as a synthetic leg beside the rows it was meant to correct, so it both keeps 75.86 of matched Thomson Reuters mass and adds a synthetic mass of 86.24, touching a combined 162.09 against the headline engine's 79.96. The verified controllers are therefore counted twice before any question of record quality arises.

After the web-verification quarantines and gate, the mechanical scan for this double counting found no confirmed collisions with current Thomson Reuters rows; the audit records the indeterminate cases separately. A full scan of all 1,381 historical graded vendor-sourced records identified a single hand-correction overlap involving a Spanish family aggregate that included a spouse already present in the roster; the overlap-audit layer moved the overlap to the quarantined tier. A companion arithmetic backstop applied across all 49 markets reverts to baseline any issuer-quarter whose Thomson Reuters and retained synthetic denominator masses would together exceed unity.

8.3 Measured quality of the vendor-sourced tier

A fresh local-language recheck measured the vendor tier's error at 26.0% (one-sided 95% upper bound 38.1%), with the demotions falling into a dozen error classes dominated by false-person attributions, superseded controllers, and custodial or state-institution rollups. This historical measurement is what motivated the record-by-record web-verification cycle reported below, and it remains the tier-wide estimate; the realized refutation rate in that cycle is not comparable, because the records it examined were exactly those prior rounds had been unable to clear. A stress simulation demotes random shares of the vendor-sourced records at rates up to 40%, beyond the measured upper bound. Every draw stays inside the graded bracket and moves toward the strict series as more records are removed, so no demotion rate the measured error could imply pushes the headline's control-chain leg outside the bracket. The verification census, the error taxonomy, and the stress simulation are reported in Appendix B, which is transitional pending the post-ship jurisdiction verification round.

The web-verification of the vendor tier has now been completed and consumed. A verification pass, run in four parallel shards, examined the 563 vendor-sourced records left unverified after the first audited round, checking each against primary and registry filings. The pass returned 216 confirmations, 283 refutations, and 64 unresolved rulings; the refutation rate was highest among records that earlier rounds could not clear, and two adversarial cross-model auditors then cross-audited a sample of the rulings. The 283 refutations, together with the eight incremental first-round refutations, form the 291-record web-verification netting layer that demotes the vendor-sourced tier from 1,352 to 1,061 at 2024Q4 while leaving the 165 verified anchors untouched. These demotions are recorded regrades that the arithmetic engine logs rather than silent exclusions from it; of the 1,061 records that remain, 997 (94%) are individually verified and 64 carry the honest-unresolved residue of 0.213 discovered-controller mass. Appendix B reports the per-round verification census and error taxonomy.

Records the roster could not corroborate account for nearly all of the demoted permissive variant's chains-leg −38.24% median movement (the composed permissive display, which folds in the dual-class leg, moves −38.26%). Every graded control-chain leg lies inside the bracket from verified records only (+0.71%) to verified plus vendor-sourced (−0.65%); after adding the dual-class leg, the composed headline median is −0.68%. The displayed headline therefore admits only the graded discovered controllers and replaces their Thomson Reuters rows without retaining them. The demoted permissive variant is reported here only as a documented comparison. This choice is made subject to co-author ratification (formal adoption of the recorded choice by the full co-author team), and the grading shows that the headline correction mainly re-weights holdings already trusted by the data, while genuine discovered controllers roughly offset in the aggregate.

9 Robustness of the control correction to modelling choices

The corrected common-ownership series rests on three modelling choices: how control is propagated through pyramid chains, how loyalty-voting regimes are treated, and how dual-class voting rights are attributed. Each is examined in a subsection below, with the effect on the corrected country-average profit weight $\bar{\kappa}$ reported market by market. The control-propagation rule leaves $\bar{\kappa}$ almost unchanged. It can be compared only within the permissive satellite, the series in which the ultimate-ownership trace and the weakest-link rule are set against each other, and the same comparison on the tier-netted headline series has not been computed. Restoring the loyalty-voting firms that the headline quarantines moves $\bar{\kappa}$ by at most about 0.001, and only in Italy and the Netherlands. The dual-class attribution rule is tested within the standalone dual-class satellites, where it matters in just two markets, Brazil and Sweden. In the composed post-verification panel, each of the three robustness legs is carried as an additional column on the loyalty-quarantined headline basis.

9.1 Control-propagation rule: ultimate-owner trace vs. weakest-link

The ultimate-ownership trace retains a controller identified through the full chain, whereas the Aminadav–Papaioannou weakest-link rule breaks a chain at its weakest intermediate layer. Table 11 sets the two rules against each other within the permissive satellite, and at the country-average level the choice between them is immaterial. Across the 49 markets, the median absolute difference in $\bar{\kappa}$ is 0.0001. The largest difference in any single market is 0.0024, reached in Ireland.

Table 11: Control-rule robustness: ultimate-owner trace vs. weakest-link

	Ultimate-owner trace		Weakest-link			Δ% vs. baseline	
	2023	2024Q4	2023	2024Q4	Rule Δ (2024Q4)	Trace	Weak
Ireland	0.1124	0.1668	0.1101	0.1644	0.0024	−17.7	−18.9
United States	0.1686	0.2822	0.1675	0.2805	0.0017	−19.9	−20.4
South Africa	0.0775	0.0583	0.0795	0.0597	−0.0014	−67.5	−66.7
Morocco	0.0118	0.0121	0.0123	0.0129	−0.0008	−40.6	−36.6
Colombia	0.0104	0.0107	0.0111	0.0115	−0.0008	−31.6	−26.5
Egypt	0.0459	0.0483	0.0464	0.0491	−0.0008	−29.7	−28.6
China	0.0332	0.0299	0.0337	0.0305	−0.0006	−32.6	−31.2
Chile	0.0145	0.0139	0.0149	0.0142	−0.0004	−29.1	−27.2
Peru	0.0046	0.0072	0.0050	0.0076	−0.0004	−55.7	−53.4
United Arab Emirates	0.0238	0.0182	0.0243	0.0186	−0.0004	−36.9	−35.7
Austria	0.0092	0.0092	0.0095	0.0095	−0.0003	−51.1	−49.3
Finland	0.0244	0.0265	0.0246	0.0267	−0.0003	−40.2	−39.6
Netherlands	0.0331	0.0472	0.0332	0.0475	−0.0003	−32.4	−32.0
Japan	0.0236	0.1253	0.0236	0.1255	−0.0002	−41.3	−41.1
United Kingdom	0.0254	0.0282	0.0256	0.0284	−0.0002	−52.0	−51.7
Median, all 49	0.0111	0.0151	0.0112	0.0153	0.0001	−38.3	−36.6

Note: Country-average $\bar{\kappa}$ under two control-propagation rules; column, base-year, and roster-attrition conventions follow Section 2.6. The trace column is the headline rule, tracing ultimate ownership through pyramid chains; the weakest-link column applies the Aminadav–Papaioannou rule, under which a chain is only as strong as its weakest layer. Both legs are the demoted permissive variant; the headline netting cannot supply a weakest-link tier, so this test does not establish the tier-netted headline result. "Rule Δ" is trace minus weakest-link at 2024Q4. Rows are the 15 countries of 49 with the largest absolute rule difference at 2024Q4; the final row is the all-49 median, whose "Rule Δ" cell is the median *absolute* difference, 0.0001 (maximum 0.0024, Ireland). Trace is presented as primary and weakest-link as robustness, pending a co-author decision.

9.2 Loyalty-voting treatment: headline quarantine vs. restored satellite

The headline quarantines the firms subject to loyalty-voting regimes, in which registered long-term holders receive additional votes. The loyalty satellite restores those firms, and Table 12 compares the two series. Only three markets contain loyalty-treated firms, so the comparison is confined to them. The remaining 46 markets are identical under both series, which makes the all-49 median effect zero. Restoring the loyalty firms lowers $\bar{\kappa}$ at 2024Q4, and the reduction is small in each of the three markets. It is largest in Italy, at 0.0011, and reaches 0.0009 in the Netherlands. In France the effect is negligible.

Table 12: Loyalty-voting robustness: headline quarantine vs. restored satellite

	Headline (quarantined)		Satellite (restored)			Δ% vs. baseline	
	2023	2024Q4	2023	2024Q4	Loyalty Δ (2024Q4)	Head.	Sat.
Italy	0.0062	0.0315	0.0059	0.0304	−0.0011	−0.9	−4.3
Netherlands	0.0452	0.0704	0.0447	0.0695	−0.0009	0.8	−0.5
France	0.0072	0.0129	0.0072	0.0129	0.0000	0.2	0.2
Median, all 49	0.0149	0.0260	0.0149	0.0260	0.0000	−0.7	−0.7

Note: Country-average $\bar{\kappa}$ under the headline, which quarantines loyalty-voting firms, and the loyalty satellite, which restores them on the headline-series basis; conventions follow Section 2.6. Under the adopted Option A the headline is loyalty-quarantined and the satellite is a supplementary display. "Loyalty Δ" is satellite minus headline at 2024Q4. Rows are the loyalty-treated markets whose satellite departs from the headline at the 2023 annual mean or 2024Q4, of 49; the final row is the all-49 median, whose cell is the median absolute effect. Italy and the Netherlands move at 2024Q4. France restores one loyalty-treated issuer at 2024Q4, but that issuer has zero net $\bar{\kappa}$ effect. Restoring the loyalty firms lowers $\bar{\kappa}$ by 0.0011 (Italy) and 0.0009 (Netherlands), because those firms carry the dual-class wedge the correction attributes to controlling holders; the effect is small, at most about 0.001. The aggregate and survival conclusions remain intact, although the Netherlands' 2024Q4 correction changes sign from +0.8% under the headline to −0.5% under the satellite.

9.3 Dual-class attribution rule: full replacement vs. holder overwrite

Table 13 compares two rules for writing dual-class voting rights into the ownership register. The comparison uses standalone dual-class satellites that carry no ownership-chain correction, so the attribution rule is the only thing that varies. The two rules differ in how much of a treated firm's ownership they rewrite. Under the headline full-replacement rule, the firm's entire control vector is replaced by the reconstructed voting-adjusted holders. Under the holder-overwrite rule, only the dual-class holders' rows are overwritten, and all other owners persist. Within these satellites the choice between the two rules is material only in Brazil and Sweden at 2024Q4, and negligible everywhere else.

Table 13: Dual-class attribution rule: full replacement vs. holder overwrite

	Full replacement (headline; standalone)		Holder overwrite (standalone)			Δ% vs. baseline	
	2023	2024Q4	2023	2024Q4	Rule Δ (2024Q4)	Repl.	Ovw.
Brazil	0.0080	0.0710	0.0086	0.0786	−0.0076	−12.4	−3.0
Sweden	0.0421	0.0385	0.0433	0.0400	−0.0015	−5.2	−1.4
Austria	0.0090	0.0175	0.0097	0.0182	−0.0007	−7.0	−3.1
Germany	0.0120	0.0172	0.0123	0.0176	−0.0005	−4.3	−1.8
Korea	0.0016	0.0014	0.0019	0.0018	−0.0003	−25.7	−9.0
Israel	0.0140	0.0163	0.0141	0.0165	−0.0002	−1.3	−0.2
Finland	0.0385	0.0439	0.0387	0.0441	−0.0002	−0.7	−0.4
Italy	0.0061	0.0317	0.0061	0.0317	−0.0001	−0.2	−0.0
China	0.0475	0.0443	0.0475	0.0443	−0.0000	−0.0	−0.0
India	0.0012	0.0013	0.0012	0.0013	0.0000	0.0	0.0
Singapore	0.0016	0.0094	0.0016	0.0094	0.0000	0.0	0.0
Median, all 49					0.0000	0.0	0.0

Note: Corrected country-average $\bar{\kappa}$ under two rules for attributing dual-class voting rights; conventions follow Section 2.6. Both columns are the *standalone* dual-class satellites, with no ownership-chain correction, so only the attribution rule varies. Under the full-replacement rule (headline) a treated firm's control vector is the reconstructed set of voting-adjusted holders; under the holder-overwrite rule the baseline register is retained and only the dual-class rows are overwritten, so other owners persist and the overwrite series stays nearer the baseline. "Rule Δ" is replacement minus overwrite at 2024Q4. Rows are the *endpoint movers*, defined as markets where either rule departs from the OSOV baseline by more than 0.001 at an endpoint, and are sorted by absolute rule difference; elsewhere the two rules coincide with the baseline (final row: all-49 median, cell the median absolute difference). Of the twelve markets treated over 2020–2024Q4, eleven are displayed movers; Denmark, treated only in 2020 (absolute rule difference ≤ 0.00008), is not. The rule matters materially only for Brazil (0.0076) and Sweden (0.0015) at 2024Q4; Italy and the Netherlands collapse toward the baseline because their loyalty-linked dual-class firms are quarantined under Option A and surface in the loyalty satellite (Table 12).

The control-propagation comparison varies only the permissive series, and the corresponding weakest-link comparison on the tier-netted headline has not been computed. The loyalty satellite moves $\bar{\kappa}$ by at most about 0.001, again only in Italy and the Netherlands. It leaves the aggregate and survival conclusions intact, although the Netherlands' 2024Q4 correction changes sign under the satellite. Within the standalone satellites, the dual-class attribution rule is material only in Brazil and Sweden. No combined series has been computed under the holder-overwrite rule, so the attribution-rule comparison is not tested against the combined levels and survival series.

10 The excluded multiclass universe: scope and voting–cash-flow separation

The preceding sections corrected control inside the panel the co-authors shipped; this section crosses to the right branch of Figure 1, the firms that never entered that panel at all. The multiclass screen removes 3,620 issuers across 41 markets before estimation begins, and multiclass structures are where voting rights depart most sharply from cash-flow rights. Describing that universe comes before measuring what its restoration does to $\bar{\kappa}$, which Section 11 takes up. Section 10.1 censuses the exclusion market by market and states how far the current reinstatement reaches. Section 10.2 then supplies the structural anatomy of the largest excluded population, the United States, where the certified DEF 14A derivation measures the separation between votes and cash-flow rights firm by firm.

10.1 The scope of the exclusion across markets

This subsection places every excluded firm on the map before any reinstatement result is read. Table 14 gives the full census of the exclusion across all markets and quantifies the remaining work; its Panel A carries the two markets with named prior certified builds, the United States and Sweden, whose reinstatement Section 11 takes up. The market-wave, variant-route, Peruvian, and Canadian appends that extend reinstatement beyond the two certified markets are accounted for separately, in Table 19 of Section 11.2.

The census starts from the full exclusion. The co-authors' construction removes 3,620 multiclass issuers across 41 markets before the panel is built. Of these, 306 reappear in the 2020–2024 panel through per-security overlap. Within that group of 306, 245 carry a mapped in-panel overlay correction vector, and the remaining 61 are present in the panel but not yet corrected by the overlay. The other 3,314 firms form the reinstatement universe. The United States and Sweden account for 1,819 of the excluded firms, and for all 1,025 firms reinstated under a prior certified build to date. The reinstated share of the reinstatement universe is therefore concentrated in the two markets where the disclosure spine was built first. The *reinstated (prior)* column of Table 14 records that earlier build, and it predates the market-wave and variant appends. The current position of every excluded firm, across all reinstatement routes, is given instead in Table 19 of the disposition accounting in Section 11.2.

Table 14: Census of excluded multiclass firms and their reinstatement status, by market

Market	Excluded (N)	In panel	Excluded, absent	Pipeline matched	Control determ.	Genuine wedge	Reinstated (prior)
Panel A. Reinstated (prior certified builds)							
United States	1,603	0	1,603	946	687	360	946
Sweden	216	0	216	128	119	119	79
Subtotal	1,819	0	1,819	1,074	806	479	1,025
Panel B. Excluded multiclass pending reinstatement (matched to the dual-class verdict pipeline)							
China	409	150	259	100	99	1	0
Canada	373	1	372	46	43	19	0
United Kingdom	179	27	152	60	20	2	0
Poland	137	0	137	60	20	20	0
Indonesia	89	31	58	7	7	0	0
Mexico	71	6	65	10	0	0	0
Switzerland	53	14	39	11	5	5	0
Peru	47	0	47	29	25	21	0

Finland	44	18	26	20	13	13	0
Denmark	41	0	41	26	15	15	0
Netherlands	29	6	23	6	5	5	0
France	28	7	21	4	4	4	0
Italy	26	13	13	13	13	7	0
Philippines	26	0	26	3	0	0	0
South Africa	25	0	25	10	4	4	0
Belgium	21	0	21	0	0	0	0
India	21	16	5	12	12	5	0
Hong Kong	17	0	17	4	0	0	0
Israel	15	4	11	3	3	3	0
Chile	12	8	4	1	1	0	0
Norway	12	0	12	5	1	1	0
Germany	11	2	9	0	0	0	0
Spain	8	0	8	2	2	1	0
Singapore	8	0	8	3	3	2	0
Austria	4	0	4	1	0	0	0
Malaysia	4	0	4	1	1	0	0
Thailand	4	0	4	0	0	0	0
Brazil	3	1	2	1	1	1	0
Greece	2	0	2	1	0	0	0
United Arab Emirates	1	0	1	0	0	0	0
Colombia	1	1	0	0	0	0	0
Morocco	1	0	1	0	0	0	0
Oman	1	0	1	1	1	0	0
Subtotal	1,723	305	1,418	440	298	129	0
Panel C. Excluded multiclass identified; voting-data derivation not yet built							
Australia	45	0	45	0	0	0	0
Japan	13	0	13	0	0	0	0
Ireland	9	0	9	0	0	0	0
Croatia	6	0	6	0	0	0	0
New Zealand	4	0	4	0	0	0	0
Egypt	1	1	0	0	0	0	0
Subtotal	78	1	77	0	0	0	0
Panel D. Multiclass firms present in the pipeline; zero construction exclusions recorded							
South Korea	0	0	0	–	–	–	–
Taiwan	0	0	0	–	–	–	–
Vietnam	0	0	0	–	–	–	–
Panel E. No excluded multiclass firms recorded; market not yet scoped							
Jordan	–	–	–	–	–	–	–
Kuwait	–	–	–	–	–	–	–
Romania	–	–	–	–	–	–	–
Russia	–	–	–	–	–	–	–
Saudi Arabia	–	–	–	–	–	–	–

Total excluded universe	3,620	306	3,314	1,514	1,104	608	1,025

Note: Series, bases, and caveats are as defined in Section 2.6. The excluded universe is the set of multiclass issuers the co-authors' sample construction removes from the estimation panel, taken from their construction-time exclusion file; it contains 3,620 distinct issuers across 41 markets, keyed on the Thomson-Reuters issuer code. *In panel* counts the excluded issuers that nonetheless appear in the built 2020–2024 panel through per-security overlap, and *excluded, absent* is the complementary reinstatement universe. *Pipeline matched* counts the excluded firms matched to a derivation record. For the non-US markets, a match is a dual-class verdict row of any value; for the United States, it is a bridged representative CIK node. The 946 United States figure uses representative nodes as its counting unit because 1,392 of the 1,603 excluded US issuers bridge to a CIK, and the usable-proxy set links 966 issuer codes to 946 CIKs after collapsing the 20 CIKs that carry two codes each. *Control determination* counts the subset for which an actual control determination exists: a non-*no data* verdict for the non-US markets, and *status* = derived for the United States (687 of the 946 nodes). *Genuine wedge* is the subset adjudicated to carry a voting–cash-flow wedge, a time-invariant verdict class distinct from whether an operational corrected vector was built or is active in any given firm-quarter; at 2024Q4, 61 of the 360 United States genuine-wedge nodes enter as dilution-only. *Reinstated (prior)* counts firms already appended under a named prior certified build: the United States under *b1-us-append-v3.1*, and Sweden under the cross-firm analog. The United States counts come from the certified skill-derived DEF 14A adjudication at representative-CIK granularity, whereas the non-US counts come from the firm-level dual-class *verdicts.csv*. The panels group markets by the correction and coverage status of their excluded firms. Panel A holds the two reinstated markets. Panel B holds markets matched to the dual-class verdict pipeline but not yet appended, where verdict coverage runs ahead of operational readiness: of the 440 verdict-matched firms only 179 currently have a built voting-control vector (and 97 of the 129 genuine-wedge firms do), and nine Panel B markets carry verdict coverage but no built vectors, so extending the correction to them requires a per-firm evidence source beyond the register: all nine are capital-only or class-implied registers whose disclosure carries no per-holder voting leg, and the build records documented honest negatives rather than imputed vectors. Panel C holds markets whose excluded firms the co-authors' screen records but for which no dual-class derivation has been built, so their pipeline, determination, wedge, and reinstated counts are construction-imposed zeros and do not reflect an empirical finding. Panel D holds markets carrying a dual-class pipeline for which the co-authors' file records no construction exclusion; entity-level in-panel retention is verified for South Korea, where 97 of 102 verdict firms match a current-panel issuer code, but for Taiwan and Vietnam it rests on pipeline presence with zero recorded exclusions, and the reinstatement columns do not apply. Panel E holds markets absent from the co-authors' exclusion screen and not yet covered by any dual-class pipeline, so their counts are shown as not-determined (–); a zero is inappropriate because a genuine absence of multiclass firms cannot yet be distinguished from an unscoped market.
For Sweden the co-authors' file records 216 excluded issuers, of which 128 match our dual-class verdict universe (the set on which reinstatement operated), 119 carry a determination and a genuine-wedge verdict, and 79 reinstate cleanly; both the raw 216 and the pipeline-matched 128 are shown. The three new Swedish genuine-wedge verdicts (Ortivus, Precio Fishbone, and Slottsviken) await vector builds. The Peruvian audit also identifies six genuine-wedge firms outside the co-authors' 3,620 exclusion spine; they are disclosed in the audited Peru sidecar and are not added to this census or the disposition ledger. Subtotals and the grand total cross-foot exactly (3,620 = 306 in panel + 3,314 absent; 1,025 reinstated = 946 United States + 79 Sweden). Full per-market figures, sources, and method notes are recorded in *table_B2census_numbers.json.*

The full census sorts the remaining markets into four groups. Panel B holds the 33 markets whose excluded firms are matched to the dual-class verdict pipeline but not yet appended. Led by China, Canada, the United Kingdom, and Poland, it contains 440 verdict-matched firms, of which 129 carry a genuine voting–cash-flow wedge. Operational vector availability is narrower still: 179 of these firms currently hold a built voting-control row, and 97 of those are genuine-wedge firms with a build vector. Panel C holds six markets, 78 firms led by Australia and Japan, whose excluded firms are recorded by the screen but not yet derived. Panel D holds the markets that carry a dual-class pipeline with zero recorded exclusions, and Panel E the markets not yet scoped. The census shows that the reinstatement covers the largest and most wedge-dense excluded populations and bounds the pending work by enumerating two defined sets: firms with derived data awaiting a build or an append, and markets awaiting derivation.

Verdict coverage runs ahead of vector coverage for a structural reason. A wedge verdict requires only class-level terms — statutes, prospectuses, and the charter's votes-per-share — whereas an operational voting-control vector requires a per-holder voting leg: each holder's stake in each class, composable into that holder's share of aggregate votes. Whether that leg exists depends on the register. The nine Panel B markets with verdict coverage and no built vectors (Chile, Colombia, Greece, Indonesia, Malaysia, Oman, the Philippines, South Africa, and Thailand) are capital-only or class-implied registers: their disclosure publishes per-holder capital percentages, or holdings without class attribution, and no per-holder voting leg anywhere. Their documents were harvested and their ownership rows and voting terms extracted, but the build leaves the voting-control panel empty by design, because constructing a voting leg from a capital-only register would be imputation, which the pipeline forbids; each such register is recorded as a documented honest negative rather than a silent skip. Two carry specific twists: Greece's excluded firms file voting-null 20-F tables, and Chile's wedge is a board-representation wedge that the voting-vector schema deliberately does not encode. The partially built markets instead reflect per-firm access gaps with recorded remediation paths. Canada's filing portal is bot-walled, so its built vectors concentrate in the EDGAR-cross-listed subset; Peru awaits a deferred design decision on the dilution treatment of its non-voting *acciones de inversión*, although the audited 2026-07-23 harvest cycle dispositioned all 34 of its genuine-wedge firms individually; and Poland, the template for closing such gaps, recovered 24 additional built firms in a single attachment-parsing wave, leaving only the missing per-holder cash-flow legs recorded in Section 12. Both kinds of gap close through bounded per-firm work whose mechanism is already validated; the rollout paragraph of Section 12 states the method and its scale.

The enlargement measured in Section 11 reverses only the co-authors' multiclass screen. Firms removed by their other sample screens remain out of both the baseline and the enlarged panels. The full screen inventory and its interpretive caveats are stated in the scope-of-the-sample-screens paragraph of Section 5.

10.2 Voting–cash-flow separation among excluded US firms

Within the excluded universe just censused, the United States is the largest single market, and its voting structures can be read directly from mandatory disclosure. The concentration is itself informative: super-voting structures are a defense controlling founders adopt against dilution of control by deep, diffuse institutional ownership, so wedge adoption clusters where index capital is thickest — in the same market whose measured common ownership is highest — and the screen's bias is accordingly largest in the market at the centre of the common-ownership debate. A bridge across the panel's two data vintages resolves the excluded US firms to 946 representative CIK nodes, where issuers that share a CIK are collapsed to one node. Of these 946 nodes, 702 are in-scope multiple-class candidates, whose control cannot be inferred from a one-share-one-vote reading of ownership. The remaining 244 fall outside the derivation's scope. Most of these 244 carry a single economic common class, and the census table details the rest.

For each in-scope firm, we read its SEC DEF 14A proxy statement and reconstruct two shares for every holder. The first is the holder's voting-control share γ, its fraction of the aggregate votes across all share classes. The second is the holder's economic-ownership share β, its share of cash-flow rights. We build β by weighting each class on its own economic terms: a non-economic voting stub counts as zero, an umbrella-partnership (Up-C) unit enters at its as-exchanged value, and an overlapping or convertible holding enters at its as-disclosed or as-converted amount. Two further adjustments apply to both shares: stock a holder can acquire within 60 days is subtracted, and overlapping ownership rows are kept as the proxy discloses them. The voting–cash-flow wedge is $\gamma - \beta$. It is positive exactly when a holder's votes exceed the matching cash-flow rights.

We assign each in-scope firm to one of four verdicts, read from the certified terms of its voting structure. A firm is a *genuine wedge* when a super-voting class makes its voting rights diverge from its cash-flow rights. It is *proportional multiclass (no wedge)* when, after normalization, voting and economic entitlements are proportional across the economic legs. It is a *single-class artifact* when the second class is authorized but unissued, converted away, or otherwise not outstanding on the record date, so that only one class remains outstanding. It is *documented unresolvable* when the proxy establishes that a second class exists but never discloses the figure needed to settle its structure.

These verdicts come from the three-tier extraction described in Section 3.4. Deterministic parsing handles the jurisdiction-invariant mathematics. A bounded, cached, and cross-model-audited model-adjudication tier settles the judgment calls that parsing cannot, such as reading voting terms, linking footnotes to rows, aggregating control groups, and applying super-vote conventions. Human triage disposes of the cases that neither tier resolves.

Table 15: The US wedge universe: structural census and controller provenance

	N	%
Panel A. Structural census (excluded US multiclass paper universe)		
US multiclass paper-bridge CIKs (two vintages)	946	–
Unscoped: dilution-only, outside adjudication scope	244	–
Scoped and structurally adjudicated	702	100.0
Genuine voting–cash-flow wedge	360	51.3
Proportional multiclass (no wedge)	261	37.2
Single voting class (vendor multiclass artifact)	66	9.4
Documented unresolvable	15	2.1
Panel B. Resolved-controller extraction provenance (production controller dataset)		
Deterministic table parsing	1,143	76.4
Model-adjudicated (LLM)	278	18.6
Concert / group aggregation	76	5.1
Total resolved-controller rows	1,497	100.0
(memo) Flagged for human triage	51	3.4

Note: Series and audit conventions follow Section 2.6. Panel A reports the certified structural census of the US multiclass firms excluded from the co-authors' panel. The two-vintage bridge holds 946 CIKs (966 issuer-code rows), of which 702 are scoped for structural adjudication and 244 lie outside classification scope: 222 whose vendor multiclass flags resolve to a single economic common class, 7 with no extractable class structure, and 15 golden-benchmark firms reserved for validation and excluded from the derivation (among them Meta, Alphabet, Ford, and Berkshire Hathaway). Of the 702 scoped firms, 360 (51.3%) are genuine wedges; the remaining 342 comprise 261 proportional multiclass firms with no normalized wedge, 66 single-class artifacts, and 15 documented but unresolvable structures. Of the proportional/no-wedge group, 25 have unequal nominal votes per economic leg; their normalized voting and economic entitlements are nevertheless proportional. Panel B reports the extraction provenance of a distinct companion artifact, the production resolved-controller dataset, at row grain (1,497 rows: 1,484 distinct non-blank accessions, 11 duplicate rows, and 2 with a blank accession): deterministic table parsing resolves 1,143 rows, a model-adjudication tier 278, and concert/group aggregation 76, with 51 flagged for human triage.

Table 16 reports the largest wedges among the census's genuine-wedge firms. Across these firms, a founder, family vehicle, or insider bloc often holds a large majority of the votes with a small share of the economic interest.

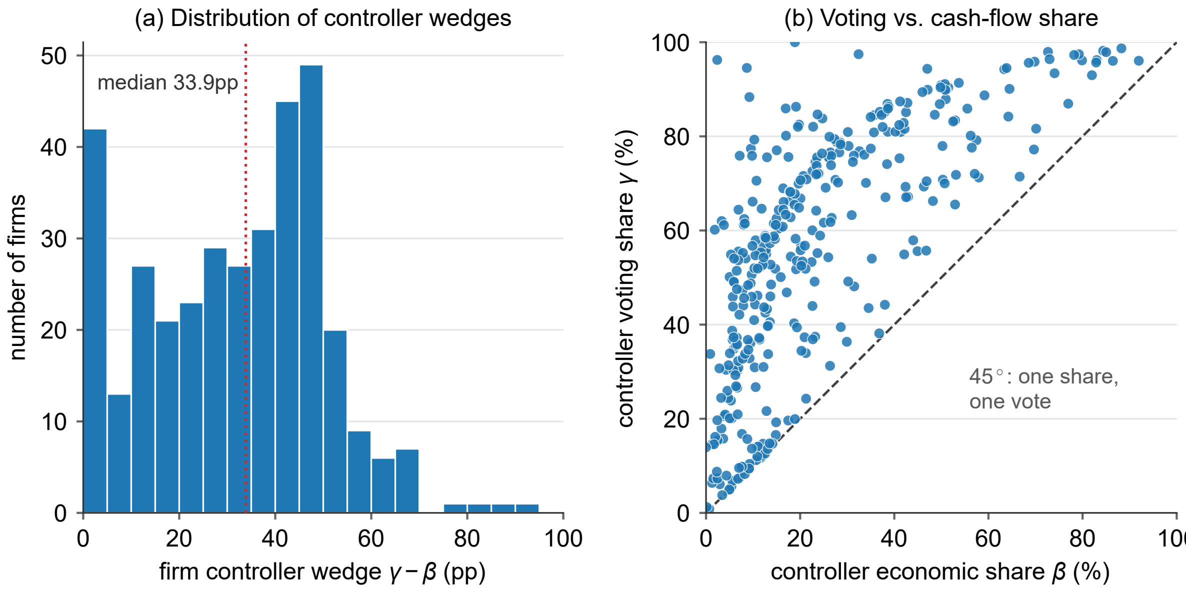
Table 16: Largest voting–cash-flow wedges among excluded US firms

Company	Controlling holder	Voting γ (%)	Economic β (%)	Wedge (pp)
MOLSON COORS BEVERAGE...	4280661 Canada Inc.	96.2	2.4	93.9
KELLY SERVICES INC	Terence E. Adderley Rev...	94.5	8.7	85.9
THE BOSTON BEER COMPA...	C. James Koch	100.0	18.9	81.1
Hims & Hers Health, I...	Andrew Dudum	88.3	9.2	79.1
DOMO, INC.	Joshua G. James	79.3	10.3	69.0
Expensify, Inc.	Expensify Voting Trust	85.9	16.9	69.0
ALPINE 4 HOLDINGS, IN...	All directors and execu...	75.9	7.1	68.7
Paramount Global	National Amusements, In...	77.4	9.5	67.9
UiPath, Inc.	Daniel Dines	86.3	19.1	67.2
Coupang, Inc.	Bom Kim	75.9	9.8	66.1
Oppenheimer Holdings...	A.G. Lowenthal	97.5	32.4	65.0
AppLovin Corporation	Shares subject to the V...	80.1	17.0	63.1
Coinbase Global, Inc.	All executive officers...	82.5	19.8	62.7
MARCHEX INC	Russell C. Horowitz	75.6	13.0	62.6
Clear Secure, Inc.	All directors and execu...	82.0	19.5	62.5
Duolingo, Inc.	All executive officers...	77.0	15.0	62.0
Hyperfine, Inc.	Jonathan M. Rothberg, P...	84.7	23.7	61.0
Coca-Cola Consolidate...	J. Frank Harrison, III,...	70.6	10.7	59.9

Note: The table reports the 18 excluded US firms with the largest certified voting–cash-flow wedge $\gamma-\beta$, drawn from the 360 genuine-wedge firms in the census (Table 15). The definitions of γ, β, and the wedge, together with the audit conventions, follow Sections 3.4 and 2.6. The controlling holder is the certified beneficial owner or voting bloc with the largest wedge in the firm's latest exact-form DEF 14A filed during 2020–2024; where a row names an aggregate ("all directors and executive officers as a group," "shares subject to the voting agreement"), the figure covers that as-disclosed bloc as a whole. Company and holder labels reproduce those in the certified derivation and are truncated to fit. Each row traces to a cited SEC accession recorded in the derivation dataset. Certified controller values are authoritative and can differ from a proxy's headline voting percentage, which often folds in 60-day options or uses a holder-specific denominator. Classic controlled firms such as Meta, Alphabet, Ford, and Berkshire Hathaway are golden-benchmark validation firms excluded from this derivation and do not appear here.

Table 16 isolates the upper tail, and Figure 6 shows the full distribution. The figure covers the 353 of the 360 genuine-wedge firms whose certified holders supply both a γ and a β value and whose largest holder-level wedge is positive. Even at the middle of this group the separation is large: the median controller holds 33.9 percentage points more of the votes than of the cash flows. At the top of the distribution, the gap reaches 93.9 points.

Figure 6: Distribution of voting–cash-flow wedges among genuine-wedge US firms



Note: The population comprises the 360 certified genuine-wedge US firms; γ, β, and the wedge are as in Section 3.4. A firm has a *computed maximum* when at least one holder in its certified legs carries both a γ and a β value (357 of 360), and a *positive maximum* when the largest holder-level wedge exceeds zero (353 of 357); the figure plots those 353 firms. Inclusion depends on holder coverage in the certified legs, regardless of whether the controller is itemized in the Item 403 table. Panel (a) shows the distribution of the controller wedge $\gamma - \beta$ in percentage points (median 33.9, maximum 93.9, Molson Coors). Panel (b) plots each firm's controller voting share γ against economic share β; the dashed 45° line is one share, one vote, and points sit above it by the wedge. Controllers routinely hold majority or near-majority votes (γ above 50%) on economic stakes of 10–40%, a separation that the control correction restores and a one-share-one-vote measure assumes away.

For these firms, a one-share-one-vote reading of the ownership register misstates the controller's power by tens of percentage points of the vote. The dataset provides a standalone census of the excluded US universe and characterizes the voting structures of its 702 in-scope multiple-class candidates. The same 946-node bridge is added back to the panel in Section 11, which measures what that restoration does to $\bar{\kappa}$.

11 Effects of reinstating excluded multiclass firms

The multiclass exclusion is a first-order limitation of the shipped panel: it disproportionately removes multiclass firms, and in doing so it biases the level of $\bar{\kappa}$ and hides real common-ownership linkage among the firms that remain. Having described that universe in Section 10, this section measures that bias. The measurement requires both universes, because the bias is the difference between the shipped panel and the panel with the excluded firms restored. The central exhibits therefore carry the shipped universe, while this section reports what changes when the excluded firms enter and decomposes that movement into a mechanical dilution and a genuine linkage gain. The reinstated control vectors carry the same per-firm adjudication as the rest of the dual-class data. For the United States that adjudication is the certified DEF 14A derivation, and for Sweden it is the register build. Reporting reinstatement separately from the central exhibits reflects the structure of the comparison and the current reach of the data. The exclusion itself is panel-wide: the co-authors' construction removes 3,620 multiclass issuers across 41 markets, censused market by market in Section 10.1 and in Table 14. The headline reinstatement calculation in Table 17 remains limited to the United States and Sweden, which together hold 1,819 of the excluded firms. The complete disposition in Section 11.2 then accounts for the remaining markets by status and route.

The co-authors' shipped κ panel excludes multiclass (dual-class) firms, which reinstatement adds back at two scopes. In the *enlarged panel*, the reinstated firms enter as new focal firms. The *incumbent* scope covers firms that were already present and whose own profit weights move once the reinstated firms join. Each reinstated firm enters carrying its corrected dual-class control vector for any firm-quarter in which one is active. A reinstated firm without an active corrected vector still enters the panel, but only as a dilution-only node: it affects the averaging denominator without contributing any new common-ownership links. Vector activity follows a strict temporal convention: a firm's holder roster anchors at its proxy's filing quarter and forward-fills at most four quarters, and it is never carried backward, because a proxy discloses ownership as of its record date and backcasting it would assert pre-listing or pre-record holdings that did not exist. Since the derivation reads current-vintage, mostly 2024-filed proxies, vector coverage is small by design in the early quarters and rises toward the sample edge. At the 2024Q4 terminal quarter, 450 of the 946 United States reinstated firms carry an active corrected vector, and the remaining 496 enter as dilution-only nodes. All 79 firms on the primary certified Swedish route carry vectors. The adopted enlarged Swedish route adds two more firms, for 81 in total, and is shown alongside the certified route with its own route provenance. Because the number of active corrected vectors rises steeply near the sample edge, the headline incumbent decomposition below is reported as the 2023 annual mean, and the 2024Q4 quarter is shown separately as a labeled sample-edge variant.

At the enlarged-panel scope, the panel-average $\bar{\kappa}$ falls sharply once the excluded firms enter. At the 2023 annual mean, the United States figure drops from 0.191 to 0.124, a change of -35.1%. The drop at the 2024Q4 sample edge is similar in proportion, at -36.1%. The incumbent scope requires a short decomposition. An incumbent firm f carries profit weight $\bar{\kappa}_f = (n-1)^{-1} \sum_{g \neq f} \kappa_{fg}$, the average of its pairwise weights over the other $n-1$ firms in a panel of n. Appending the reinstated firms changes this weight by an amount that splits exactly into a mechanical dilution term and a linkage term,

$$\Delta\bar{\kappa}_f = \underbrace{\left(\frac{1}{n_{\text{tot}}-1} - \frac{1}{n_{\text{corr}}-1}\right) \sum_{g \in \text{inc},\, g \neq f} \kappa_{fg}}_{\text{dilution}} + \underbrace{\frac{1}{n_{\text{tot}}-1} \sum_{r \in \text{reinst}} \kappa_{fr}}_{\text{linkage}},$$

where n_{corr} is the incumbent-only panel size and n_{tot} the enlarged panel size. The dilution term is purely mechanical. Each incumbent's $\bar{\kappa}_f$ averages its pairwise weights over the other panel firms, so enlarging the panel spreads the incumbent's unchanged link sum over more peers and lowers the average. This fall happens even if the entrants bring no common-ownership links at all. The linkage term is the incumbent-to-reinstated link sum $\sum_{r \in \text{reinst}} \kappa_{fr}$, divided by $n_{\text{tot}} - 1$. It is never negative, because every κ_{fr} is itself non-negative. The two terms work against each other, and the mechanical one dominates. Averaged over incumbents at the 2023 mean, the total United States change is -0.0372, and the dilution term alone accounts for -0.0377. The linkage term measures the common-ownership connection that incumbents acquire through the reinstated firms' owners, and it is positive in every one of the twenty quarters in both countries. Its 2023-mean value is $+0.0005$ for the United States and $+0.0009$ for Sweden. Excluding the multiclass firms therefore overstates the panel-average level of $\bar{\kappa}$, while suppressing a small but systematic amount of real linkage among the firms that remain.

The asymmetry has an economic reading. A reinstated multiclass firm is commonly owned while not commonly controlled: the wedge holds the institutions' cash-flow stakes apart from the founder-locked control rights. On the ownership side, such a firm enters the matrix as a full column, because incumbents place genuine profit weight on it through the index and 13F managers that hold both sides of each pair. On the control side it enters as a nearly empty row, because the profit weights it places on rivals follow its controller's portfolio, and United States founders hold essentially no rival stakes. Full columns and empty rows compress the enlargement into one image: the empty rows pull the panel average down, and the full columns supply the linkage. The empty row is a finding about the identity of the United States controllers rather than a property of multiclass structure as such. Where the controller is itself a diversified vehicle, as in the Swedish investment spheres, the controlled firm becomes a common owner of the intra-group kind. The correction therefore substitutes each firm's measured controller vector and lets the data decide, instead of assuming that multiclass firms leave the common-ownership network.

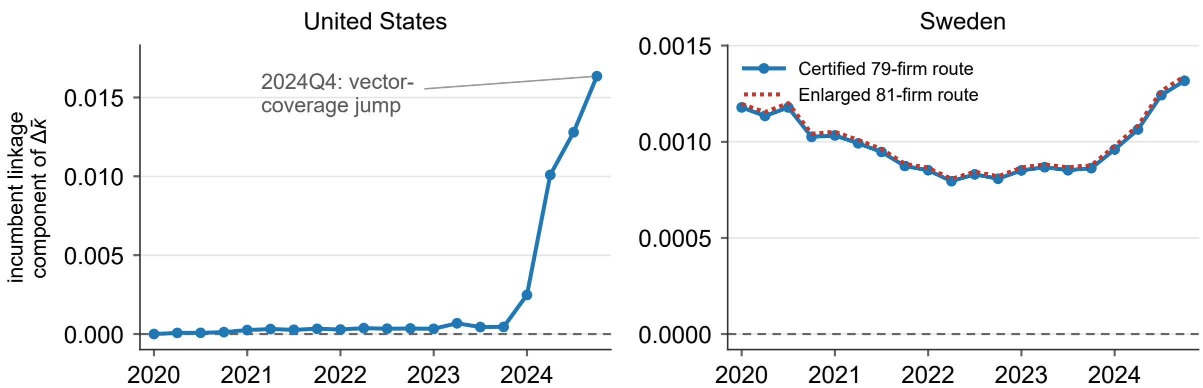
Table 17: Reinstatement of excluded multiclass firms: two scopes

	United States	Sweden certified (79)	Sweden enlarged (81)
Reinstated multiclass firms (N)	946	79	81
Reinstated-firm vector coverage (2024Q4)			
Carrying corrected control vector (N)	450	79	81
Entering as dilution-only node (N)	496	0	0
Enlarged-panel scope			
Shipped-panel firms (N, 2024Q4)	3,087	432	432
Enlarged-panel firms (N, 2024Q4)	4,033	511	513
Shipped-panel $\bar{\kappa}$ (2023 mean)	0.191	0.043	0.043
Enlarged-panel $\bar{\kappa}$ (2023 mean)	0.124	0.034	0.034
$\Delta\bar{\kappa}$ level (2023 mean)	−0.067	−0.009	−0.009
$\Delta\bar{\kappa}$ % (2023 mean)	−35.1	−21.1	−21.5
Shipped-panel $\bar{\kappa}$ (2024Q4)	0.353	0.041	0.041
Enlarged-panel $\bar{\kappa}$ (2024Q4)	0.225	0.031	0.031
$\Delta\bar{\kappa}$ level (2024Q4)	−0.127	−0.010	−0.010
$\Delta\bar{\kappa}$ % (2024Q4)	−36.1	−23.8	−24.2
Incumbent scope: 2023 mean (headline)			
Total movement	−0.0372	−0.0048	−0.0049
Dilution (mechanical)	−0.0377	−0.0057	−0.0058
Linkage (genuine)	0.0005	0.0009	0.0009
Incumbent scope: 2024Q4 (sample-edge variant)			
Total movement	−0.0664	−0.0050	−0.0051
Dilution (mechanical)	−0.0827	−0.0063	−0.0064
Linkage (genuine)	0.0164	0.0013	0.0013
Linkage: quarters positive	20/20	20/20	20/20

Note: Series, bases, and caveats as defined in Section 2.6. Reinstatement adds the excluded multiclass firms back at two scopes. *Enlarged-panel scope* lets them enter as new focal firms and compares the panel-average profit weight $\bar{\kappa}$ before (shipped panel) and after (enlarged panel); *incumbent scope* restricts attention to firms already in the shipped panel and reports the equal-weighted change in their own $\bar{\kappa}$, decomposed as total = dilution + linkage by the displayed identity above, with *linkage* positive in all twenty quarters in both countries. Coverage (2024Q4): 450 of the 946 United States reinstated firms carry an active corrected control vector and 496 enter as dilution-only nodes, whereas all 79 Swedish firms on the primary certified route carry vectors. The third column reports the adopted enlarged Swedish route with 81 firms; the certified 79-firm column remains primary. The decomposition is reported as the 2023 annual mean (headline) and, separately, at the 2024Q4 sample edge, whose United States linkage is inflated by the terminal-quarter jump in coverage (Figure 7) and is shown only as a variant. The United States reinstates 946 representative CIK nodes bridged from its 1,603 excluded multiclass issuers and Sweden 79 of 216; levels are shown to three decimals, the decomposition to four. The United States series is the certified temporal build *b1-us-append-v3.1* and Sweden the *b1-sweden-v2* cross-firm analog.

Figure 7 traces the linkage component quarter by quarter. It remains positive throughout the five-year window, including before the terminal-quarter coverage jump that increases its United States magnitude at the edge.

Figure 7: Incumbent linkage over time, 2020Q1–2024Q4



Note: Series, bases, and caveats as defined in Section 2.6. The linkage component of the incumbent $\bar{\kappa}$ movement, by quarter, for the United States (left) and Sweden (right), each panel on its own vertical scale with a dashed zero line. In the Sweden panel, the solid blue line is the primary certified 79-firm route and the dotted red line is the adopted enlarged 81-firm route. Linkage is the non-mechanical component of the incumbent decomposition in Table 17, measuring the common-ownership connection incumbents gain through the reinstated firms' owners; it is positive in every one of the twenty quarters in both countries, so reinstating the excluded firms adds common-ownership linkage in addition to diluting denominators. The United States terminal quarter (2024Q4) shows a pronounced increase driven mainly by vector coverage, which rises from 169 reinstated firms carrying an active corrected control vector in 2023Q4 to 450 in 2024Q4 as the 2024 proxy season arrives: each roster activates at its proxy's filing quarter and forward-fills at most four quarters, never backward, so early-quarter coverage is small by design. All 946 reinstated nodes are present in every quarter; what rises across the window is measured coverage, not the number of multiclass firms. Each quarter's linkage is therefore a lower bound on true linkage, and the series' slope reflects disclosure vintage — partly genuine absence (many reinstated firms first listed in 2021–2022, with no earlier proxy to read) and partly a measurement floor for firms listed throughout. The slope is accordingly not an economic trend; the period-correct backfill that would replace the floor with a true time profile is an open item in Section 12. The Swedish series shows no such jump, because its evidence object is an event stream rather than a snapshot — dated register notifications and annual per-holder class tables supply period-correct rosters in every quarter, and all 79 certified Swedish firms carry active vectors throughout — and because its excluded firms are long-listed A/B-share structures with no recent-listing wave. The always-positive sign of the linkage does not depend on that quarter.

The audited Peru and Canada extensions provide the same comparison on smaller rosters, 17 firms for Peru and 16 for Canada. At the 2023 annual mean, Peru's enlarged-panel $\bar{\kappa}$ falls by 40.7%. Canada's moves by only −1.1%. In both markets, the incumbent linkage term is positive in all twenty quarters. The extension table and series figure are reported in Appendix A (Table 23, Figure 9).

11.1 Panel-enlarged variant: the headline series composed with the reinstated United States firms

The reinstatement above appends the excluded firms to the co-authors' uncorrected panel. A companion series instead composes the reinstatement with the headline series. In this companion series, the in-panel United States firms carry their corrected headline control weights, and the 946 bridged excluded firms are then appended on top. Because it enlarges the estimation panel beyond the co-authors' current screen, this series is reported as a labeled variant with per-firm route provenance. The route was adopted on 2026-07-22 as an admissible measurement of voting control, and its citability remains subject to the verification cycle.

Whether the dual-class append and the United States correction interact depends on the two roles that any firm plays in $\bar{\kappa}$. A firm can appear as a *portfolio* firm, meaning a focal row in the average over other firms that defines its own $\bar{\kappa}$. It can also appear as an owner, through the stakes it holds *in* other firms. Panel membership governs only the first role, since a firm's stakes in others sit on the owner side of the data whether or not the firm is itself in the panel. The in-panel control-chains correction already handles the corporate-conduit case, in which one listed firm holds another. This distinction explains why the in-panel United States headline does not move with the dual-class leg. The co-authors exclude United States multiclass firms from the panel, so there is no in-panel wedge to correct, and the entire United States correction runs through the control-chains channel instead. Reinstating the excluded firms adds *new* focal-firm pairs, but it does not rewire the incumbent-incumbent block of the κ matrix. The incumbent movement therefore decomposes into a mechanical dilution, from the weakly linked new pairs, and a positive linkage, from the common ownership that incumbents genuinely share with the reinstated firms.

The in-panel headline correction and the reinstatement do not combine by simple addition. The headline series replaces in-panel firms' control vectors, which changes those firms' links to the reinstated firms. The enlarged panel must therefore be recomputed over the corrected firm set, and it cannot be recovered by summing the two effects separately. The size of this interaction is small. At 2024Q4, naively adding the reinstatement's panel-enlargement change to the headline level would imply $\bar{\kappa} = 0.2331$. The exact joint recompute instead gives 0.2301. The gap between them, −0.0030, is the interaction that the recompute captures; it is an order of magnitude smaller than the corresponding interaction under the demoted permissive variant, and of the opposite sign.

Table 18 and Figure 8 report the United States series under three lenses: the co-authors' baseline, the headline in-panel correction, and the panel-enlarged variant. The level moves in two steps as the lens changes. At 2024Q4, the panel-average $\bar{\kappa}$ starts at the baseline value of 0.353. The headline in-panel correction raises it to 0.361, a move of +2.3% across 2,518 treated firms, which reverses the sign of the demoted permissive variant. Appending the excluded firms then lowers the level to 0.230, a fall of −0.130, or −36.2% relative to the corrected level. The incumbent firms can also be examined on their own, measured against the headline-series panel. At 2024Q4, the variant moves their average $\bar{\kappa}$ by −0.0678, which decomposes into a dilution of −0.0846 and a linkage of +0.0168. At the 2023 annual mean, the movement is smaller, −0.0386, decomposing as −0.0391 of dilution and +0.0005 of linkage. The linkage is positive in all twenty quarters, the same pattern as the baseline reinstatement. Enlarging the panel therefore adds real common-ownership linkage, even as the reinstated firms' low profit weights lower the overall panel level.

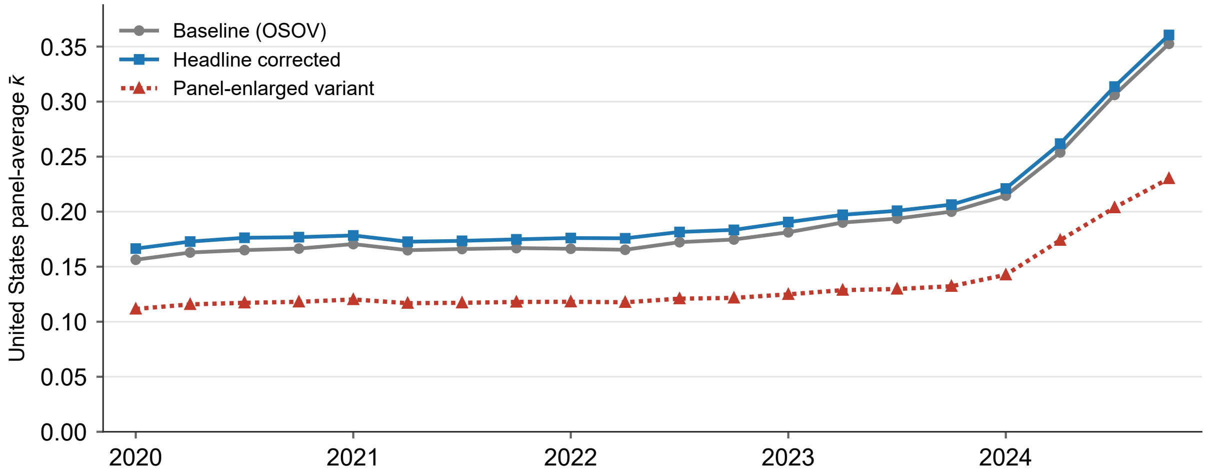
The ownership underlying the positive linkage predates the enlargement. Every incumbent-to-reinstated link was already carried in the co-authors' Thomson–Reuters ownership data. What the multiclass screen had deleted was the *pairs* that those links populate, and reinstatement simply restores them. A cross-link between two firms forms only through an owner that the two firms share. Here the shared owners are the ordinary institutional cross-holders, the index and 13F managers that hold both the incumbents and the reinstated, index-held dual-class mega-caps. A provenance decomposition of the 2024Q4 link mass confirms this account. Owners already present in the Thomson–Reuters data carry 100.000% of the incumbent-to-reinstated link mass, and overlay-introduced controllers carry 0.0000%. Consistently, the 123 reinstated firms whose disclosed holders are exclusively overlay-introduced controllers carry no incumbent links at all. The discovered multi-firm controllers of the Gates–Cascade type do form a genuine, second-order additive layer. That layer operates within the in-panel control-chains channel, outside the enlargement channel, and the enlargement channel introduces no new ownership.

Table 18: United States panel-enlarged variant: headline series composed with reinstated excluded firms

	2024Q4	2023 mean
Panel-average $\bar{\kappa}$		
Baseline (co-authors' OSOV)	0.353	0.191
Headline corrected	0.361	0.199
Panel-enlarged variant	0.230	0.129
Incumbent movement (variant vs. headline-corrected panel)		
Total movement	−0.0678	−0.0386
Dilution (mechanical)	−0.0846	−0.0391
Linkage (genuine)	0.0168	0.0005
Reinstated-firm vector coverage (2024Q4)		
Carrying corrected control vector (N)	450	
Entering as dilution-only node (N)	496	
Linkage: quarters positive	20/20	

Note: Series, bases, and caveats as defined in Section 2.6. The labeled panel-enlarged variant composes the headline in-panel control correction (the headline estimator on the pre-verification netting basis, applied to in-panel United States firms) with the reinstatement of the 946 excluded multiclass firms, recomputed jointly over the enlarged, corrected firm set. *Panel-average $\bar{\kappa}$* reports the United States average profit weight under three lenses: the co-authors' baseline (OSOV), the headline correction of in-panel firms, and the panel-enlarged variant. *Incumbent movement* reports the equal-weighted change in the in-panel firms' own $\bar{\kappa}$ from appending the reinstated firms, measured against the headline-series panel and decomposed as total = dilution + linkage by the displayed identity above, with linkage positive in all twenty quarters. The two corrections do not compose additively, because correcting in-panel weights changes those firms' links to the reinstated firms; at 2024Q4 the exact recompute (0.2301) differs from the naive additive value (0.2331) by −0.0030. Levels are shown to three decimals and the decomposition to four, and the corrected in-panel average reproduces the pre-verification headline netting basis to machine precision. All numbers on this variant remain subject to the verification cycle.

Figure 8: United States panel-average $\bar{\kappa}$ under the three lenses, 2020Q1–2024Q4



Note: Series, bases, and caveats as defined in Section 2.6. The United States panel-average profit weight $\bar{\kappa}$ by quarter under three lenses: the co-authors' baseline (OSOV), the headline correction of in-panel firms, and the panel-enlarged variant that additionally appends the 946 reinstated excluded firms. The in-panel correction shifts the level slightly upward in every quarter; enlarging the panel lowers it below the baseline. The panel-enlarged series is drawn as a dotted red line. The gap between the corrected and enlarged series widens toward the sample edge as active-vector coverage of the reinstated firms rises, whereas the baseline-to-corrected gap reflects the in-panel correction alone and does not depend on reinstatement coverage. All numbers on this variant remain subject to the verification cycle.

Table 22 in Appendix A restates the two reinstatements in the channel-decomposition format of the levels section, placing the panel-enlargement channel beside the in-panel channels on the same footing. For the United States, it reports a move from a reference $\bar{\kappa}$ of 0.3606 to an enlarged 0.2301, a total change of −0.1304. Of that total, −0.0678 is incumbent movement, which itself splits into −0.0846 of dilution and +0.0168 of linkage. For Sweden, the corresponding move is from 0.0406 to 0.0309. Because the reinstated firms are dual-class-routed by construction, their ownership-chain layer is zero by construction rather than by measurement.

11.2 The complete disposition of the excluded universe

The exclusion bias measured in Table 17 is the object of this section. How to interpret that bias depends on how much of the excluded universe the reinstatement has actually reached. That reach is stated completely in Table 19, which assigns every one of the 3,620 excluded issuers exactly one disposition and partitions the universe into five classes. The measured bias can then be read against an explicit account of what remains unmeasured. The account is deterministic: each firm's status follows from the artifacts of the reinstatement build under a fixed precedence, with no matching judgment introduced at this stage.

The canonical reinstated class is the largest of the five and contains 1,340 issuer links or firms. Most of these come from the two certified builds: 966 excluded United States issuer links, represented by 946 CIK nodes, and 79 Swedish firms in the certified register build. The remainder comes from the later extensions: 59 firms in the audited market wave, 16 audited Peruvian

Table 19: Complete disposition of the excluded multiclass universe

	Firms (N)	Share
Panel A. Disposition class		
Reinstated	1,340	37.0%
In-panel overlay correction	306	8.5%
Evidenced infeasible	790	21.8%
In progress	153	4.2%
Historical-vintage, pending	1,031	28.5%
Total excluded universe	3,620	100.0%
Panel B. Status detail (within class)		
Reinstated		
United States, certified (966 issuer links; 946 CIK nodes)	966	26.7%
Sweden, certified	79	2.2%
B1 market wave, audited	59	1.6%
Peru audited increment (17-firm final roster)	16	0.4%
Canada audited increment (Fairfax; 16-firm final roster)	1	0.0%
Adopted control-chains route	169	4.7%
Adopted proportional equal-vote route	47	1.3%
Adopted non-genuine-verdict built-vector route	3	0.1%
In-panel overlay		
Correction vector mapped	245	6.8%
Correction vector pending	61	1.7%
Evidenced infeasible		
United States bridge shortfall	637	17.6%
No control-chain, no TR identity	110	3.0%
Control-chains vector infeasible	21	0.6%
Capital-only register or not listed in window	18	0.5%
Equal-vote, dispersed-only roster	3	0.1%
Verdict failed current-vintage verification	1	0.0%
In progress		
Harvestable, queued	95	2.6%
Matched, awaiting build/append	42	1.2%
Round-2 evidence, needs review	8	0.2%
Identity under review	8	0.2%
Historical-vintage, pending		
Historical-only vintage, shelved	1,031	28.5%

Table 19: Complete disposition of the excluded multiclass universe (continued)

Market	Excl. (N)	Rein- stated	Over- lay	Infea- sible	In prog.	Hist. pend.
Panel C. Largest excluded populations by market						
United States	1,603	966	0	637	0	0
China	409	137	150	23	0	99
Canada	373	38	1	60	35	239
Sweden	216	81	0	2	18	115
United Kingdom	179	7	27	8	39	98
Poland	137	11	0	2	18	106
Indonesia	89	0	31	4	0	54
Mexico	71	6	6	6	9	44
Switzerland	53	5	14	6	6	22
Peru	47	22	0	0	5	20
Australia	45	5	0	1	0	39
Finland	44	13	18	2	2	9
Denmark	41	10	0	1	13	17
Other markets (28)	313	39	59	38	8	169
Total	3,620	1,340	306	790	153	1,031

Note: Series, bases, and caveats as defined in Section 2.6. Complete deterministic disposition of the 3,620 excluded multiclass issuers: every firm carries exactly one status, and the five classes of Panel A partition the universe. *Reinstated* contains 966 US issuer links to 946 representative CIK nodes, 79 certified Swedish firms, the 59-firm audited wave, and 219 firms measured through the three routes adopted on 22 July 2026 (169 control-chains, 47 proportional equal-vote, and 3 non-genuine-verdict built-vector), plus 16 audited Peru increments and one audited Canada increment (Fairfax), for 1,340 reinstated firms. Route provenance is retained per firm; all 219 route vectors remain non-citable pending the next certification cycle. *In-panel overlay* are the firms that reappear in the panel through a per-security overlap and are corrected in place (306; 245 carry a mapped vector). *Evidenced infeasible* records an evidenced terminal outcome, chiefly the United States bridge shortfall (637: 509 with no usable 2020–2024 proxy and 128 with no CIK bridge). *In progress* and *historical-vintage, pending* bound the work that remains. Panel C aggregates the per-firm statuses into the five classes for the largest excluded populations. Row sums reproduce each market's excluded count. Full status vocabulary and per-market matrix in `DISPOSITION.md`. Citability remains subject to the verification cycle.

increments, the audited Fairfax Canadian increment, and 219 firms entering through the three adopted measurement routes described below. The in-panel overlay class carries the 306 firms that reappear in the panel through a per-security overlap, of which 245 already hold a mapped correction vector. The evidenced-infeasible class of 790 firms records a terminal outcome supported by evidence rather than an open task. That class is dominated by the United States bridge shortfall of 637 issuers, of which 509 have no usable 2020–2024 proxy statement and 128 have no CIK bridge. The remaining mass is prospective. It comprises 153 firms still in progress and 1,031 historical-vintage issuers, whose listings fall outside the 2020–2024 estimation window and which are shelved pending the vintage decision recorded in Section 12.

The append engine has been executed across 19 Panel-B markets plus Sweden. That run restored 74 Panel-B firms and 81 Swedish firms alongside the United States links. The Peruvian audit closes the disposition of all 34 genuine-wedge firms, seventeen of them appended. The supporting records for these appends are collected in the reinstatement audit notes of Appendix A: the per-firm identity recoveries, register harvests, owner-identity adjudications, and evidenced-infeasible case notes, together with the Peruvian per-firm outcomes and six out-of-spine disclosures.

The three adopted measurement routes extend coverage into markets and firm types that the primary voting-derivation pipeline does not yet reach. The first is a control-chains route, which appends 169 firms across nineteen markets and carries the ORBIS ultimate-owner divergence between control and cash-flow rights through pyramids and holding chains. The second is a proportional equal-vote route, which appends 47 Chinese equal-vote firms under the votes-equal-capital identity. The third appends the built two-leg vectors of three adjudicated non-wedge firms. All three routes were adopted on 2026-07-22 as admissible measurements of voting control, and each is reported with per-firm route provenance. Control is defensibly measured for about 114 of the 219 route firms. Every adopted-route vector nonetheless remains *non_citable* pending the next certification cycle. The verification ledger's status taxonomy, per-status counts, and engine-gate outcomes are likewise recorded in Appendix A.

The geography of the recovery deserves a direct statement. The certified United States and Swedish builds contribute 1,047 of the 1,340 reinstated issuer links, and the other 39 markets contribute the remaining 293. Three facts place these magnitudes in context. First, the excluded universe itself concentrates where multiclass structures concentrate: of the 3,620 excluded issuers, 1,603 (or 44%) are American and 216 are Swedish, which leaves 1,801 in all other markets. Second, recovery outside the two certified markets runs through two channels that together account for 599 firms. Reinstatement through the market wave, the adopted routes, and the audited Peruvian and Canadian increments supplies 293 of them, and the in-panel overlay supplies the other 306, all lying outside the United States and Sweden. Third, most of the firms that remain outside the two certified markets are closed to current-vintage measurement by construction. Of those 1,202 firms, 916 are historical-vintage issuers whose listings fall outside the 2020–2024 window, and 151 record an evidenced terminal outcome. The addressable population is thus 734 firms, of which 599 (or 82%) are recovered through reinstatement or the overlay and 135 remain in the harvest queue.

The largest apparent gaps are dominated by historical-vintage listings, notably in Canada, Poland, and Indonesia. China, the largest excluded market outside the United States, is substantially covered: of its 409 excluded firms, 137 are reinstated and 150 are overlay firms. The two certified markets dominate in level for a structural reason. The United States and Sweden are the two jurisdictions with authoritative, machine-readable whole-market registers, the DEF 14A proxy statements and the Swedish FI ownership registers, which allowed full-market certified builds. Elsewhere the same evidence sits in heterogeneous per-firm filings, so the routes, the market wave, and the in-progress queue constitute that per-firm harvest.

Read against this disposition, the exclusion bias of Table 17 is measured on the firms that the reinstatement actually reaches. The classes that remain open bound what the current measurement cannot yet see. These are chiefly the 1,031 historical-vintage issuers outside the estimation window and the 790 firms whose evidence records a terminal outcome, most of which have no per-holder voting leg to restore.

12 Open decisions and next steps

The exhibits in Sections 5–11 present the control correction on one chosen basis, the headline series, beside three comparators: the strict series that admits the verified discovered controllers alone, the demoted permissive variant, and the specification robustness checks. This section records the editorial decisions taken in the present revision, along with the decisions and upstream tasks that remain open. The author made the decisions in the first list as a co-author, subject to full co-author ratification (each decision stands as recorded until the whole team formally adopts or revises it).

The figures throughout are drawn from the current build, though they do not all sit on the same netting basis. The summary-statistics, census, levels, attribution, and robustness exhibits are regenerated on the post-verification netting basis. The remaining exhibits still reproduce the frozen pre-verification netting basis and await regeneration. Figure 10 in Appendix B is a deliberate exception, retained as an earlier-pool sensitivity simulation. Citability remains subject to the verification cycle.

12.1 Decisions taken in this revision, subject to ratification

The present revision made the following decisions, all of which remain reversible because labelled variants retain the alternatives. Ratification therefore amends a display choice without changing the computation.

1. **Displayed corrected series.** The displayed series is the headline series, which combines the verified and the vendor-sourced discovered controllers while excluding the held-out candidate controllers. The vendor-sourced tier's records have since been verified individually (Section 8 and Appendix B). At 2024Q4 this estimator moves the all-country median $\bar{\kappa}$ by −0.68%. It also moves the United States from 0.353 to 0.361 (Section 6). This choice replaces the former practice of displaying the permissive series.

2. **Strict verified-only bound.** The tier of verified discovered controllers (165 records) is reported beside the headline as a strict robustness bound. At 2024Q4 it moves the all-country median $\bar{\kappa}$ by +0.71%. Because the headline estimator moves the median in the opposite direction, the sign of the median correction depends on whether the vendor-sourced tier is admitted. The two figures are therefore presented together.

3. **Demotion of the permissive variant.** The estimator displayed in earlier drafts is demoted to a labelled variant. That estimator retained all candidate records and moved the median by −38.26% at 2024Q4. The attribution exhibit (Section 8) retains and explains it as the transparency benchmark for how much of the earlier movement rested on records the hand-coded roster cannot corroborate. The overlap audit reported there supports the demotion. The

records the variant retains beyond the headline are unverified rather than refuted: within the applied quarantined tier, only the web-verification layer's 291 refuted records and the eight overlap-audit demotions are affirmatively dismissed, and the reason codes on the remainder establish no row-level refutation. The demotion accordingly rests on two grounds, neither of which asserts that the held-out records are false: the variant draws nearly all of its movement from unadjudicated assertions, and its construction double-counts even the verified controllers against their own Thomson Reuters rows (Section 8).

4. **Loyalty-share treatment.** Loyalty-voting firms (33 issuers in France, Italy, and the Netherlands) are held out of the headline and restored in a separate loyalty-share satellite panel. The robustness table (Table 12, Section 9) quantifies the difference. Italy and the Netherlands move, while one restored French issuer has zero net effect. The Netherlands' 2024Q4 correction changes sign in the satellite. This treatment implements Option A of the loyalty decision below. The upstream defect in propagating the loyalty flag, which had blocked this variant, is now fixed, and the combined build is available.

5. **Record de-duplication from the overlap audit.** The overlap audit led to eight vendor-sourced records being demoted. These are three duplicates, four pension-plan records reclassified to the co-authors' institutional owner class, and one Chinese record pending identity resolution. The two New Zealand Superannuation Fund records are retained, by contrast, as a sovereign-wealth pair under the government-institution class. The issuer-level arithmetic gate that reverts overfull issuers to baseline (§3.6) applies here as the arithmetic backstop.

6. **Vintage anchoring of the corrected series.** Because the control overlay is of current (2020–2024) vintage, citable correction evidence is anchored to 2020Q1–2024Q4. The panel is recomputed over its full available 2005Q1–2024Q4 span for display, but the pre-2020 display remains under audit. As the panel runs back before 2020, the vendor-sourced control vectors become arithmetically incompatible with historical holding sums in progressively more issuers. The time profile of the overfull-issuer gate (§3.6) quantifies this growing incompatibility.

7. **Period-correct proxy backfill for the reinstated United States firms.** The temporal reinstatement anchors each roster at its proxy's filing quarter and forward-fills at most four quarters, so pre-2024 linkage in Figure 7 is a measured floor: at 2023Q4 only 169 rosters are in-window, against 450 at 2024Q4. Each reinstated firm's own 2020–2023 DEF 14A filings exist on EDGAR for the quarters in which the firm was public, so a bounded per-firm harvest of the audited-cycle shape, executed through the same certified derivation, would anchor period-correct rosters in the earlier quarters, separate genuine pre-listing absence from the measurement floor, and turn the linkage series into a true time profile. Deferred to the historical lane; the always-positive sign of the linkage does not depend on it.

12.2 The co-author decision list, in priority order

The decisions below belong to the co-authors. They are ordered by how much each one changes what the paper claims: Tier 1 defines the paper's stated results, Tier 2 fixes measurement conventions with visible movement, and Tier 3 settles record-level and presentational matters. A final list collects the requests — data, specifications, and access registrations — that call for co-author action rather than a ruling.

12.2.1 Tier 1: decisions that define what the paper claims

1. **Ratification of the series choice.** The recorded configuration (the headline series, the strict series, and the demoted permissive variant) is the author's recommendation; the co-authors decide whether to adopt it formally or to promote the strict series or the demoted permissive variant to the headline. This is the largest lever among the open decisions, because it sets the sign of the all-country result — the headline moves the 2024Q4 median $\bar{\kappa}$ by −0.68%, the strict series by +0.71%, and the demoted permissive variant moved it by −38.26% — and every downstream exhibit is displayed on the chosen basis. The decision has also become cheap to make well: promoting or retaining the vendor-sourced tier once required a record-by-record re-verification, which has now been executed (Section 8 and Appendix B), leaving 997 of 1,061 records verified and 64 unresolved.

2. **The default United States estimand.** The United States DEF 14A voting vectors are delivered as the labelled panel-enlarged variant (Section 11.1), which composes the headline correction with the reinstated excluded firms and recomputes the two jointly over the enlarged firm set; the integration overlay itself carries no United States dual-class rows, so the in-panel exhibits reach the United States through ownership-chain tracing only (Sections 5 and 6). The open choice is whether the enlarged panel becomes the paper's default United States estimand or remains a labelled variant beside the headline: whether the stated United States level reads 0.361 (corrected, shipped universe) or 0.230 (corrected, full universe including the 946 reinstated multiclass firms). The measurement is complete and audited, so what remains is an estimand-and-presentation choice together with the certification cycle that clears the adopted-route vectors' *non_citable* status. The author's recommendation is the enlarged panel as the default, with the shipped-universe series retained beside it for comparability with the published draft: an average for the United States that excludes its largest multiclass issuers does not describe the United States market. The route-split attribution build in the upstream list below is designed to inform exactly this discussion.

3. **Reinstatement vintage target.** The open question is whether reinstatement targets the 2020–2024 estimation window only or the full 2005–2024 history. The answer determines whether the 1,031 shelved historical-vintage issuers ever enter a κ panel, and with it whether the paper's time-series claims are corrected or only its recent cross-section. The question interacts with three recorded items: the vintage anchoring of citable evidence to 2020Q1–2024Q4, the pre-2020 display still under audit, and the period-correct proxy backfill, which is the natural first step of any historical extension. The working default adopts the complete-disposition framing within the current window and defers the historical-vintage rebuild; because the co-authors' published growth claims span the long window, the choice should be made deliberately rather than inherited from the default.

12.2.2 Tier 2: measurement conventions with visible movement

1. **Non-voting preferred convention.** Non-voting preferred classes (Korean preferred shares, Brazilian PN, German Vorzüge, Peruvian investment shares) follow the working convention of inclusion at their statutory zero votes, with a voting-common-only variant computed alongside. The convention governs the largest dual-class markets inside the panel — Brazil, Korea, and Germany — so its ratification locks in the in-panel dual-class results. The recommendation is to retain it: the proportional-control assumption fails in exactly these structures, and the voting-common-only variant already covers the narrower reading. Implementation is a one-line toggle in the build without re-extraction.

2. **Control-propagation rule.** The corrected series traces ultimate ownership through pyramid chains and retains a controller identified through such a chain; the alternative is the Aminadav–Papaioannou weakest-link rule. Section 9 shows the difference between the two rules to be small in the permissive satellite where they can be compared: across the 49 markets the median absolute difference in $\bar{\kappa}$ is 0.0001, with a maximum of 0.0024 in Ireland. The standing recommendation uses the ultimate-owner trace as primary and the weakest-link rule as a labelled robustness variant, because the weakest-link rule applies a second attenuation to the control side, and because deleting a pyramidal controller removes every common-ownership link it spans. Given the measured difference, this should be a fast ratification whose function is to fix the reported convention.

3. **Voting-power robustness leg.** The corrected γ is a measured voting share, and the substitution-overlay discussion of Section 2 discloses that proportionality in votes is directionally conservative at controlled firms. A Banzhaf or Shapley–Shubik weighted variant of κ, computed from the same shipped per-holder vote distributions, is the natural quantification of that modelling margin. The open decisions are whether to adopt such a variant as a labelled companion exhibit, which power index to use, and whether the co-authors' existing voting-power robustness checks already span the register-based correction. The recommendation is to build the companion exhibit once the series choice is ratified; the expected direction is disclosed — deeper reductions at founder-controlled firms, stronger intra-group links where the controller is itself diversified — so the leg brackets the headline rather than replacing it.

4. **Consumption of verified voting stakes.** The web-verification round produced observed voting stakes for the 216 records it confirmed, and the audit recommends replacing the vendor's claimed stakes with the verified observed stakes. Because that changes the engine's control inputs rather than only a record's tier, a rebuild follows; the natural sequencing takes this decision together with the series ratification, so the ratified series is rebuilt once, on the better inputs.

5. **Loyalty-share regime membership.** The settled measurement mechanism is the recorded loyalty-share treatment above: quarantine from the headline with a restoring satellite. The open question concerns the sample definition, namely whether the 33 loyalty firms ever enter the headline sample. The paper's original screen never removed them, so admitting them

would extend the sample definition and should be a deliberate co-author choice; the movement involved is bounded, although the Netherlands' 2024Q4 correction changes sign under the satellite. The recommended working default keeps them as an explicit opt-in satellite, which leaves the headline's sample identical to the co-authors' own.

12.2.3 Tier 3: record-level and presentational decisions

1. **Disposition of the unresolved residue.** The verification round could neither confirm nor refute 64 vendor-sourced records. They remain in the SECONDARY tier under benefit of the doubt and carry 0.213 of discovered-controller mass, a small fraction of the admitted total. Their final disposition awaits either targeted primary-source evidence or a co-author ruling to hold them out.

2. **The three uncertain wedge verdicts.** One Indian issuer whose depository-receipt flag conflicts with a partly-paid-share reading, and two United Arab Emirates listings whose literal super-vote classes may be de-SPAC artifacts, remain genuinely uncertain. Short web-evidence dossiers are the pending deliverable feeding the decision; the conservative default — outside the corrected set until a dossier positively confirms a genuine wedge — is already applied.

3. **Attribution conventions.** The construction (Section 3) adopted working conventions for cases in which the registers admit more than one defensible reading. One convention decides whether a stake held through a personal holding vehicle is attributed to the vehicle or to the person behind it. A second set, applied in the United States derivation (Section 10.2), governs issuer-specific proxy conventions: aggregate-vote rules, as-converted and as-exchanged treatments, and the tolerance for divergence between beneficial-ownership disclosure and economic stakes. Each convention is documented with its adjudication record. The recommendation is to ratify the adopted defaults as a block, revisiting individual conventions only where an exhibit turns on one.

4. **Formal regime classification.** The market ranking in Section 6 uses illustrative groups of four pyramid and four dispersed-ownership anchors, and its note records that the groups overlap in the middle of a continuous ordering. Whether the paper adopts a formal pyramid-versus-dispersed classification, and on what criterion, is a presentational choice; the levels result does not depend on any particular grouping.

12.2.4 Requests: data, specifications, and access registrations

1. **Regression specifications and estimation data.** The cross-sectional counterpart to the survival analysis re-estimates the co-authors' regression tables with corrected $\bar{\kappa}$ as a drop-in dependent variable, reporting coefficients side by side. The build proceeds on the author's side once the specifications and estimation data arrive, under the series-and-rule decisions above; the control-propagation alternative of Section 9 is the natural robustness variant.

2. **Raw multi-security Thomson–Reuters delivery.** The disposition covers the 41 markets present in the delivered exclusion record. Russia and Romania are absent from that record

and are a probable screen gap; confirming the screen's country scope and obtaining the raw multi-security delivery would let the multiclass screen run for those markets.

3. **Register access registrations, assignable to a research assistant.** Four disclosure channels are gated on free registrations that must be completed by a person. The Japanese EDINET v2 API key from the Financial Services Agency blocks the entire 30-firm Japanese harvest and is the one hard blocker in this list; Japan is also the one Panel C market whose 13 excluded firms warrant a full derivation (Table 14). The Korean OpenDART key upgrades a harvest that currently works through slower page-fragment parsing; the United Kingdom Companies House API key adds exemption checks; and the Norwegian VPS shareholder-register login opens the full register. Each registration takes minutes and is suited to a research assistant or doctoral student; they are listed here so they can be assigned.

12.3 Internal audit items and upstream work

A further set of decisions remains for the internal cross-model audit to resolve before any affected figure is cited. The **Barclays Wealth anchor** (issuer code 1355025) is a scope violation that would alter a verified-tier record, so it was not acted on in the present build. The **pre-2020 survival display** requires a choice, for the window before 2020 in which the channel legs are not separately isolable, between the mirrored, vintage-flagged corrected line and the clean ownership-chain line. Three **engine-hardening** corrections come from the overlap audit: an owner-role guard against reading an asset-manager or institutional row as a controller, porting the oversum gate into the recompute stage, and correcting a legacy label. The **Chinese overfull attribution** concerns two issuers caught by the arithmetic gate, which require manual adjudication of whether the state-tracing overlay and the co-authors' state-owned-enterprise consolidation double-count the same control.

The remaining work concerns upstream data, code, and post-shipping verification.

1. **Jurisdiction verification of the parked mass.** The quarantined records fall predominantly in the categories the pipeline marks unresolved by default. The headline is therefore a conservative lower bound on the coverage of newly discovered controllers, because genuine controllers may remain quarantined for want of verification. A verification round ordered by consequence is planned after shipping, to promote the records that verification confirms. It takes Singapore, Hong Kong, India, Indonesia, and the Philippines first. This quarantined-tier round is distinct from the completed secondary-tier web-verification (Section 8 and Appendix B), which resolved the vendor-sourced records themselves.

2. **Historical-quarter verification.** The web-verification round covered the 2024Q4 roster. A further 5,735 SECONDARY keys appear only in pre-2024Q4 quarters and were never web-checked. Verifying them is required before the corrected series is cited on the pre-2020 window.

3. **Amanet family identity adjudication.** The Amanet family key (*fam:ASSAF*), flagged by the verification cross-audit, must be adjudicated between Ruth and Avraham Assaf, with a

possible Assaf–Bar-Or concert to resolve.

4. **Dror Ortho blocker de-duplication.** The Dror Ortho block, whose shared Schedule 13D/G filing links it across issuers, must be de-duplicated so the same blocker is not counted twice in any κ overlay.

5. **Reinstatement rollout.** The multiclass exclusion is panel-wide, covering 3,620 issuers across 41 markets (Table 14). Table 19 records the complete disposition of every excluded firm, of which 1,340 issuer links or firms have been reinstated to date. Three alternative voting-control derivations — the control-chains, proportional equal-vote, and adjudicated non-wedge routes of Section 11.2 — add a further 219 firms. They were adopted on 2026-07-22 as admissible measurements of voting control, with per-firm provenance retained. All 219 route vectors remain *non_citable* pending the next certification cycle, and the route-audit ledger is reported in Appendix A. Completing the rollout requires several further steps: building voting-control vectors for the remaining verdict-matched Panel B firms; collecting the missing cash-flow data for Poland, where 14 of its 20 genuine-wedge firms still lack a per-holder cash-flow stake; recovering the two South African super-voting firms by harvesting their share-class ratios; scoping the markets where exclusion or pipeline coverage is not yet determined (Panels C and E); and holder-netting the 32 incumbent vectors that are built but not yet appended because they fail the engine's gates. The method for each step is established rather than open. Markets whose registers publish no per-holder voting leg take an audited per-firm harvest cycle — reading each firm's annual report or prospectus and extracting the holder–class structure directly — of the kind that dispositioned all 34 Peruvian genuine-wedge firms and the Canadian additions; built or appended vectors awaiting certification take record-by-record verification against primary documents, the same shape as the United States web-verification layer applied to its vendor-sourced tier. The populations are dozens of firms for the harvest cycles and a few hundred vectors for the certification pass, so the remaining rollout is bounded in scale; the chains-leg quarantine, by contrast, holds ~46,000 records, which is why record-by-record review is feasible here and infeasible there. This rollout is the stated limitation of the reinstatement analysis and its priority follow-on work.

6. **Route-split attribution of the enlargement.** Figure 1 tags the reinstatement routes by mechanism, and the disposition census backs those tags with firm counts; no exhibit yet decomposes the enlargement's κ movement by route. The open build is the natural co-author question: how much of the United States enlargement movement ($\bar{\kappa}$ 0.3606 $\rightarrow$ 0.2301 at 2024Q4) is carried by the certified DEF 14A firms and how much by the firms entering through the three adopted measurement routes. Producing the split requires route-subset recomputes of the enlargement rather than a re-reading of existing output, and its citability is sequenced behind the certification of the adopted-route vectors, which remain *non_citable*; a computed version carrying that caveat could precede certification if the co-author discussion needs it.

7. **Audited harvest disposition.** The 2026-07-23 cycle accounts for all 34 Peruvian genuine-wedge firms and adds Fairfax Financial as the sixteenth Canadian reinstatement. Its per-firm outcomes are recorded in the reinstatement audit notes of Appendix A, together with two

items still outstanding: the three Swedish genuine-wedge verdicts not yet built (Ortivus, Precio Fishbone, Slottsviken) and the Kinnevik AB holder-netting quarantine.

8. **Wedge time-invariance: results so far.** Each firm's voting-control vector is a single register snapshot applied across the 2020–2024 window, whereas the holdings side varies quarterly. The design matches measurement frequency to the frequency at which each object actually moves. A wedge is generated by charter class terms and a controller's class holdings, both of which change only at discrete corporate events such as unification, recapitalization, or conversion on sale. Institutional holdings, by contrast, churn quarterly and are measured quarterly in the TR panel. An annual wedge panel is in addition unobservable for most markets, since disclosure is annual at best and historical filings survive selectively. A panel constructed from them would import survivorship selection, concentrated in the concentrated-control tail. The conversion screen is complete for all 894 firms and identifies 214 in-window events. Of these, 40 in-panel firms show genuine, disclosed mid-life changes that invalidate a single snapshot. A cross-model audit of the completed DEF 14A annual panel qualifies its raw movement statistics. The median firm-year shows zero change against the snapshot. The large apparent year-to-year movements concentrate in firms whose charters are verifiably static. In those firms, fresh-year extraction on the permanent super-voting stratum intermittently fails to vote-weight the controlling class. Raw counts of moving firm-years therefore reflect extraction inconsistency rather than structural change, and they enter no κ computation. The genuine in-window changes the panel captures are a small, verified set of sunsets and recapitalizations, such as Twilio and Veeva in 2023 and U-Haul in 2022. These are the event class the conversion-event treatment is designed to absorb, so the exercise supports the register-snapshot design. Recorded follow-ups are a United States conversion-event screen, the extraction-consistency defect class on the super-voting stratum, and a magnitude review of the snapshot on that same stratum.

9. **Further register reconstructions.** The register-based reinstatement analysis of Section 11 reports the United States and Sweden as its primary two-market comparison, while the rollout and adopted routes now reach further markets. Extending primary register derivations to the remaining usable markets is scheduled after the decisions above.

The intended sequence follows the tiers: ratify the series choice, settle the United States estimand and the vintage target, then the Tier-2 conventions, with the Tier-3 items and the batch ratifications closing the list; the requests can be issued immediately and in parallel. The internal audit resolves the Barclays anchor, the pre-2020 survival display, and the engine-hardening items before any figure is cited. The loyalty satellite, the United States merge, and the regression analogs then proceed under the resulting configuration. A comprehensive cross-model verification of the present draft now includes the complete-disposition and route delta audit. Certification of the adopted route vectors nonetheless remains owed before any of those vectors is cited.

A Reference tables

The tables in this appendix provide reference material for selective consultation. The totals and argument-bearing numbers appear in the main text; each table below contains the full per-market detail behind a main-text excerpt or cross-market summary, and every note begins with the series, bases, and caveats defined in Section 2.6.

Table 20: Coverage and treatment of the control correction, by market

	Firms covered	Treated firms (by mechanism)			Total treated	Treated firms (%)	Treated mkt. cap (%)	Mean $\lvert\gamma-\beta\rvert$
		Chains	Dual-cl.	Both				
China	11979	3505	78	0	3583	29.9	52.5	0.158
United States	7195	2064	0	0	2064	28.7	17.4	0.085
India	6149	2464	3	2	2469	40.2	71.1	0.125
Japan	4024	1222	0	0	1222	30.4	25.1	0.122
Korea	2745	1172	17	32	1221	44.5	84.9	0.149
Canada	2475	426	1	1	428	17.3	12.9	0.124
Taiwan	2122	772	2	0	774	36.5	13.4	0.093
Australia	1763	731	0	0	731	41.5	16.7	0.080
Vietnam	1574	775	1	0	776	49.3	77.7	0.151
United Kingdom	1506	886	0	1	887	58.9	39.8	0.098
Malaysia	1074	644	0	0	644	60.0	63.7	0.158
Indonesia	953	522	0	0	522	54.8	59.8	0.222
Thailand	873	373	0	0	373	42.7	43.3	0.181
Sweden	836	217	83	54	354	42.3	–	0.128
Russia	746	331	0	0	331	44.4	71.1	0.204
Poland	715	373	0	0	373	52.2	69.4	0.156
France	628	351	13	2	366	58.3	–	0.202
Germany	609	315	5	10	330	54.2	70.0	0.208
Singapore	562	335	12	0	347	61.7	56.0	0.182
Israel	457	171	1	2	174	38.1	40.0	0.168
Brazil	449	183	86	58	327	72.8	73.0	0.225
Italy	381	270	4	14	288	75.6	70.5	0.206
Romania	338	163	0	0	163	48.2	30.7	0.194
Saudi Arabia	306	86	0	0	86	28.1	9.8	0.112
Philippines	276	61	0	0	61	22.1	29.9	0.212
Spain	272	191	0	0	191	70.2	63.8	0.193
Norway	268	172	0	1	173	64.6	35.2	0.142
Switzerland	239	85	0	0	85	35.6	35.0	0.132
South Africa	228	130	0	0	130	57.0	64.0	0.100
Hong Kong	226	111	0	0	111	49.1	61.0	0.168
Egypt	224	84	0	0	84	37.5	31.2	0.176
Jordan	193	81	0	0	81	42.0	65.6	0.134
Chile	177	119	0	0	119	67.2	83.7	0.330

	Firms covered	Treated firms (by mechanism)			Total treated	Treated firms (%)	Treated mkt. cap (%)	Mean $\lvert\gamma - \beta\rvert$
		Chains	Dual-cl.	Both				
Denmark	169	96	10	3	109	64.5	50.5	0.139
Finland	167	91	6	4	101	60.5	93.8	0.105
Peru	165	80	0	0	80	48.5	70.3	0.404
Netherlands	144	69	3	2	74	51.4	72.5	0.171
U.A.E.	143	69	1	0	70	49.0	55.5	0.246
Kuwait	134	35	0	0	35	26.1	11.2	0.222
Greece	132	48	0	0	48	36.4	21.8	0.132
Belgium	128	74	0	0	74	57.8	52.2	0.186
Oman	128	70	0	0	70	54.7	53.3	0.173
Mexico	127	60	0	1	61	48.0	69.8	0.230
New Zealand	124	58	0	0	58	46.8	31.9	0.077
Ireland	83	42	0	0	42	50.6	14.1	0.068
Croatia	82	26	0	0	26	31.7	25.1	0.127
Morocco	74	36	0	0	36	48.6	62.2	0.278
Austria	66	38	3	2	43	65.2	74.5	0.207
Colombia	60	38	0	0	38	63.3	69.1	0.361
All countries	54488	20315	329	189	20833	38.2	50.3	0.144

Note: Series, bases, and caveats are as defined in §2.6. Markets are ordered by the number of firms covered. *Firms covered* counts distinct listed firms for which the correction computes a corrected control vector; the census adopts this cross-sectional *gamma_overlay* universe as its covered definition. That universe is not nested within the overlay panel's per-quarter universe, which covers more firms than the cross-section in some markets and fewer in others. *Treated firms* are those whose control vector the correction alters, identified from an owner-level wedge $\gamma - \beta$ between corrected control (γ) and cash-flow rights (β) that exceeds a 10^{-6} tolerance. A firm is chains-treated when ownership-chain tracing moves control, so that traced control differs from the cash-flow chain; it is dual-class-treated when it carries a genuine voting–cash-flow wedge; and it is *both* when both mechanisms apply. Dual-class firms are matched to their ownership-chain identity through the recorded partner identifier. *Treated firms (%)* is treated firms over covered firms. *Treated mkt. cap (%)* is the treated firms' share of covered-firm market capitalisation, computed within each market over its modal reporting currency; the All-countries entry is the unweighted mean of the market-level shares over the 47 joinable markets. A mechanical capitalisation-weighted global share would evaluate to 69.6% but would sum local-currency capitalisations without exchange-rate conversion. Two markets whose firm identifiers do not join to the market-cap source (France, Sweden) are shown as "–". *Mean* $\lvert\gamma - \beta\rvert$ is the largest owner-level control-share reallocation within a treated firm, averaged over treated firms, in control-share units. In this integration vintage the dual-class layer covers 22 markets, and the United States enters only through ownership-chain tracing; its DEF 14A voting vectors are reported separately as a labeled panel-enlarged variant in the reinstatement exhibit of Section 11.1 and remain outside the overlay. The reinstatement effect of the two-vintage CIK bridge is quantified there and omitted from the integration-layer reinstatement table. Panel cross-checks use the demoted permissive variant at reference quarter 2023Q4. The applied-under-headline counts on the post-verification netting basis are recorded in the applied-recompute artifact, and the full per-market figures, sources, and discrepancies are recorded in *census_numbers.json*.

Table 21: Overlap between the correction data and the co-authors' TR panel, by market

	TR-panel firms	Covered firms	Matched to TR	Match (%)	Treated firms	Overlay applied	Applied/ TR (%)	Treated cap (%)
China	5610	11979	4654	38.9	3583	3646	65.0	66.9
India	3947	6149	4348	70.7	2469	3580	90.7	81.1
United States	3757	7195	5454	75.8	2064	2935	78.1	95.6
Japan	3681	4024	3744	93.0	1222	3268	88.8	84.9
Korea	2440	2745	2283	83.2	1221	2137	87.6	95.2
Canada	1779	2475	2257	91.2	428	1177	66.2	69.0
Taiwan	1723	2122	1621	76.4	774	1545	89.7	80.7
Australia	1545	1763	1591	90.2	731	1212	78.4	75.4
Hong Kong	1219	226	193	85.4	111	140	11.5	22.9
United Kingdom	1205	1506	1198	79.5	887	900	74.7	92.4
Malaysia	918	1074	944	87.9	644	719	78.3	66.4
Vietnam	873	1574	125	7.9	776	92	10.5	15.6
Thailand	842	873	573	65.6	373	531	63.1	69.4
Indonesia	603	953	857	89.9	522	575	95.4	81.4
Poland	564	715	601	84.1	373	474	84.0	68.0
Singapore	531	562	479	85.2	347	396	74.6	54.0
Sweden	524	836	683	81.7	354	387	73.9	63.0
Israel	512	457	411	89.9	174	304	59.4	68.5
France	509	628	517	82.3	366	408	80.2	80.2
Germany	431	609	471	77.3	330	347	80.5	90.0
Italy	380	381	304	79.8	288	259	68.2	81.4
Romania	301	338	238	70.4	163	220	73.1	39.3
Brazil	261	449	304	67.7	327	207	79.3	87.5
Saudi Arabia	247	306	6	2.0	86	5	2.0	0.3
Norway	230	268	226	84.3	173	184	80.0	93.6
Spain	224	272	203	74.6	191	169	75.4	81.2
Philippines	217	276	248	89.9	61	185	85.3	92.3
Egypt	205	224	154	68.8	84	137	66.8	75.6
Switzerland	197	239	210	87.9	85	164	83.2	91.5
South Africa	181	228	193	84.6	130	140	77.3	75.7
Jordan	178	193	38	19.7	81	31	17.4	41.3
Chile	147	177	147	83.1	119	101	68.7	76.1
Greece	146	132	89	67.4	48	80	54.8	62.5
Finland	145	167	156	93.4	101	118	81.4	96.6
U.A.E.	144	143	110	76.9	70	101	70.1	84.1
Kuwait	137	134	94	70.1	35	81	59.1	42.7
Russia	128	746	2	0.3	331	1	0.8	0.5
Netherlands	124	144	123	85.4	74	88	71.0	86.7
New Zealand	112	124	115	92.7	58	85	75.9	79.5
Denmark	101	169	133	78.7	109	76	75.2	76.7
Oman	101	128	94	73.4	70	77	76.2	87.4
Belgium	96	128	105	82.0	74	84	87.5	77.2
Croatia	75	82	51	62.2	26	36	48.0	44.0
Ireland	65	83	74	89.2	42	59	90.8	99.9

	TR-panel firms	Covered firms	Matched to TR	Match (%)	Treated firms	Overlay applied	Applied/ TR (%)	Treated cap (%)
Morocco	64	74	52	70.3	36	46	71.9	84.1
Mexico	62	127	106	83.5	61	48	77.4	85.0
Austria	55	66	57	86.4	43	49	89.1	95.0
Colombia	54	60	51	85.0	38	40	74.1	73.6
Peru	49	165	106	64.2	80	41	83.7	72.5
All markets	37639	54488	36793	67.5	20833	27685	73.6	71.5

Note: Series, bases, and caveats are as defined in Section 2.6. The table reports market-level overlap between the correction datasets and the co-authors' TR holdings panel at the reference quarter 2023Q4. *TR-panel firms* is the overlay-panel firm count. *Covered firms* is the cross-sectional correction universe, and *Treated firms* is the covered firms whose control vector the correction changes (total 20,833); both columns are identical to Table 20. *Matched to TR* counts covered firms whose crosswalk row matches a TR issuer at the ISIN, exact-name, or fuzzy-name tier, and *Match (%)* scales that count by covered firms. *Overlay applied* counts TR firms receiving the demoted permissive variant's corrected vector at 2023Q4 (total 27,685; the headline series applies to 27,179), and *Applied/TR (%)* scales that count by TR-panel firms. *Treated cap (%)* is the overlay-applied firms' share of TR-panel market capitalisation; the All-markets row gives the unweighted cross-market mean for this column and column sums otherwise. Because this share uses the TR-panel denominator, it is not comparable to the census's covered-universe treated share (50.3%).

Table 22: Panel-enlargement channel: reinstated multiclass issuers (2024Q4)

	Reference $\bar{\kappa}$	Enlarged $\bar{\kappa}$	Δ total	Incumbent movement		
				Total	Dilution	Linkage
United States	0.3606	0.2301	−0.1304	−0.0678	−0.0846	0.0168
Sweden (OSOV)	0.0406	0.0309	−0.0097	−0.0050	−0.0063	0.0013

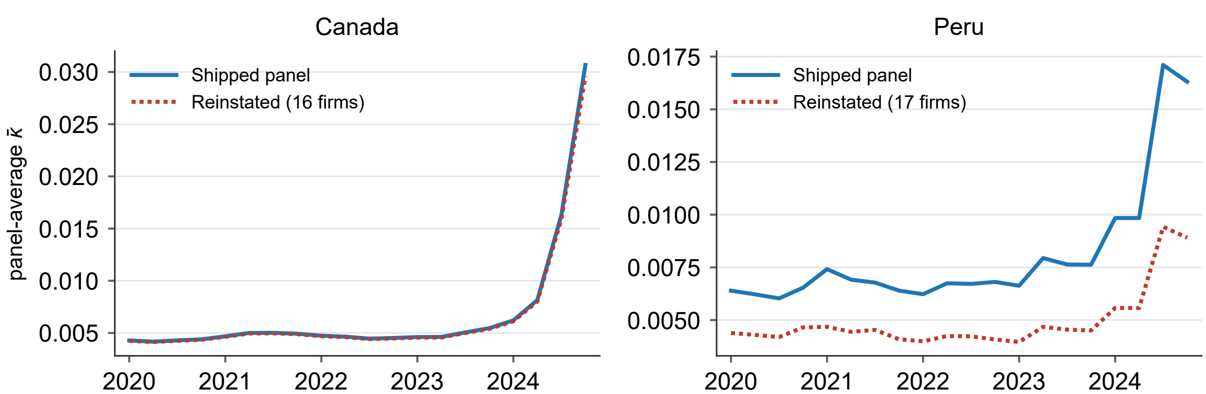
Note: The United States rows are drawn from the pre-verification panel-enlarged build (the labeled panel-enlarged variant, with citability subject to the verification cycle); the Sweden row is drawn from the OSOV reinstatement series (*b1-sweden-v2*). The table reports the second population of the correction, in which firms excluded from the baseline panel are reinstated and average $\bar{\kappa}$ is recomputed over the enlarged panel. The reference column is the panel average before reinstatement: for the United States it is the headline-series panel (the headline correction of the in-panel firms), and for Sweden it is the baseline of the OSOV reinstatement series. The enlarged column is the average after reinstatement, and the delta-total column is their difference. The delta total is the sum of an incumbent-movement component and a composition component. The composition component is the implied residual, contributed by the reinstated firms' own κ, while the incumbent-movement component is the change borne by firms already in the panel. The incumbent-movement total decomposes further into a dilution component, which is the denominator effect of enlarging the firm count, and a linkage component, which is the new common-ownership links from reinstated firms to incumbents. For the United States, 946 excluded multiclass firms (representative CIK nodes) are reinstated, 450 with full recomputed control vectors and 496 as dilution-only, moving the panel average from 0.3606 to 0.2301. The reinstated firms are dual-class-routed by construction, and the reinstatement engine applies no ownership-chain correction to any reinstated firm (the applied legs are recorded in the build manifest), so their ownership-chain layer has a construction-imposed value of zero and does not reflect measurement. Provenance and gate statistics are recorded in *reinstatement_numbers.json* and the build manifest.

Table 23: Audited Peru and Canada reinstatement extensions

	Canada (16)	Peru (17)
Reinstated multiclass firms (N)	16	17
Enlarged-panel scope		
Shipped-panel $\bar{\kappa}$ (2023 mean)	0.005	0.007
Reinstated-panel $\bar{\kappa}$ (2023 mean)	0.005	0.004
$\Delta\bar{\kappa}$ level (2023 mean)	−0.000	−0.003
$\Delta\bar{\kappa}$ % (2023 mean)	−1.1	−40.7
Incumbent scope: 2023 mean		
Total movement	−0.0000	−0.0016
Dilution (mechanical)	−0.0000	−0.0019
Linkage (genuine)	0.0000	0.0003
Linkage: quarters positive	20/20	20/20

Note: Final audited rosters contain 17 Peruvian firms and 16 Canadian firms. Levels and the exact incumbent decomposition use the same construction as Table 17; linkage is positive in every quarter. The Peruvian roster includes Alicorp, and the Canadian roster includes Fairfax Financial.

Figure 9: Peru and Canada reinstatement series



Note: Shipped-panel baseline in solid blue and the audited reinstated series in dotted red, preserving the convention used for the enlarged Swedish route.

A.0.0.1 Reinstatement audit notes.

These notes record the per-firm evidence behind the disposition accounting of Section 11. *Market wave.* The 59-firm audited wave includes three Polish, eight Canadian, and two Swedish round-two additions whose vectors and appends passed an adversarial cross-model audit on 2026-07-22. *Swedish recoveries.* The Swedish build was enlarged beyond its certified 79 firms in two ways: a coverage-gap harvest of Telefonaktiebolaget LM Ericsson, whose controllers hold 24.5% (Investor AB) and 15.1% (Industrivärden) of the votes against 9.3% and 2.6% of the capital, and a web-adjudicated identity recovery (Midway Holding). The certified 79 firms reproduce with identical control vectors inside the enlarged series. Three audited genuine-wedge determinations (Ortivus, Precio Fishbone, Slottsviken) remain at the *genuine_verdict_unbuilt* stage. IFS and Iptor are resolved grade-A negatives because neither was listed in the estimation window. The merger-successor match of the pre-2004 Kinnevik entity to today's Kinnevik AB is recorded in the spine, with its vector held back by the over-sum quarantine. *Danish and Finnish appends.* These rest on a web-grounded owner-identity adjudication that resolves the ownership registers' synthetic holder strings to the co-authors' Thomson–Reuters owner codes, anchored on two verified appends: the Danish holding behind Ambu and the Finnish pension holding behind Metsä Board and KONE. With the verified owners in place, the Danish and Finnish reinstated levels rise above the synthetic-holder counterfactual the audit had flagged. *Evidenced-infeasible cases.* The class includes one firm whose frozen wedge verdict failed current-vintage verification (Alimentation Couche-Tard, whose 2025 conversion collapsed the dual class). *Peruvian per-firm outcomes.* Of the 34 genuine-wedge firms, seventeen are appended (including Alicorp); one is routed to the overlay class and recorded in the audited sidecar because it is not an excluded-spine issuer; one remains at *identity_review*; three resolve to no holder at or above 15%; and six remain in *needs_review*. A further six genuine-wedge firms outside the co-authors' 3,620 exclusion spine are preserved in the audited sidecar without entering the disposition ledger. *In-progress split.* The 153 in-progress firms divide 95 + 42 + 8 + 8 across the four tracked stages. *Route verification ledger.* Each of the 219 adopted-route firms carries one primary status, assigned under the precedence *vector_flagged* > *nominee_fixed* > *corrected* > *vintage_refreshed* > *coverage_limited* > *route_confirmed* > *verified_clean* > *unverifiable*, alongside a separate seven-firm vector-quality-pending class. The respective primary counts are 25, 11, 3, 8, 39, 65, 49, 12, and 7, and the raw float-only coverage finding is 40 because one such firm is also vector-flagged. In substantive terms, control is defensibly measured for about 114 firms — 44 clean control-chain, 65 identity-confirmed chain-semantics, and five clean proportional cases — whereas 40 proportional cases are float-only, 25 are vector-flagged, 12 are unverifiable, and nominee defects were fixed for 11; audit findings can overlap even though the ledger primary status cannot. *Engine gates.* Of the equal-vote route's candidates, 48 are struck by the in-panel guard because they already appear in the built panel, and 3 carry only a dispersed roster. Of the non-wedge route's 35 candidates, three survive the gates (ATCO, Canadian Utilities, and Vend Marketplaces), while 18 lack a usable two-leg named vector, 10 fail the over-sum guard, 3 are dispersed-only, and one is an alias; each rejection is recorded with its reason.

B Measurement of the vendor-sourced tier (transitional)

This transitional appendix documents the measurement apparatus behind the vendor-sourced tier that the headline series admits. The record-by-record web-verification cycle described below has resolved all but 64 of the tier's records; once the remaining jurisdiction verification (pending-work item 1) clears that residue, the apparatus collapses into the record-grade ledger and this appendix is retired.

B.1 Construction detail

The passages below give the full construction mechanics behind the correction's three derivation legs and the China vehicle-level overlap, moved here from the main-text construction section (§3.2, §3.3, §3.4, and §3.6), whose paragraphs point here.

Control-chain reconstruction.

Deterministic parsing converts the vendor's percentage codes and holder-type codes to numeric stakes and typed owners. Line items are then consolidated into ultimate-owner blocks: individuals are aggregated into families under keys assigned offline by the judgment tier, corporate holders are rolled up to their global ultimate owner, sibling special-purpose vehicles are merged, and free float, treasury positions, and unnamed residuals are dropped from the block structure. A concert tier resolves, with cached verdicts, which legally distinct co-investors constitute a single control group and which are rivals to be left apart. Three identity passes supplement the base build. A named-beneficial look-through re-keys nominee register entries that carry their beneficial owner inline to that owner and drops the rest as float. A deduplication pass then removes double-counted owners. Its trigger is an impossibility argument: when one firm's named blocks sum to more than 100% of voting rights, at least two of those blocks must overlap. The pass deterministically generates candidate same-entity pairs, and the judgment tier adjudicates each pair against web evidence, because the distinction between a genuine alias and a coincidental collision between unrelated institutions requires judgment. A confirmed alias folds two blocks into one owner—for example, a translation variant, an acronym and its expansion, or a person and their personal holding company—and the absorbed block's cash-flow leg is re-attributed to the survivor. Throughout, structural guards protect the build from model failure: a rate alarm blocks any build in which the tier marks an implausible share of named individuals as dispersed, and population-level movement guards verify after each pass that country control shares moved only within tight bounds. Validation is a method-level comparison, because the vendor's current vintage cannot replicate Aminadav and Papaioannou's 2012 cross-section; the vintage gap is stated explicitly. Time-stable named pyramids provide a sharper check: the cascades behind Hermès, LVMH, and L'Oréal resolve to the correct controlling families with economically sensible wedges, near zero where the family wholly owns its holding chain and large where control rests on a thin cash-flow stake.

Non-US dual-class construction.

Construction proceeded in four stages, with the first mapping sources of per-holder and per-class disclosure in each jurisdiction, including registers maintained by securities regulators, notifications of threshold crossings, stock-exchange filings, and annual reports. Negative findings document a jurisdiction's absence from the panel, and every selected document was archived under a checksum manifest before parsing, making the build reproducible against the documents fetched. Deterministic, register-specific adapters then handle table walking, footnote linkage, locale and number-format handling, and PDF parsing, and map each register into a common schema. Each adapter is routed by register archetype. The archetypes are structured machine-readable filings; semi-structured filings that print both ownership legs; non-Latin-script PDF regimes; EDGAR 20-

F filings of foreign private issuers; and notification streams that carry only the voting leg, joined to a separately sourced capital leg. Most non-US registers publish each holder's percentage of capital and percentage of votes side by side; the published figure is taken as primary, the recomputation from class terms provides a cross-check, and the divergence between the two becomes a quality instrument attached to every row. Two final checks apply to the committed panel. The first is a hard universal bound: it excludes an entire firm whenever any computed leg leaves $[0, 1]$ or a wedge reaches one in absolute value, treating the violation as a parse or denominator error rather than a data point. The second is a statute-derived envelope, computed per register from the lawful vote-ratio cap and audited statute by statute, which moves individual rows above the legal ceiling into review. Validation uses named controller anchors, including the Wallenberg sphere at SKF and Porsche SE at Volkswagen; verification against the primary register is repeated on every rebuild, with any movement treated as a regression. A round-trip gate required the generalised schema to reproduce the US-derived wedge set exactly before any foreign number was trusted. The remainder of the adjudicated candidates that carry no genuine wedge are equal-vote multiclass structures (dual listings, nationality classes, par tranches), instrument artifacts (warrants, preferred share lines, de-SPAC leftovers), board-seat rather than voting wedges, or firms with no extractable holder data. The adjudication spans 37 of the 48 covered non-US markets, and the eleven markets without adjudicated candidates carry recorded coverage reasons in the pipeline ledger: the Japanese harvest (30 candidate firms) is written but gated on the EDINET disclosure-API key; the Russian register is bot-walled and withholds part of its disclosures; the Australian and New Zealand multiple-class counts reflect trust and stapled-security structures rather than founder dual-class equity; Jordan and Kuwait surfaced zero candidate firms; and Croatia, Romania, Ireland, Egypt, and Saudi Arabia carry near-zero candidate counts whose registers were never promoted to harvest. Each entry is a coverage record, and no market-level absence of wedges is adjudicated from it. The integration overlay remains frozen at 616 firms and 3,854 holder rows. The current verdict census includes Ericsson, whose live Sweden build contains 15 holder rows. The committed refresh path nonetheless preserves the frozen genuine-wedge set and does not document a safe crosswalk-and-eligibility rebuild, so Ericsson is a known integration seam rather than a hand-appended overlay row.

United States DEF 14A derivation.

The universe is the excluded-issuer set, bridged to EDGAR filings by CIK and collapsed to 946 representative CIK nodes. Of these nodes, 702 are in-scope multiple-class candidates. Within that in-scope set, the derivation resolves 360 genuine wedges, 261 proportional multiclass structures with no normalized wedge, and 66 single-class artifacts, and documents the remaining 15 as unresolvable. Twenty-five of the 261 have unequal nominal votes per economic leg but proportional normalized voting and economic entitlements. The extraction has three tiers. The first is a deterministic layer that resolves merged and spanning Item 403 headers, preserves each footnote's attachment to its cell, and anchors votes-per-share extraction to the narrative spans where it lives. The second is a bounded language-model fallback that decides the cases the deterministic layer cannot; it returns output against a fixed schema, with every numeric validated by pattern and range and frozen in a provenance-tagged cache, so the dataset remains deterministic. The third is a review-flag escalation: every value carries an extraction-method tag, and rows that fail even the fallback's validation escalate to that flag. Two conventions are fixed in code. Voting power is emitted both as the issuer's as-reported percentages, which use holder-specific denominators and are not summable across holders, and as a single-denominator consistent measure suitable for aggregation, with the divergence between the two surfaced as a column. The wedge is computed at the consolidated control-group level, both legs built from the group's summed holdings before differencing, matching the convention of the dual-class literature; accuracy was assessed against a reconciled benchmark, with a set of golden-benchmark firms reserved for validation and excluded from the derivation entirely.

China vehicle-level overlap.

China is the closest overlap case, because the co-authors consolidate state holders under one government owner while the reconstruction traces the same holders through provincial vehicles. There the discovered mass is 58.8% quarantined, 41.0% replacement, and 0.16% added, and the cross-model overlap audit judged the residual state-vehicle matches indeterminate, holding the eight arithmetically overfull rows for manual adjudication instead of demoting them automatically.

B.2 Record-grade definitions

We grade every candidate record by the strength of the evidence behind it into three tiers, and we report each tier's population at three stages: the graded stage, the frozen cache, and the 2024Q4 applied roster. The verified discovered controllers (internally MAIN) hold steady at 165 anchors across all three stages. Each anchor is mapped to a named row in the hand-coded Thomson Reuters roster, individually reviewed, and certified through repeated adversarial cross-model audits. A documented ten-record spot check of these anchors against primary filings returned no errors, at a one-sided 95% upper bound of 25.9%. The vendor-sourced controllers (internally SECONDARY) are taken from the ownership vendor's chain data. Their count moves across the three stages from 1,381 graded records to 1,373 in the frozen cache and then to 1,061 in the 2024Q4 applied roster, the last step reflecting the 291 refuted records that the web-verification layer demotes. A fresh adversarial sample of the graded tier, checked against local-language primary filings, failed at a measured rate of 26.0%, at a one-sided 95% upper bound of 38.1%. That historical rate motivated the record-by-record web-verification cycle, after which 997 of the 1,061 applied records (94%) are individually verified and 64 remain unresolved, the unresolved residue carrying 0.213 of discovered-controller mass. The quarantined tier (internally QUARANTINE) holds records set aside as identities the roster could not corroborate. Its count progresses from 45,846 graded to 45,854 cached and 46,129 applied at 2024Q4, so the full applied ledger across the three tiers is 165 / 1,061 / 46,129. The tier carries reason codes. Under the current two-code convention, about 62% of its discovered-controller denominator mass lies in the default unresolved classes and 38% falls outside them, and those codes do not establish row-level refutation.

B.3 Measured quality of the vendor-sourced records

Table 24 documents the disclosed error rate behind the vendor-sourced tier and the web-verification round that resolved it. It reports a verification census of the web-checked records, an error taxonomy of the demotions, and a fresh local-language recheck of sampled vendor-sourced records. The recheck provides the 26.0% figure quoted in the main-text attribution section.

Table 24: Vendor-sourced control records: measured quality

Panel A. Verification census		
Records in the round-9 verification worklist	1,656	
Initial web-check flag (true / false / unset)	1,341 / 304 / 11	
First audited web-verification round	315	records
Confirmed / refuted / unresolved	21 / 13 / 281	
Demotes already in earlier audits / incremental	5 / 8	
Updated web-check flag (true / false / unset)	1,375 / 281 / 0	
Audited SecVerify round (residual records)	563	records
Worklist extension (never-in-corpus)	+70	
Vendor records examined across rounds	1,726	
Confirmed / refuted / unresolved	216 / 283 / 64	
Prior-unresolved (278): C / R / U	81 / 151 / 46	
Kept low-confidence (215): C / R / U	86 / 114 / 15	
Never web-checked (70): C / R / U	49 / 18 / 3	
Refuted → demoted (folded into the netting layer)	283	
Post-round SECONDARY tier (2024Q4)	1,061	records
Individually verified	997	
Prior faithful / SecVerify-confirmed	781 / 216	
Honest-unresolved (mass 0.21)	64	
Demoted from tier (8 first-round + 283 SecVerify)	291	

Panel B. Error taxonomy of the frozen earlier-audit demotes

Error class	Detection mechanism	Count	Example
False-person attribution	Primary filing lists no such holder	178	Lim Ah Cheng / Stamford Land (SG)
Superseded controller	Filing shows the controller divested	47	Icahn / Cheniere: exited 2023
Custodian / manager as controller	A manager's rolled-up position, not personal	39	McBirney / Topline Capital (US)
State-institution rollup	State-institution block → sovereign	28	EPF / ViTrox (pension block)
Wrong-issuer link	Holder belongs to a different issuer	19	Malayandi / SMRT Holdings (MY)
Sovereign custodial rollup	Sovereign rolled up across micro-caps	14	Government of China / D&O (MY)
Sovereign chain artifact	Chain ends spuriously at a sovereign	9	Government of France / Catana (FR)
Corrupt identity key	Malformed identity as a controller	6	Johnson Matthey keyed as MATTHEY
Activist-fund principal	Fund / LP capital, not personal	5	Shulkin / Brainsway (Valor Equity)
Deceased ultimate owner	Owner deceased; not succeeded	5	Enderle / City Lodge (deceased 2019)
Duplicate-alias key	Same holder recorded under two keys	3	Prommasutra vs Phrommasut, one holder (TH)
Public authority as controller	A government body, not an owner	1	Legislative Assembly / SPAC (US)
Total demoted in earlier audits		354	

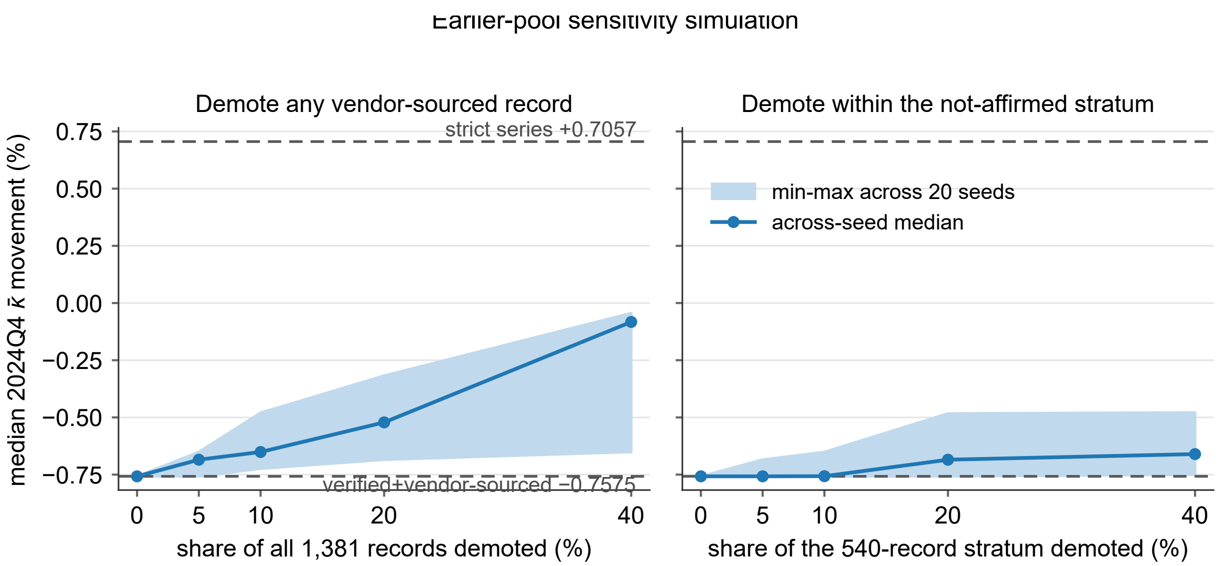
Panel C. Measured residual error rate		
Vendor-sourced controllers sampled (pre-verification)	50	records
Failed vs primary filing	13	(26.00%)
One-sided 95% upper bound		38.13%
Verified discovered controllers sampled	10	records
Failed vs primary filing	0	(0.00%)
One-sided 95% upper bound		25.89%

Note: *Panel A* traces the verification of the vendor-sourced tier across rounds in chronological order. The worklist starts at 1,656 owner–company records. The first audited web-verification round examined 315 of them and returned 21 confirmed, 13 refuted, and 281 unresolved; eight of its refutations were incremental beyond the earlier-audit demotes. That round closed its flag census at 1,375 true and 281 false. A four-shard SecVerify round came next: it examined the 563 records left unverified, and the addition of 70 never-in-corpus records extended the worklist to 1,726 vendor records examined across rounds. The SecVerify round returned 216 confirmations, 283 refutations, and 64 unresolved rulings; its refutation rate was highest among the previously unresolved records, at 151 of 278, and lowest among the never-checked records, at 18 of 70. The 283 refutations were folded into the web-verification netting layer. After this round the SECONDARY tier holds 1,061 records at 2024Q4. Of these, 997 are individually verified, comprising 781 carried forward as primary-source faithful and 216 newly confirmed, and 64 remain unresolved. The 291 demotes comprise the eight incremental first-round refutations and the 283 SecVerify refutations. *Panel B* preserves the frozen 354-record earlier-audit error taxonomy. *Panel C* rechecks a fresh sample of 50 vendor-sourced records against local-language primary filings, such as SSE/cninfo, EDINET, or the annual report. Of the 50, 13 failed, a 26.00% error rate with an exact one-sided 95% upper bound of 38.13%, while 10 verified discovered controllers returned no failures. This 26.00% is the historical tier-wide measurement that motivated the verification round; the round's own refutation rate is higher because it examined only the adversely selected residual that earlier rounds could not clear.

B.4 Stress robustness of the headline under vendor error

A simulation tests whether the headline's control-chain leg remains within its graded bracket when vendor-sourced records fail at the measured error rate of 26% and are removed. Figure 10 demotes random shares of the vendor-sourced records up to 40%, beyond the measured upper bound, and recomputes the median movement at each rate.

Figure 10: Earlier-pool simulation: stress robustness of the control-chain leg under random demotion



Note: This is a sensitivity simulation on the earlier record pool, preceding the overlap-audit and web-verification demotions, and retained for context rather than regenerated from the current pool. Each panel plots the all-49 median 2024Q4 $\bar{\kappa}$ movement of the verified + vendor-sourced series as a random share of the vendor-sourced records is demoted to the quarantined tier. The two panels demote from different pools. The left panel removes a random share of all 1,381 web-checked vendor-sourced records. The right panel restricts demotion to the 540-record stratum not affirmed against a primary source, so its 5, 10, 20, and 40% demotion rates correspond to 27, 54, 108, and 216 records, equal to 1.96, 3.91, 7.82, and 15.64% of the full 1,381. The shaded band spans the minimum to maximum across 20 fixed seeds at each rate and the marker is the across-seed median; the dashed lines are the graded bracket, the strict series (+0.7057%) above and the verified + vendor-sourced base (−0.7575%) below. Every draw remains inside the bracket and moves toward the strict series as more records are removed, so no demotion rate the measured error could imply pushes the series outside the graded bracket. The 8 records the overlap audit demoted are a realized subset of this stress.